Model Structure (26.1_1Reg):

This is a panmictic model (do_spatial == 0).
There are 2 sexes, 2 fishing fleets, 3 survey fleets, and 1 regions being modeled.
The model has 67 years, starting in 1960 and ending in 2026.
There are 30 ages, where recruitment occurs at age−2 and the plus group occurs at age−31.
There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model estimates new growth curves for the post−1996 time block (upd_growth == 1, use_new_bio == 2) and uses the 2021 SAFE growth estimates for the pre−1996 time block (use_hist_grwth_blk == 1, do_rec_growth == 0). A new weight curve is estimated with a single block (do_rec_WAA == 0).
For fishery composition data aggregation catch−weighting, the newly estimated sex−disaggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 1).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) −− no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2026, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations were NOT included in the LLS RPN index (drop_WAI_LLS == 1).

The longline survey used carryover values for off−year BS (even years) or AI (odd years) area RPNs in all index years (do_LLS_carry == 1).

The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

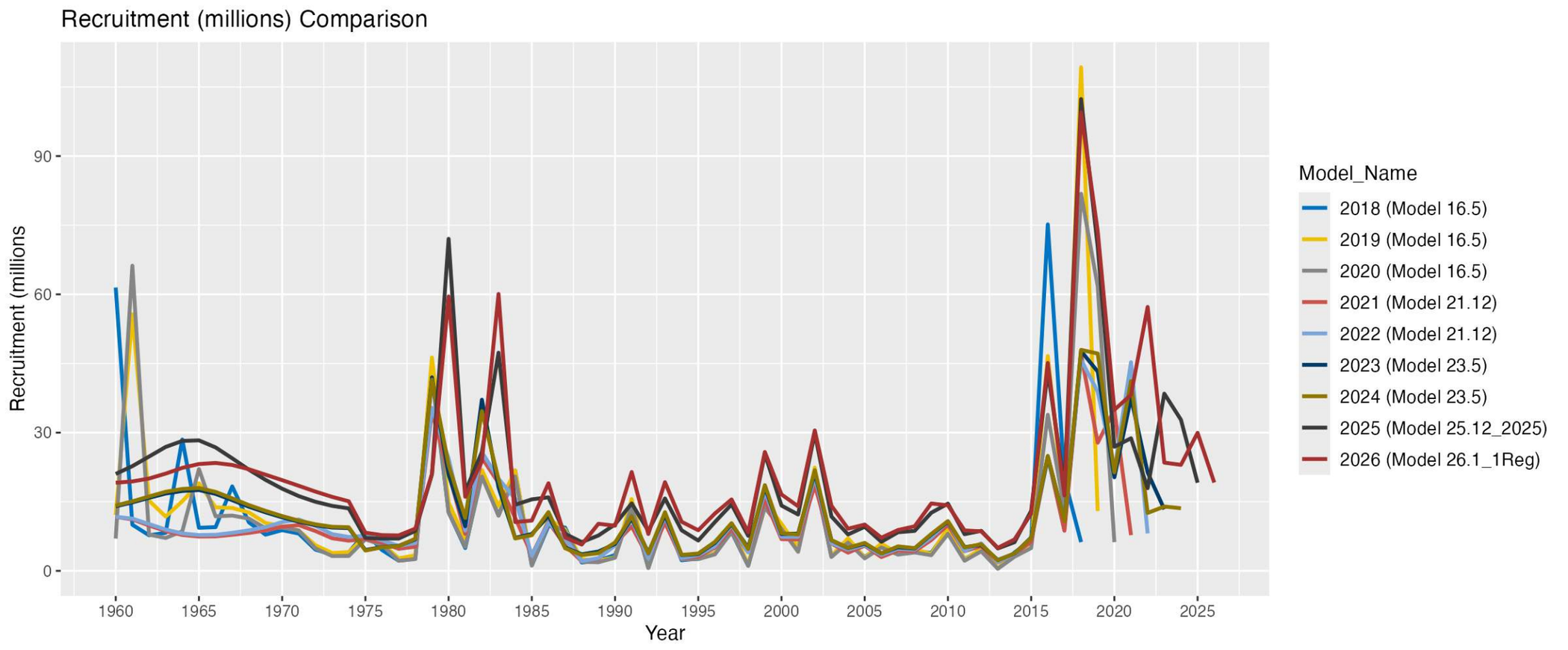
Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years) (do_spatial == 0 & sex_age_comps == 5).
The Input Sample Size (ISS) for composition data is scaled to a maximum value across years/sexes to address interannual variability (do_spatial == 0 & do_scale_ISS == 1).

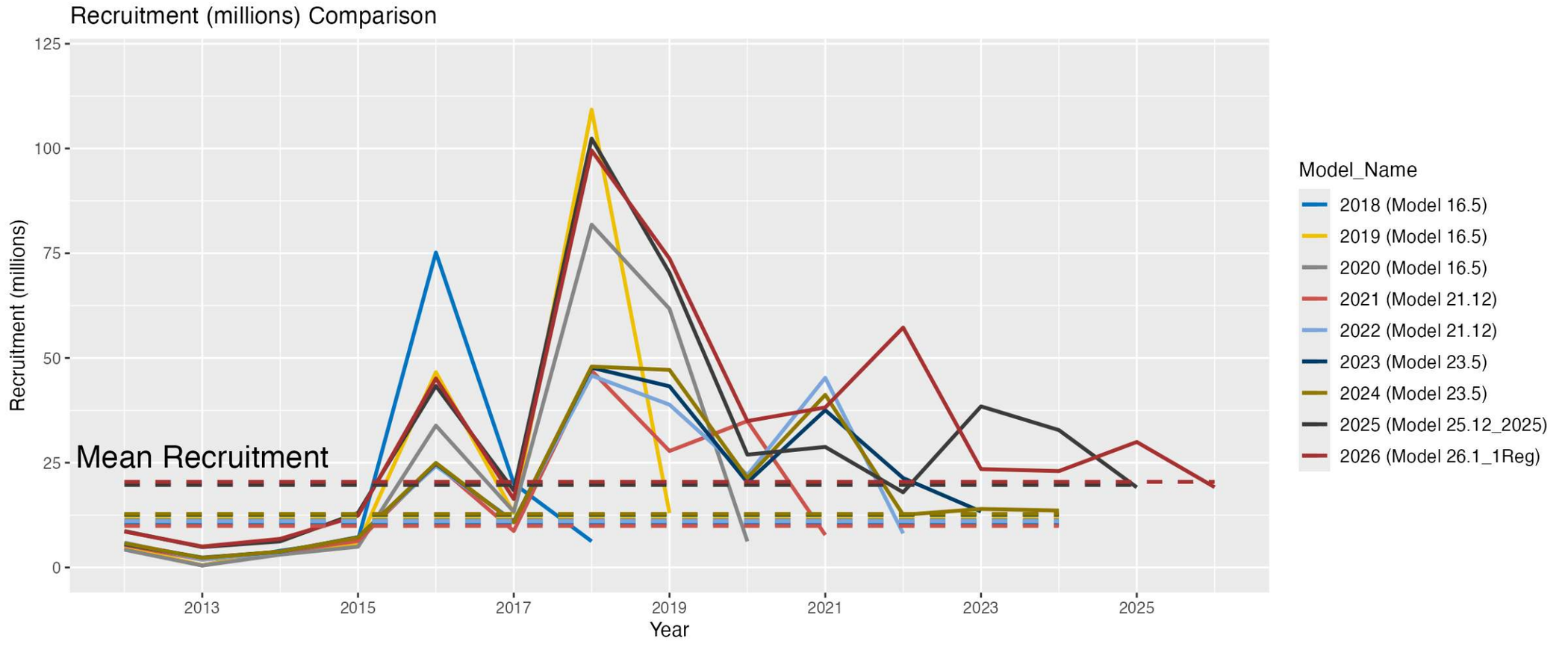
Length Composition Data Use:

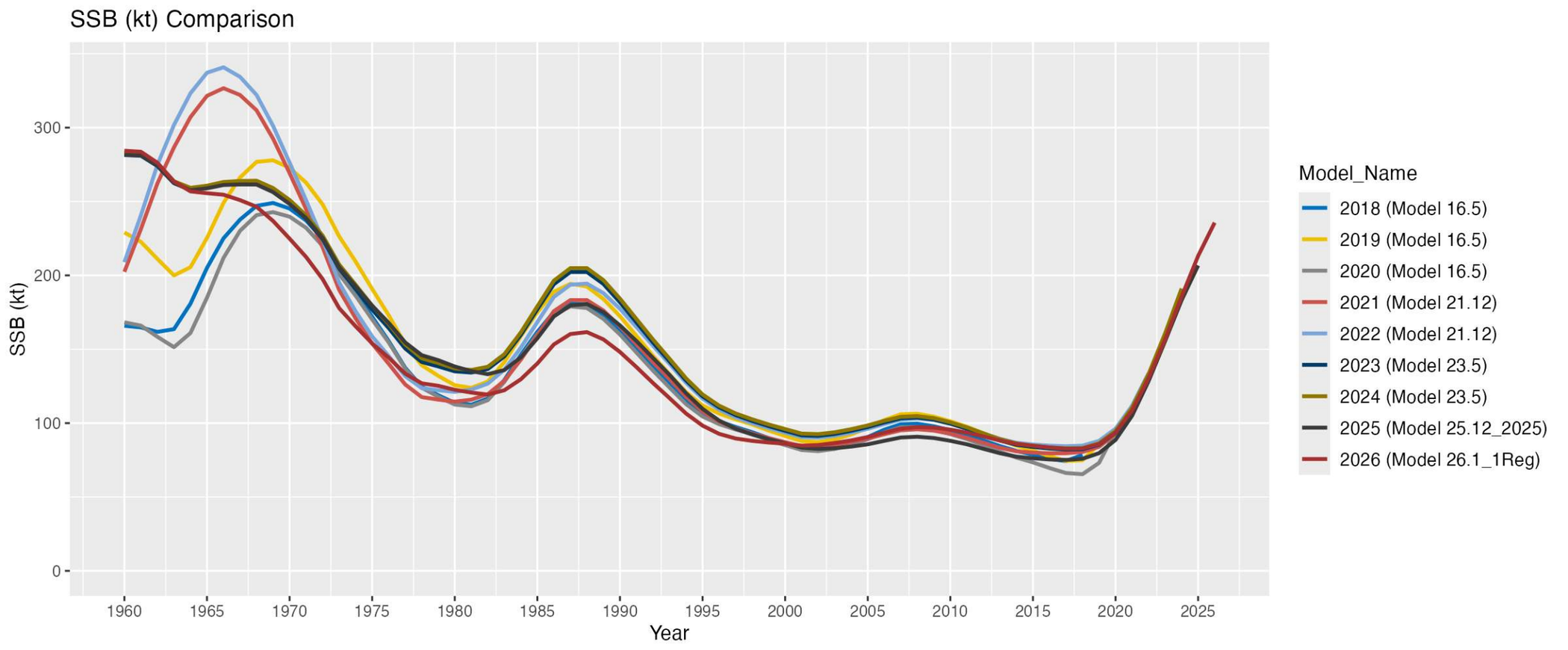
The longline survey length composition data is weighted/calculated using the raw length data and weighted by area RPNs consistent with the age composition data (LLS_length_comp_source == 1).
No longline survey length data are fit in the model (this does not apply to JPN LLS lengths) −− only age data are used (fit_LLS_len == 0).
No fixed gear fishery length data are fit in the model (this does not apply to trawl fishery lengths) −− only age data are used (fit_LLF_len == 0).

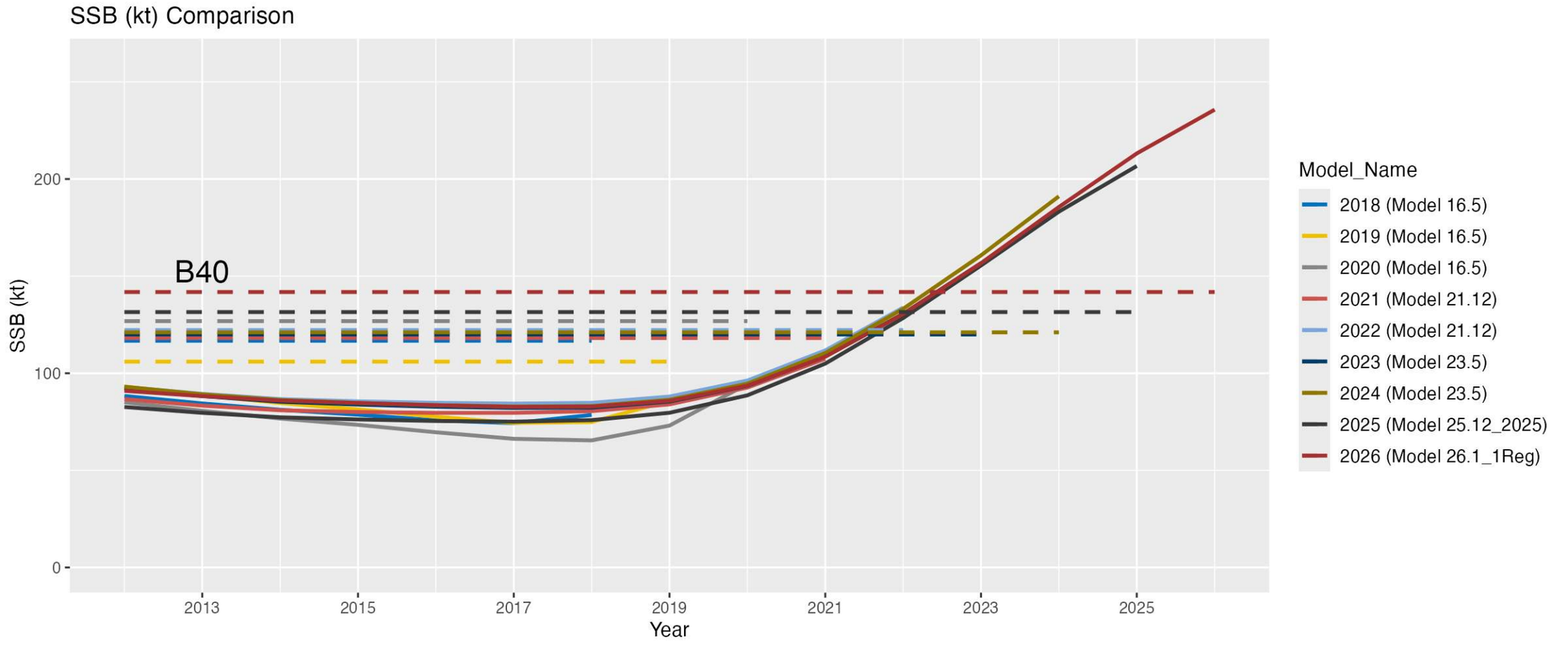
General Data Fitting Notes:

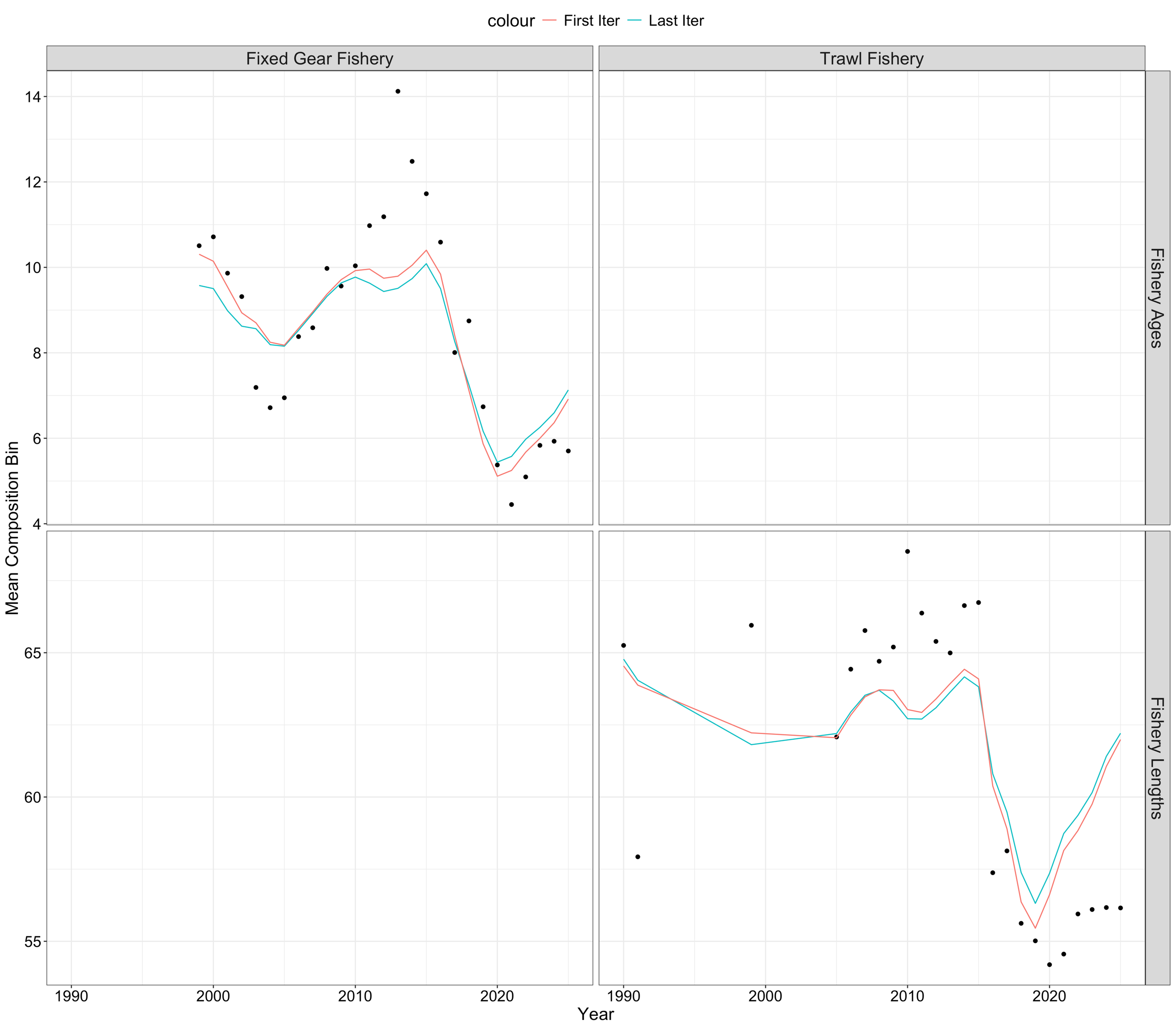
Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).
The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).
The fishery CPUE index was NOT fit in this model (use_CPUE == 0).
Non−fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where all sources of removals were available for terminal year (add_non_fish_catch == 1 & non_fsh_catch_updated == 1).


Recruitment (millions) Comparison
Recruitment (millions
0
30
60
90
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg)


Recruitment (millions) Comparison
Recruitment (millions)
0
25
50
75
100
125
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg)


SSB (kt) Comparison
B40
SSB (kt)
200
100
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.1_1Reg)


colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year

colour — First Iter — Last Iter

Japan LL Survey

NOAA LL Survey

Survey Ages

Survey Lengths

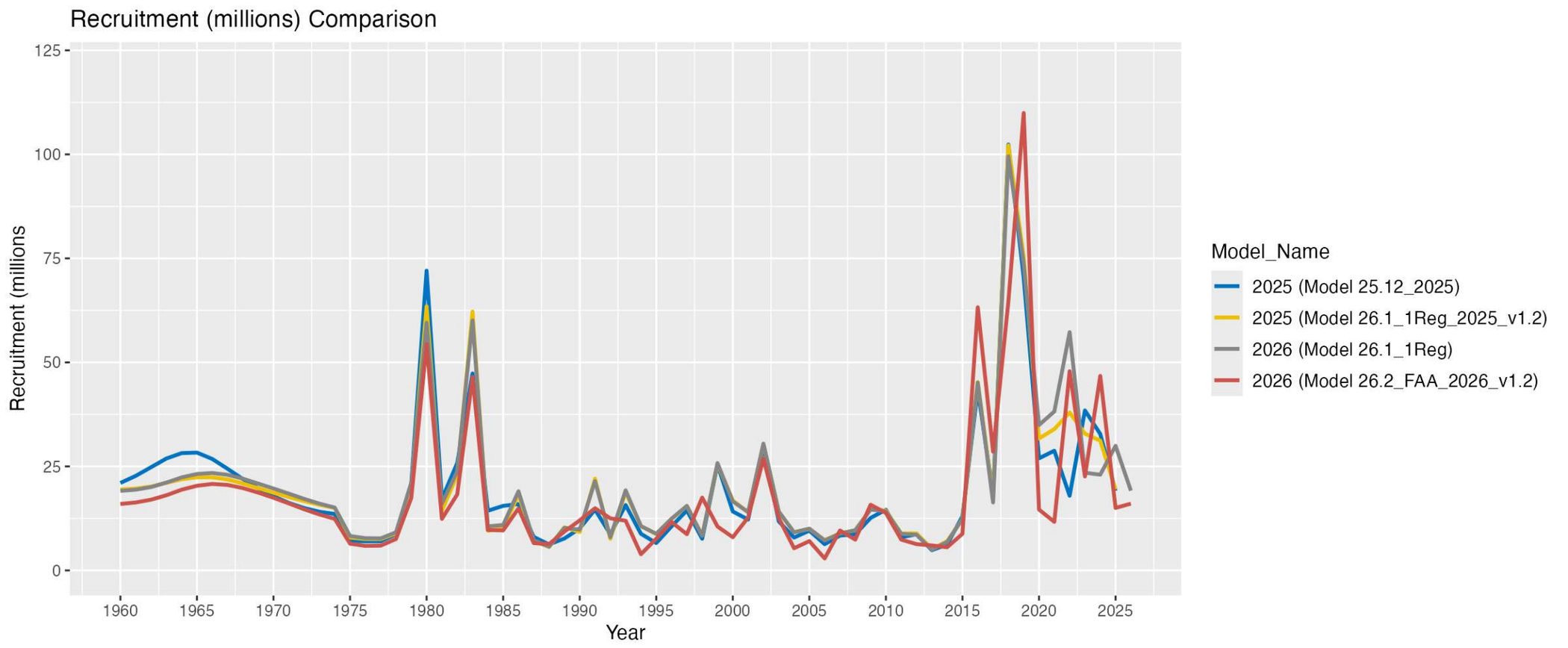
Mean Composition Bin

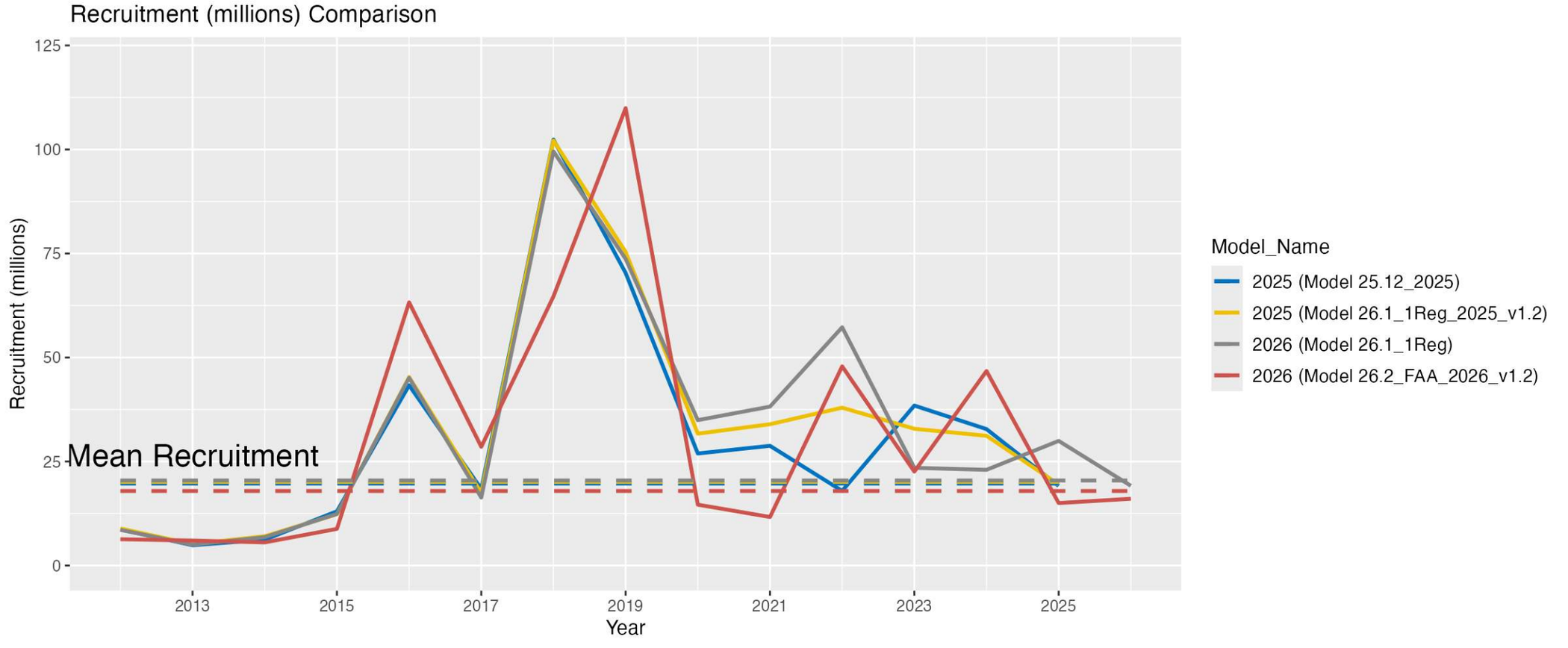
12
10
8
6
4

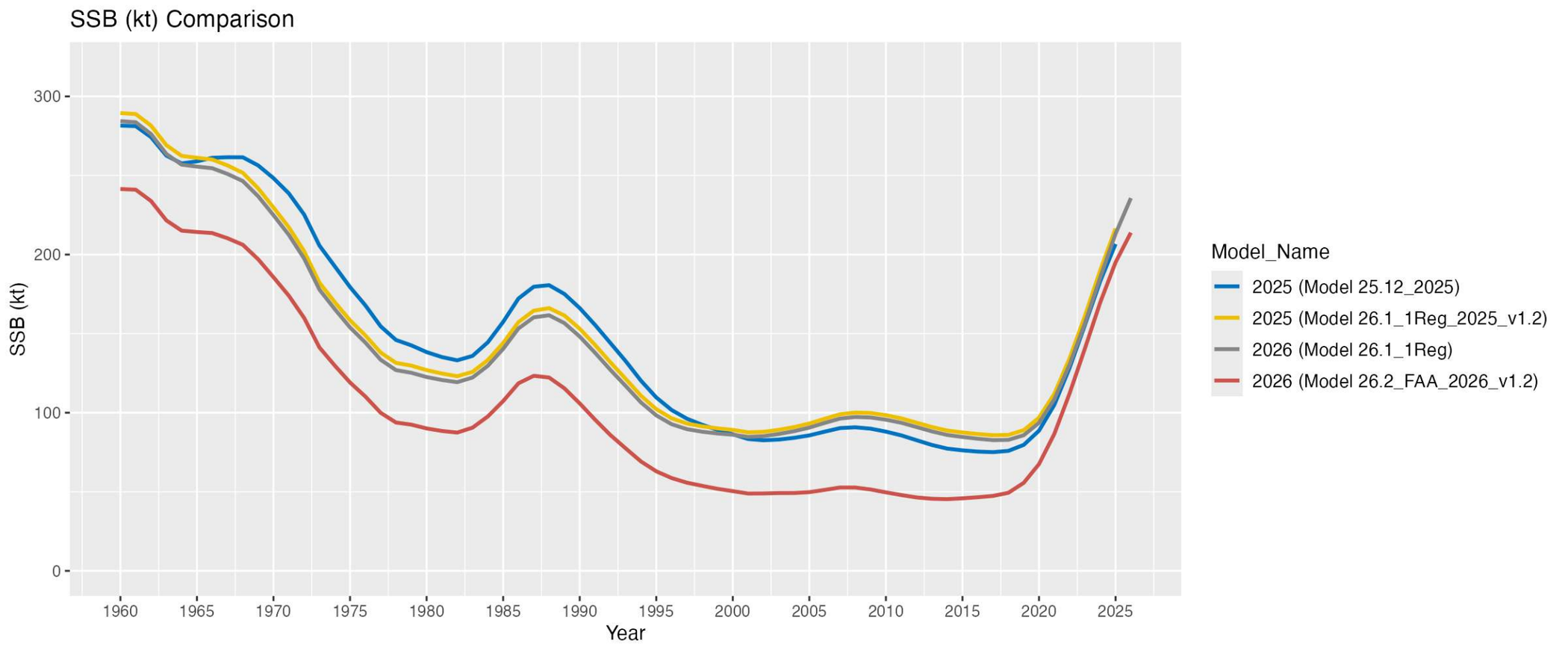
67.5
65.0
62.5

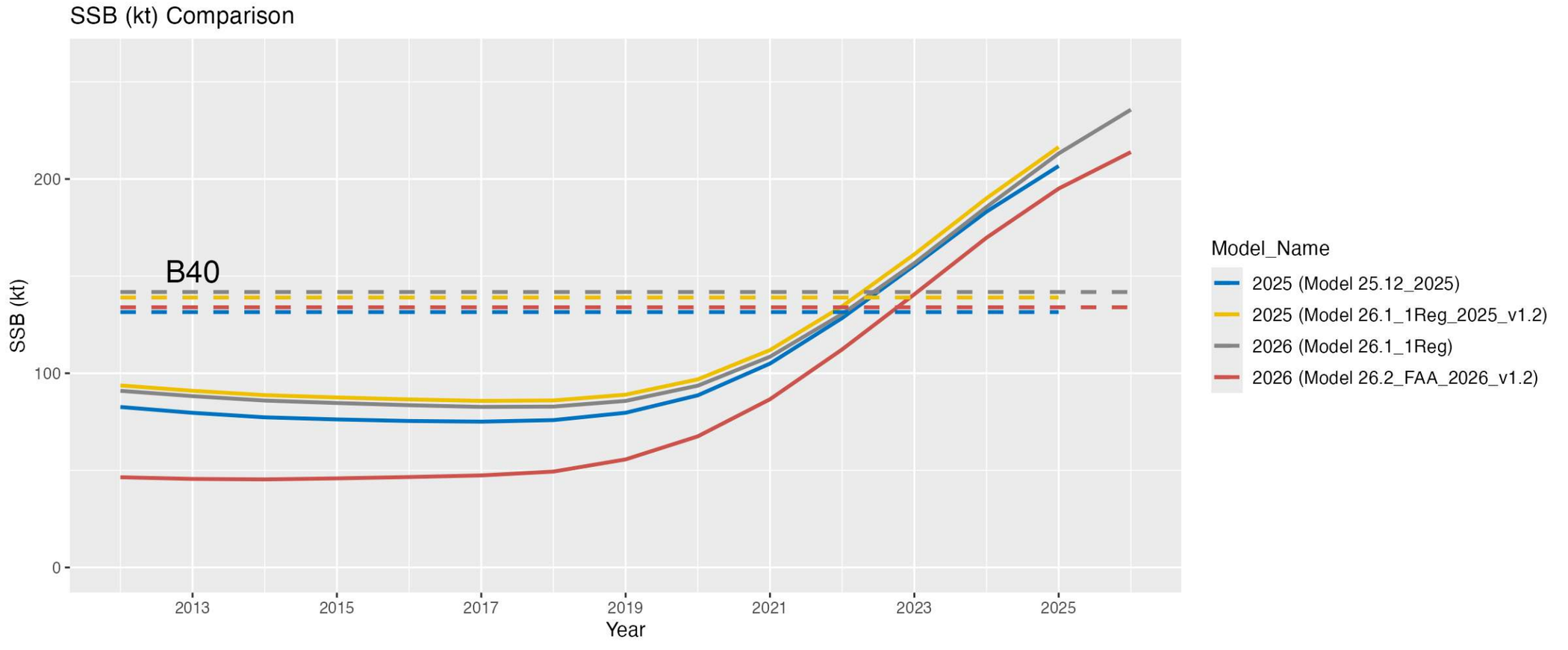
1980
1990
2000
2010
2020

Year


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg)
2026 (Model 26.2_FAA_2026_v1.2)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg)
2026 (Model 26.2_FAA_2026_v1.2)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg)
2026 (Model 26.2_FAA_2026_v1.2)


SSB (kt) Comparison
B40
SSB (kt)
200
100
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2026 (Model 26.1_1Reg)
2026 (Model 26.2_FAA_2026_v1.2)

Model 26.1_1Reg Population Population 1

Female | Male | AK

Spawning Weight-at-Age (kg)

0.075
0.050
0.025
0.000

0 10 20 30

Age

Model 26.1_1Reg Population Population 1

Female

Male

AK

Maturity at Age

1.00

0.75

0.50

0.25

0.00

0

10

20

30

Age

Model 26.1_1Reg Population Population 1

Female

Male

AK

Spawning Weight at Age

6

4

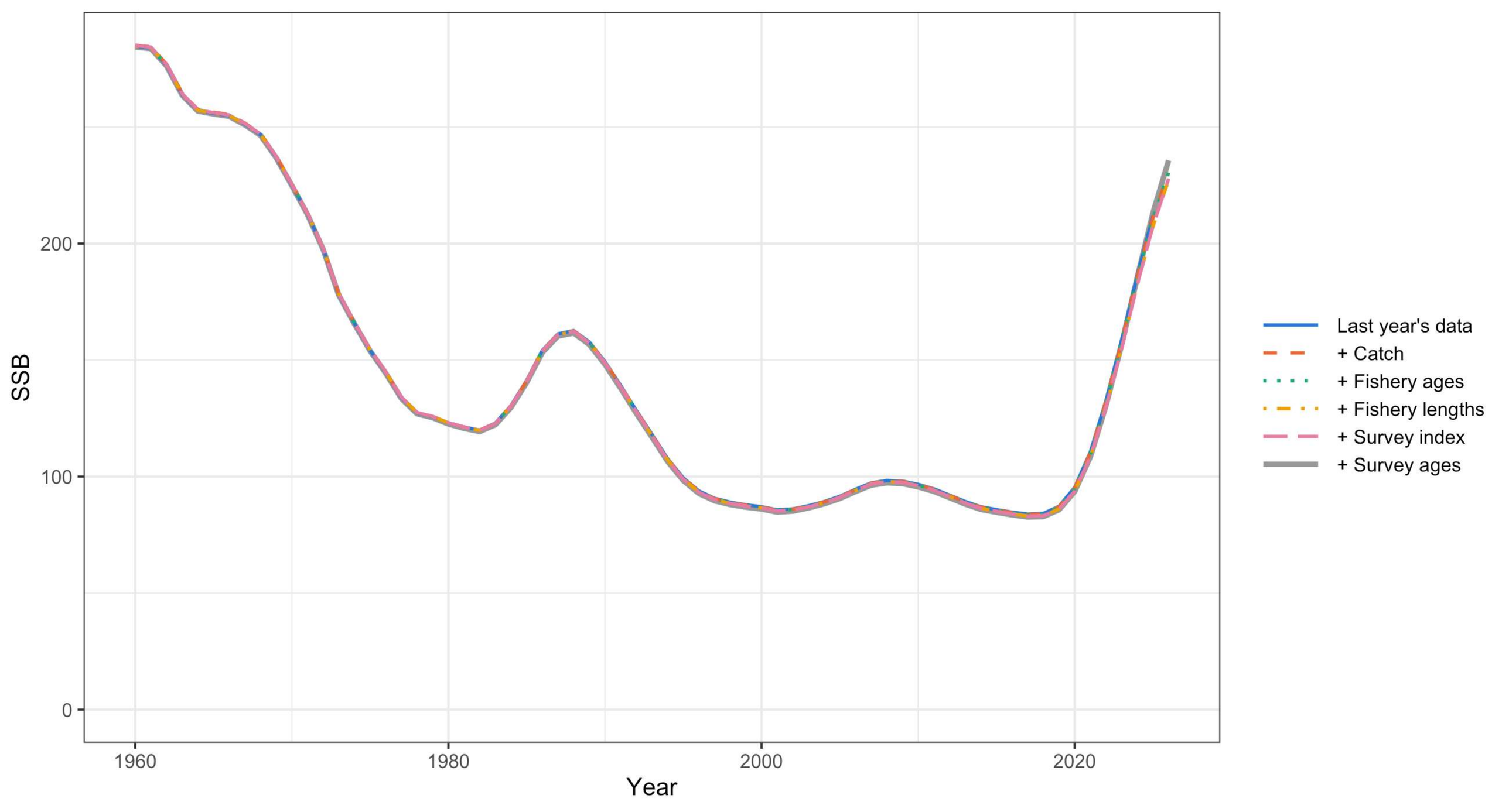
2

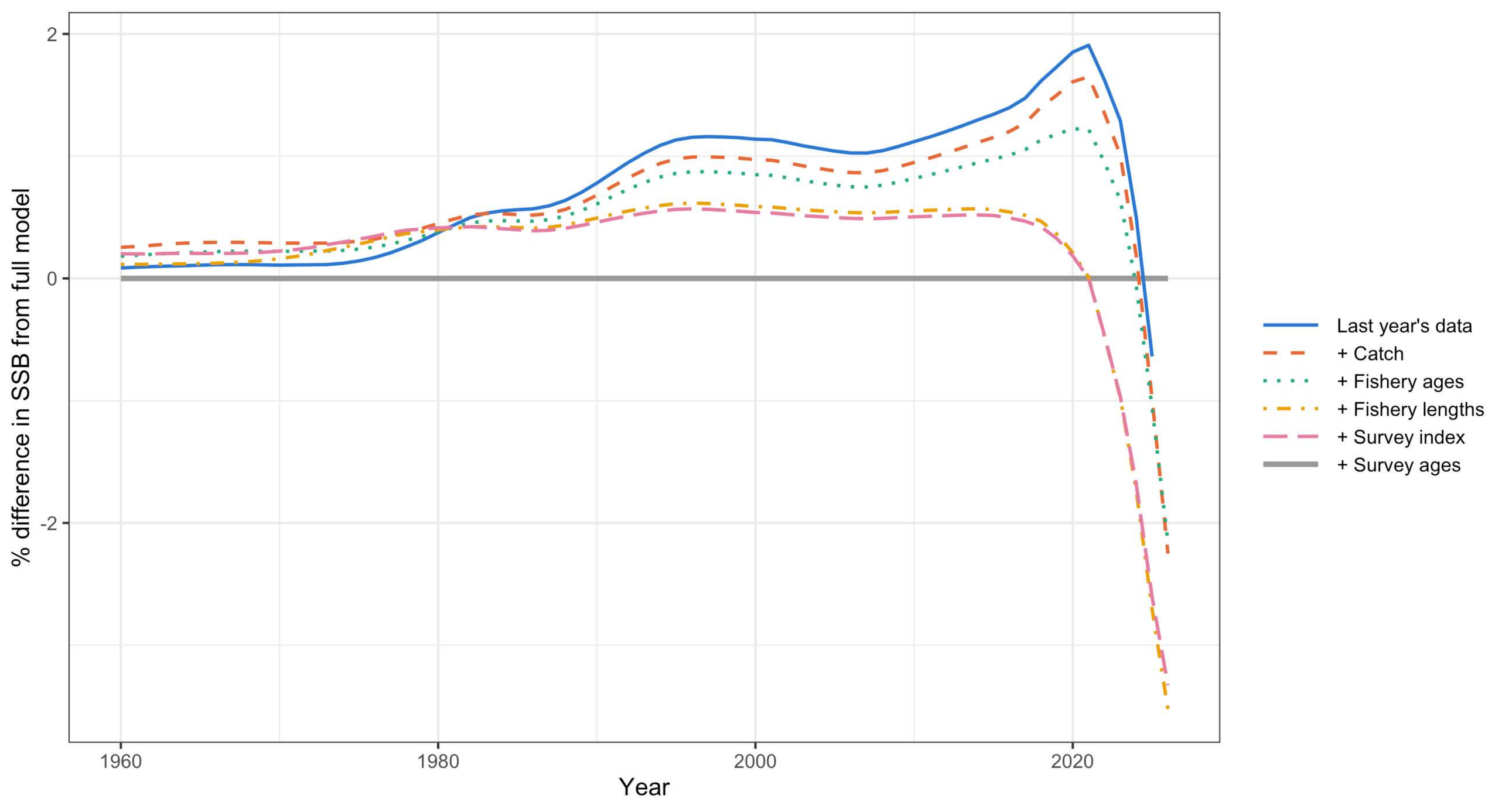
0

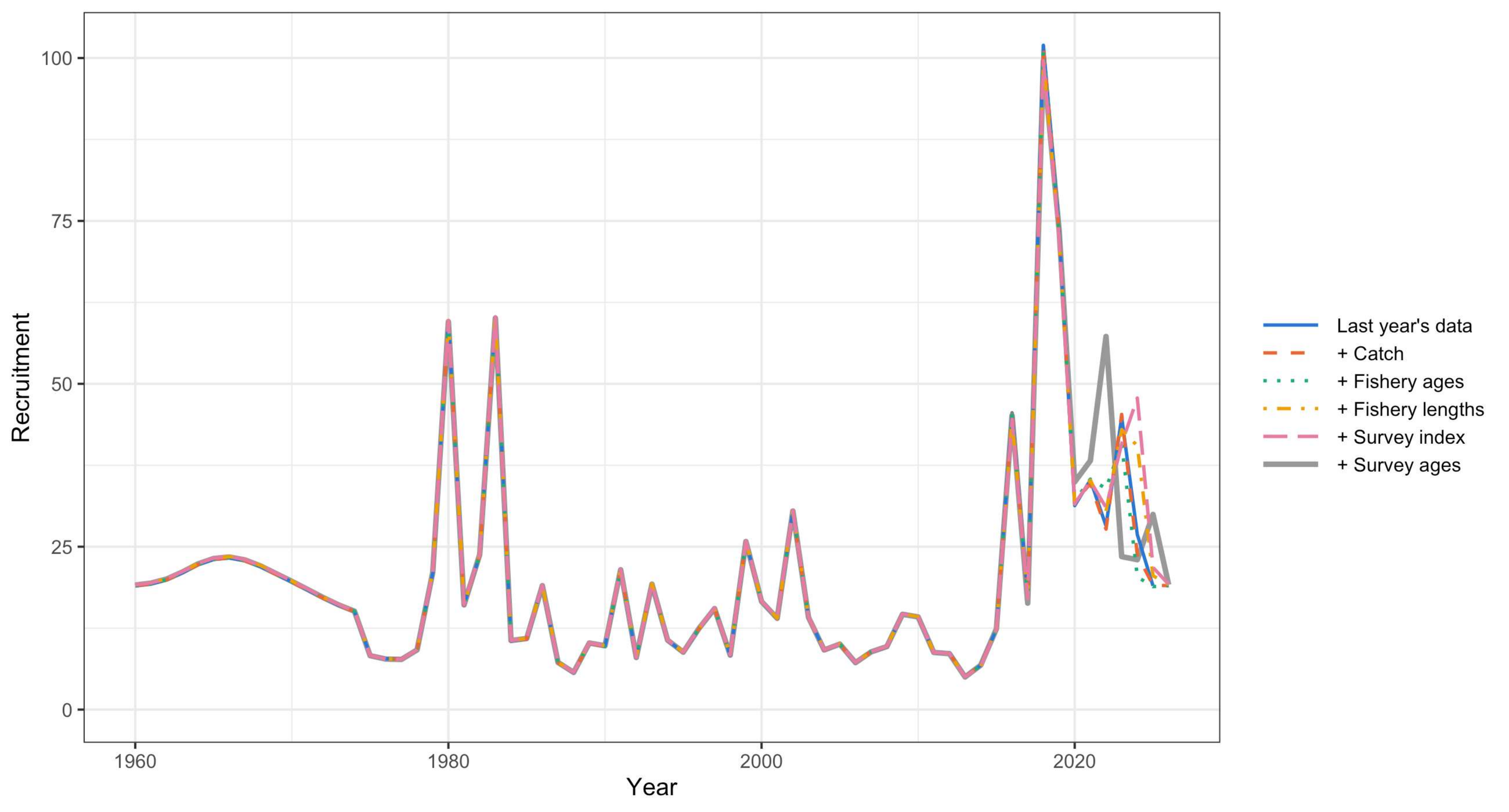
0 10 20 30

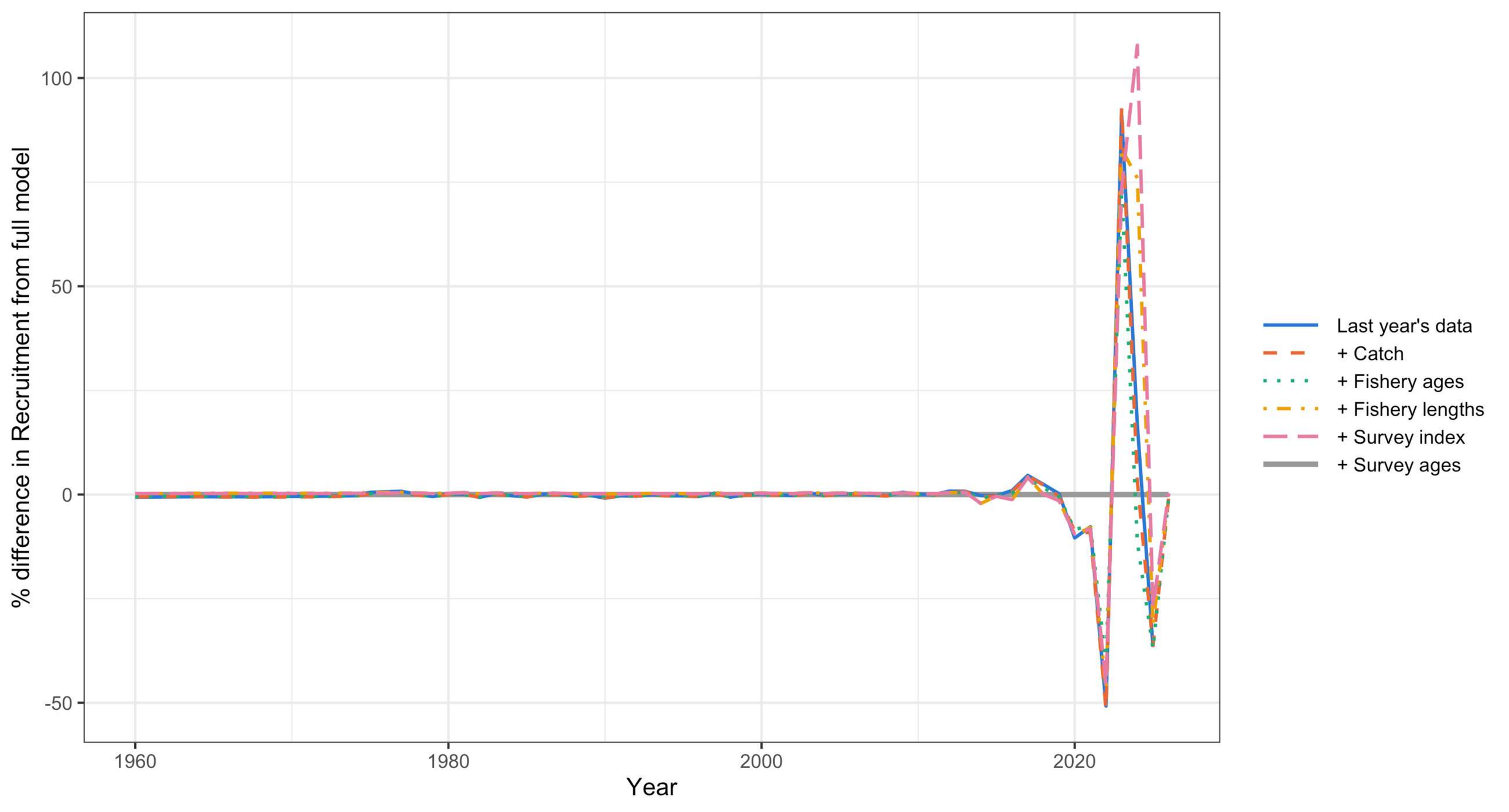
0 10 20 30

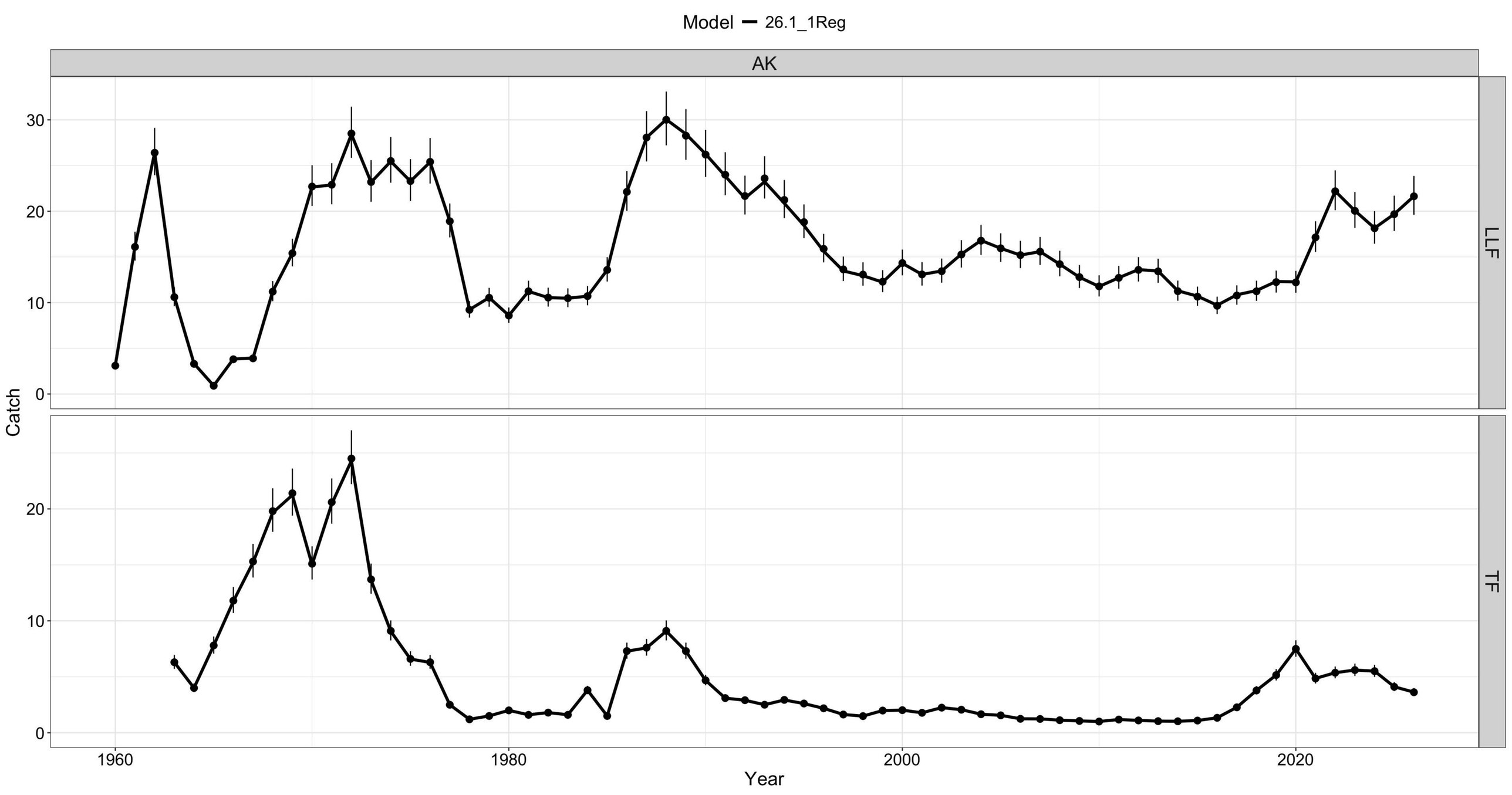
Age


SSB
200
100
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


2
0
-2
% difference in SSB from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


Recruitment
100
75
50
25
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


100
50
0
-50
% difference in Recruitment from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


Model — 26.1_1Reg
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

AK

JPN LLS Lengths

JPN LLS Index

LLS Index

JPN LLS Ages

LLS Ages

TF Lengths

TF Catch

LLF Catch

LLF Ages

26.1_1Reg

1960

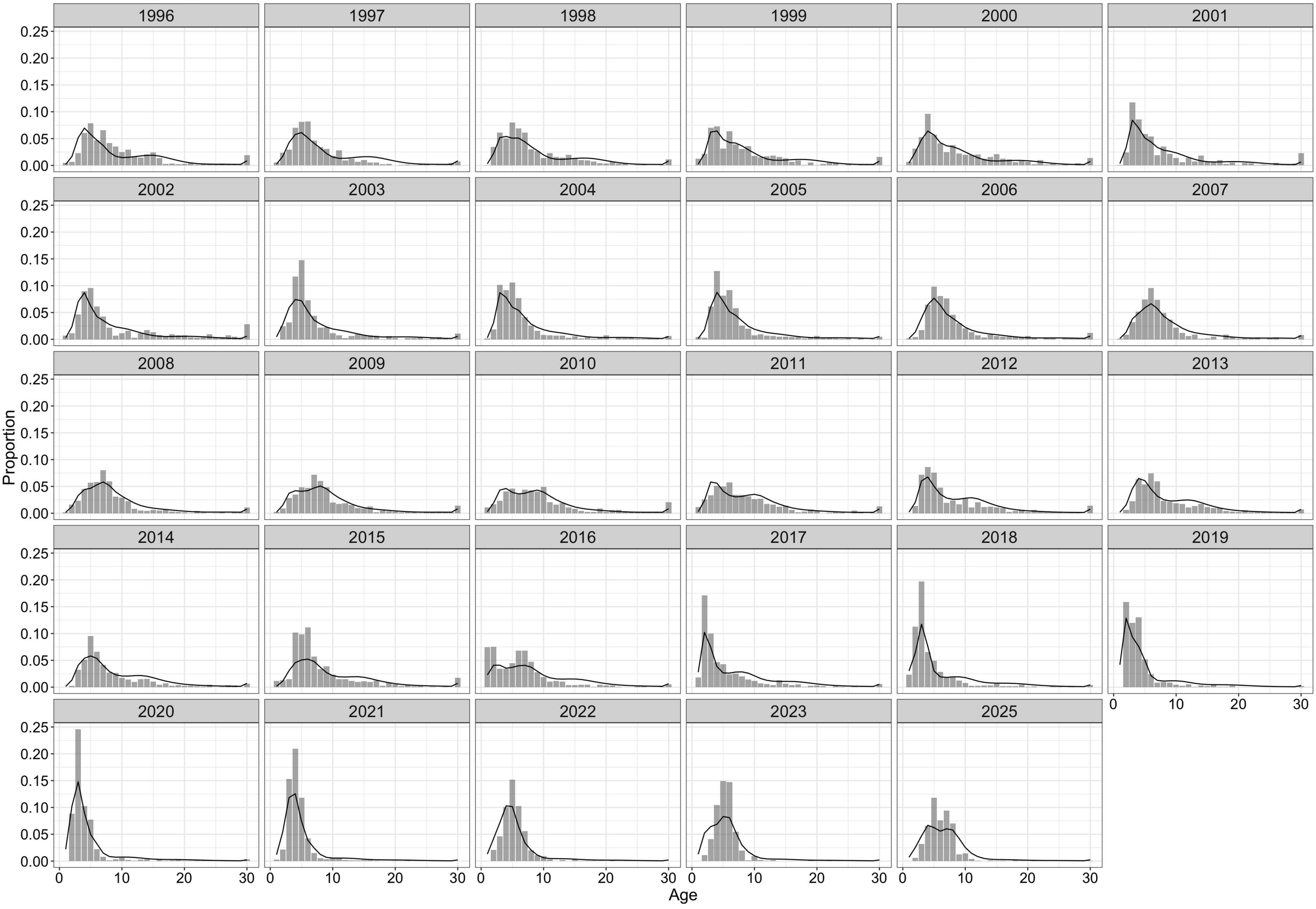
1980

2000

2020

Year

LL Survey Age Comps Female AK



LL Survey Age Comps Male AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

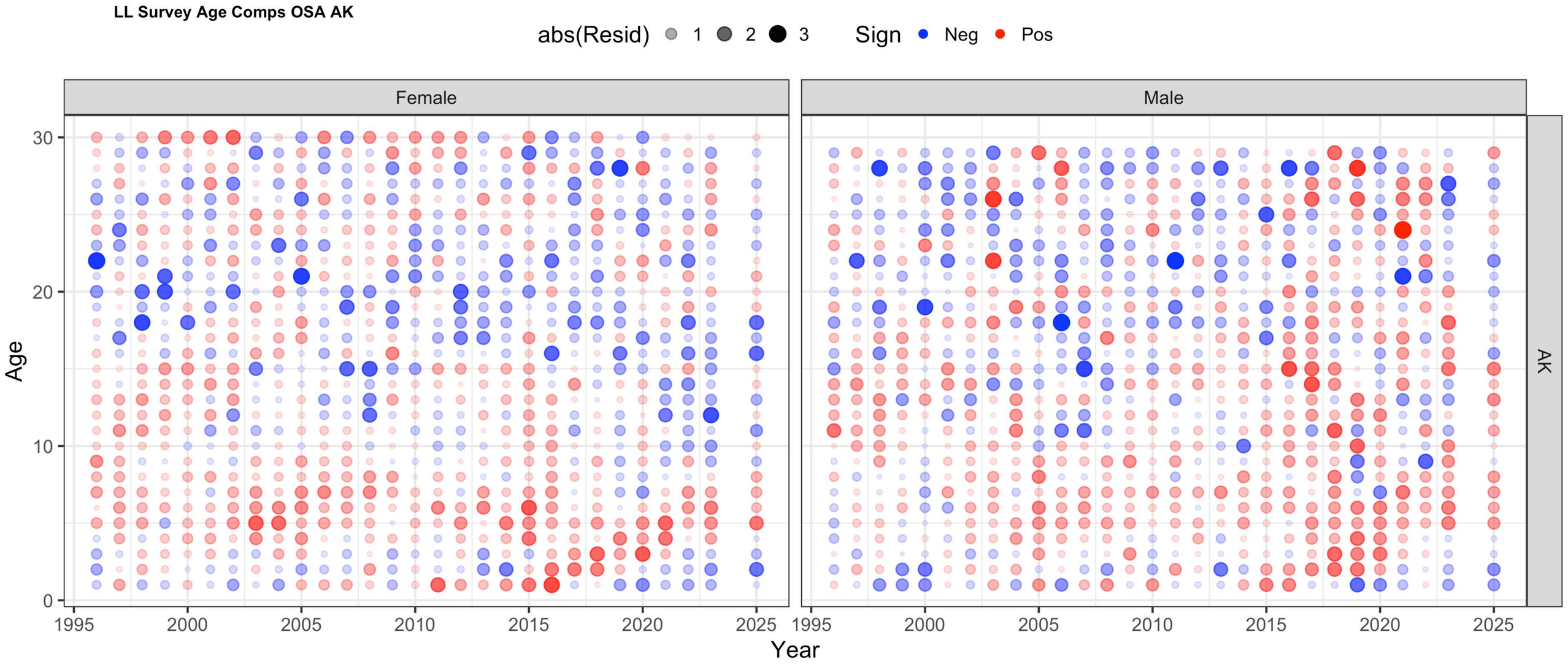
Proportion

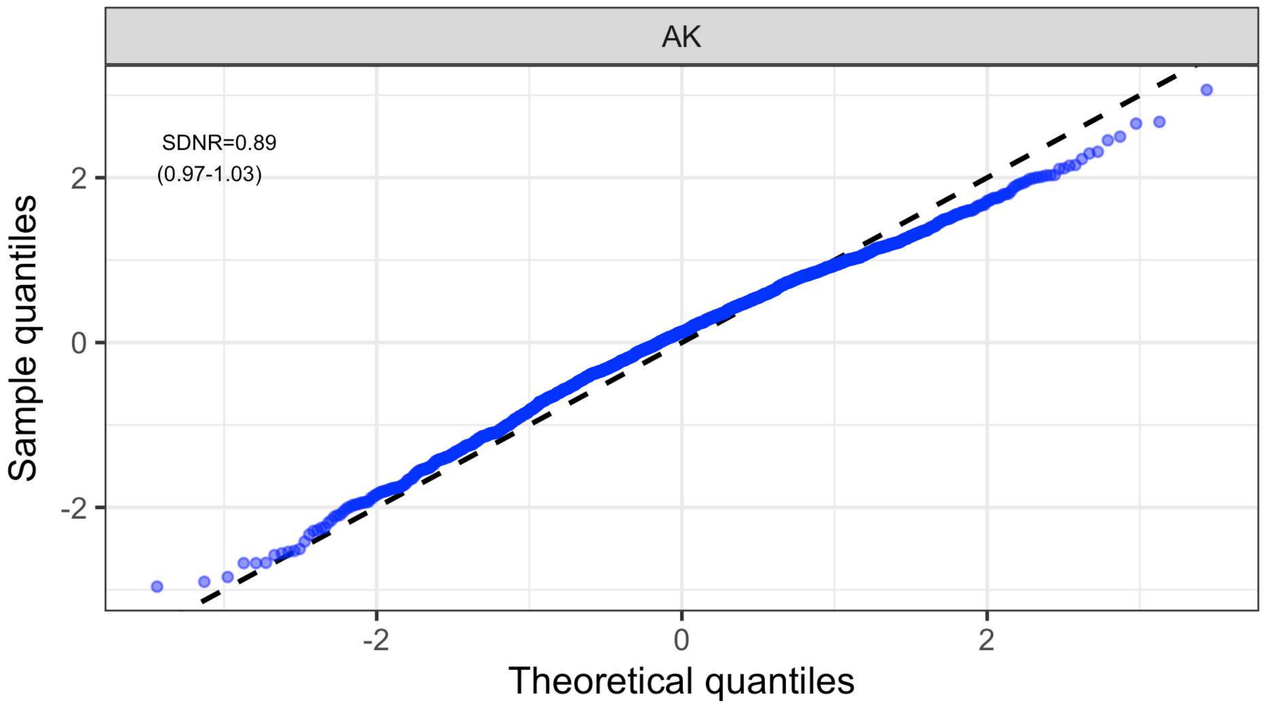
0.00 0.05 0.10 0.15

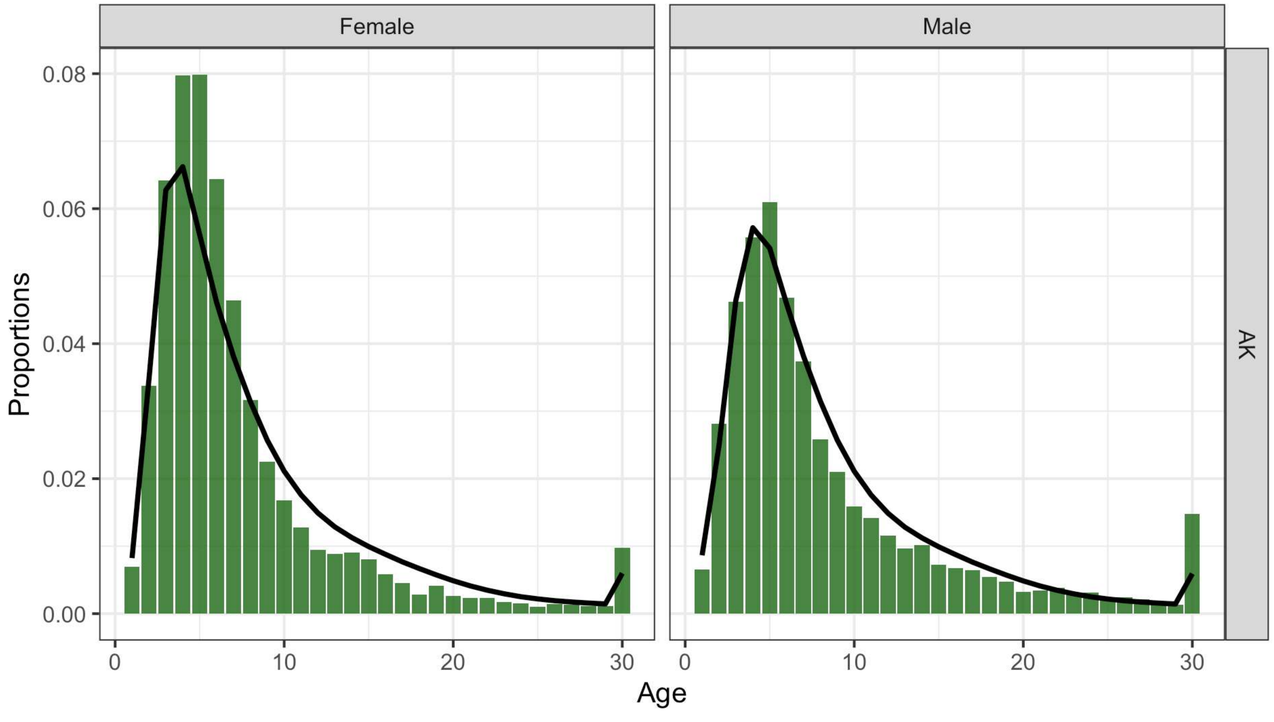
0 10 20 30

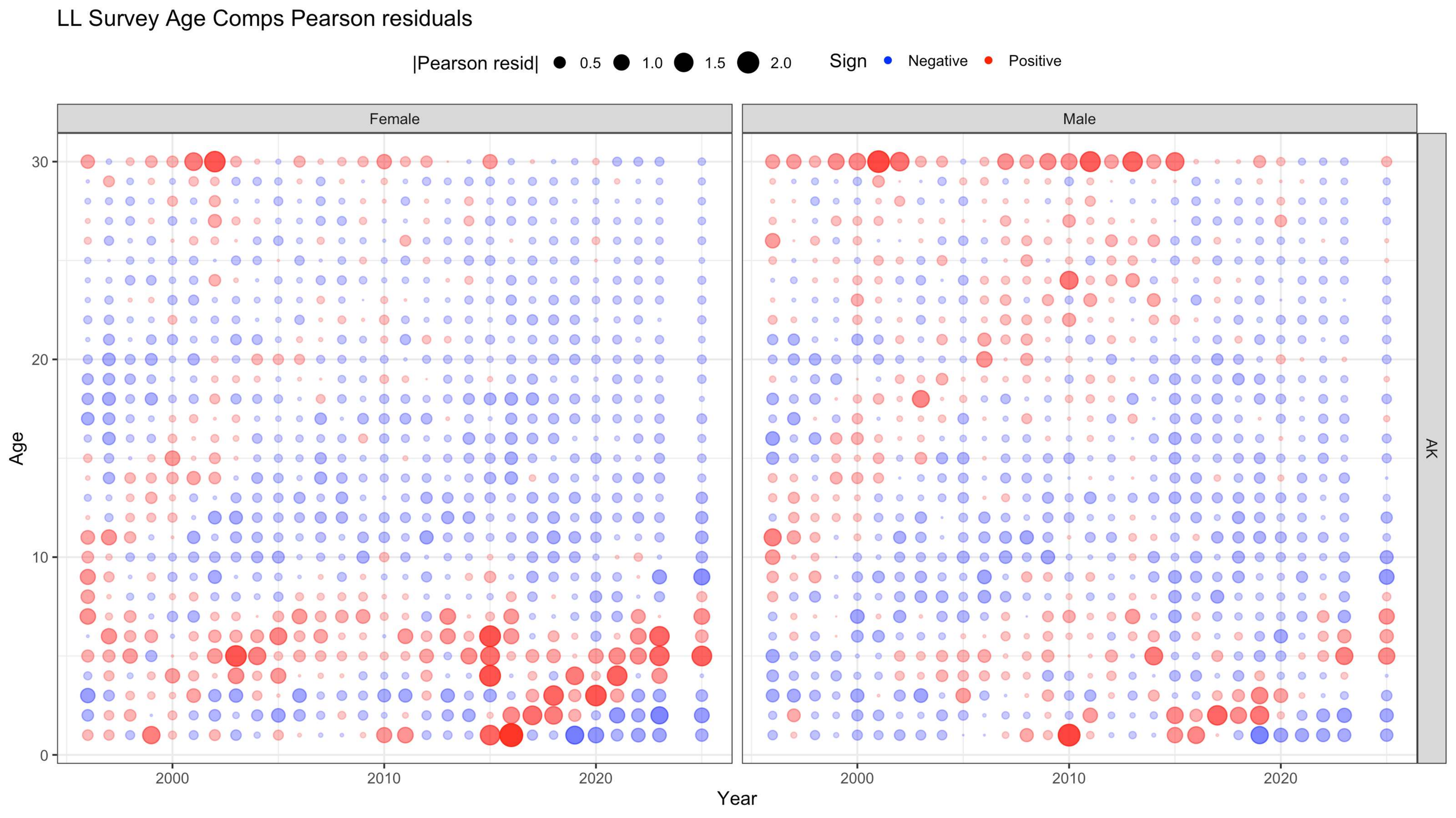
Age

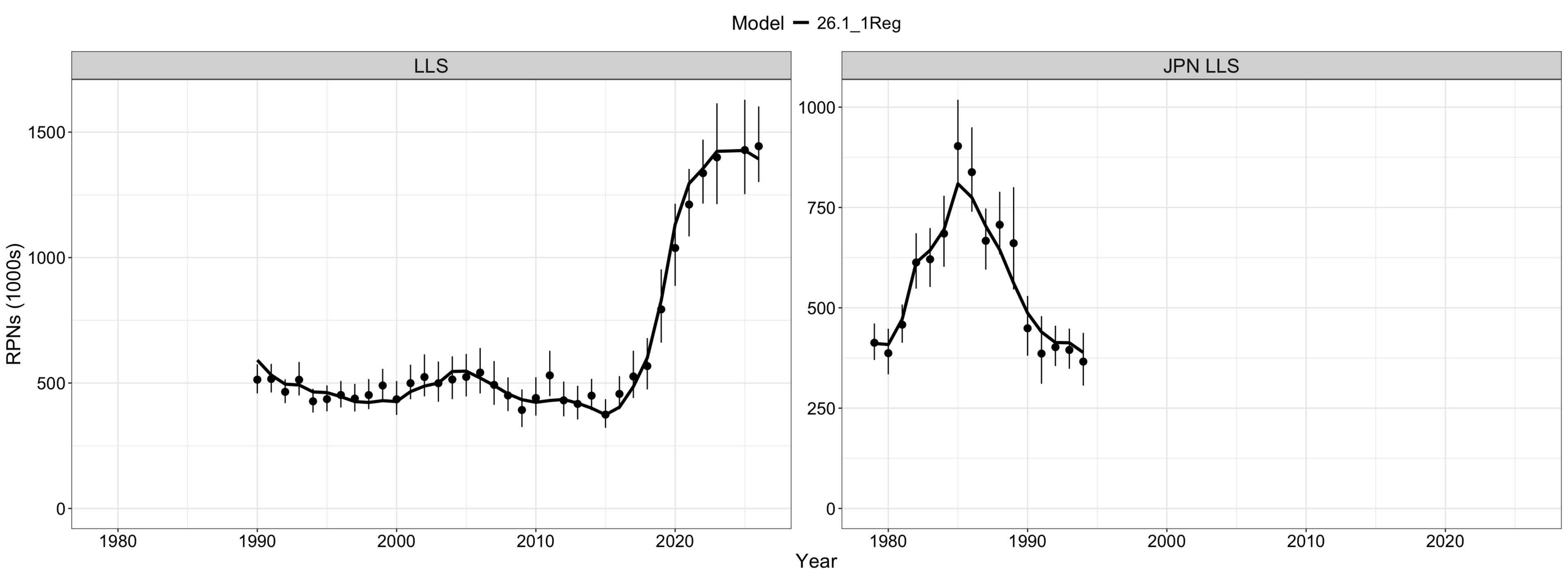
LL Survey Age Comps OSA AK



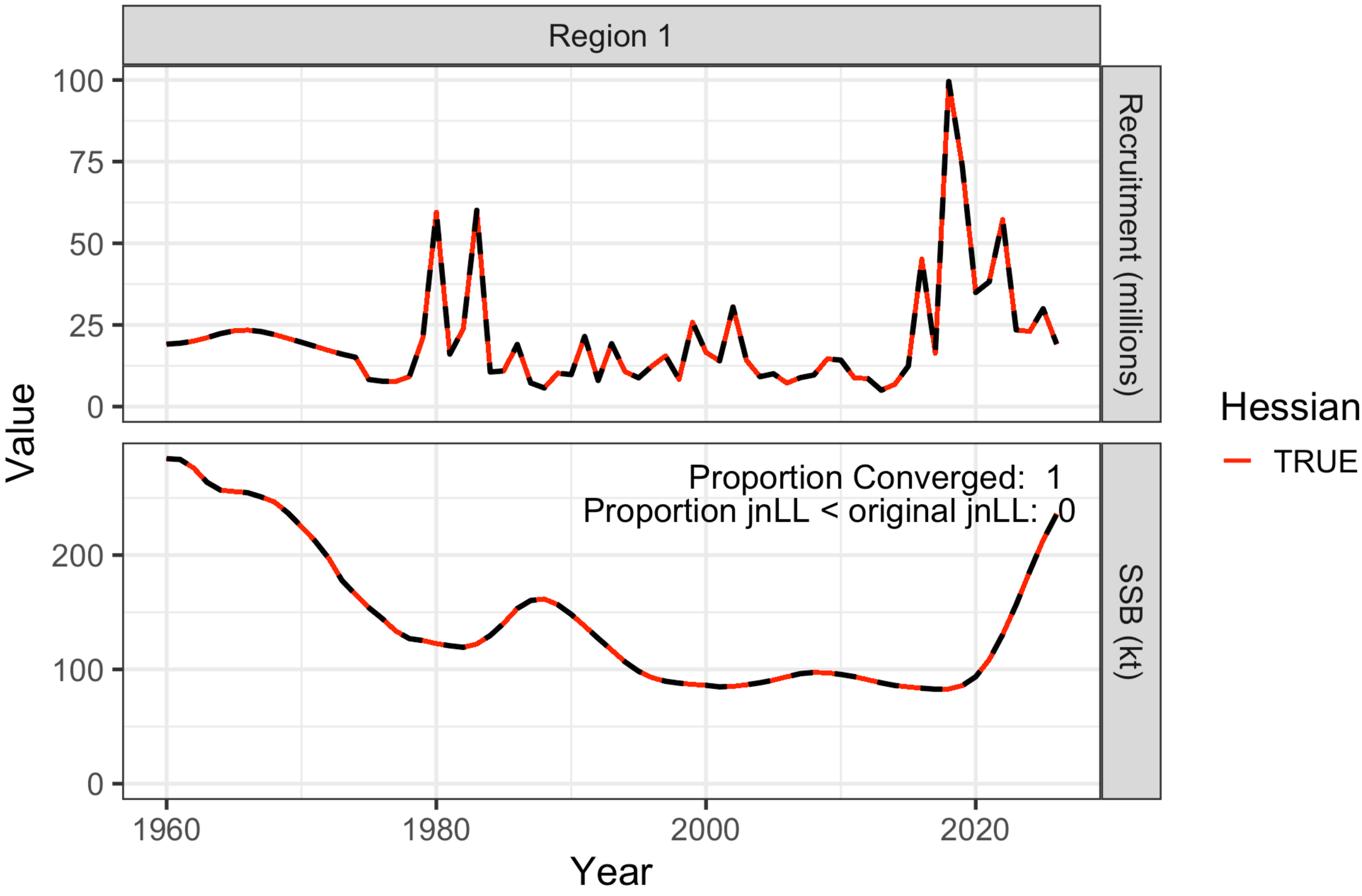




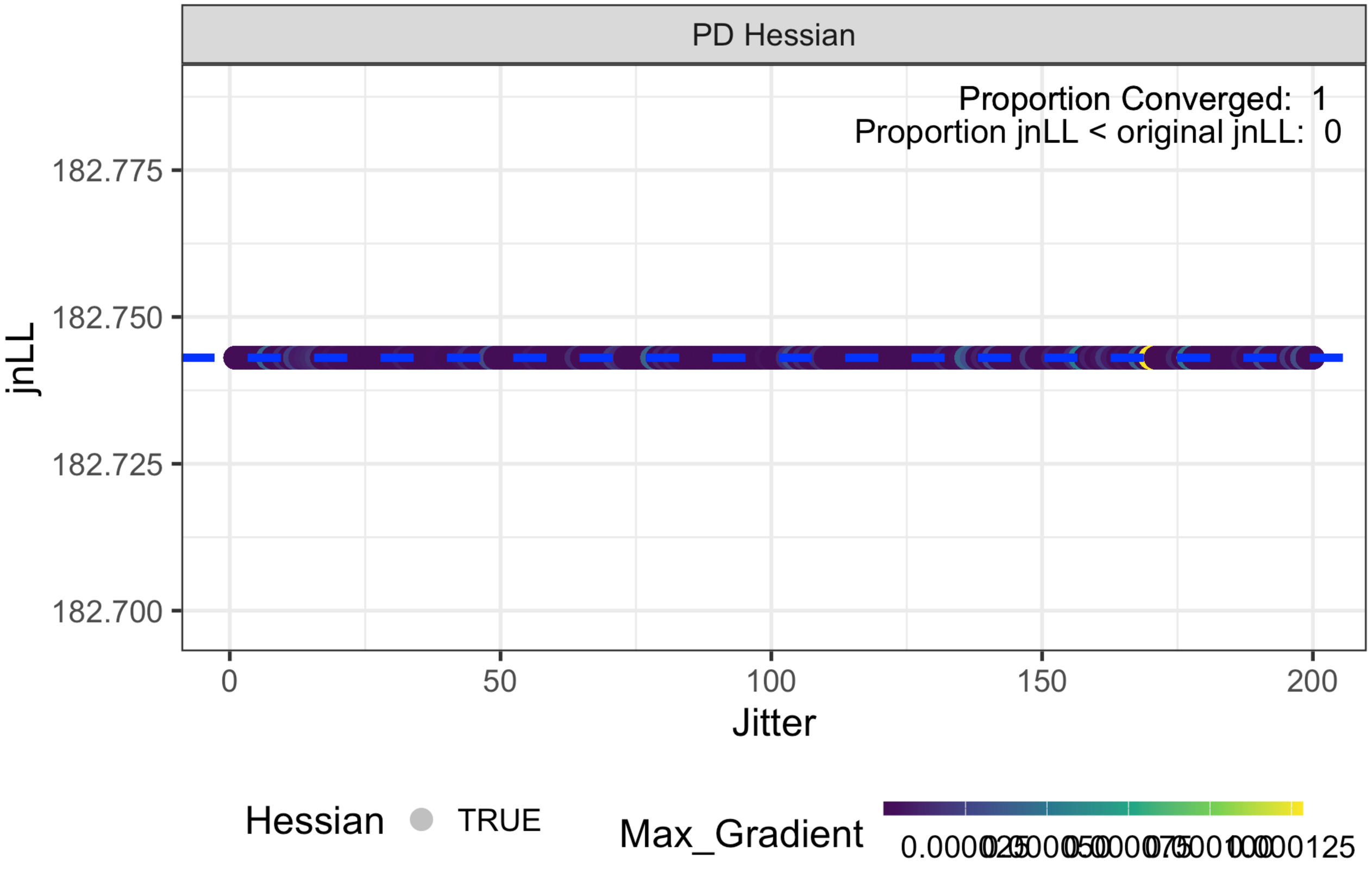

LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

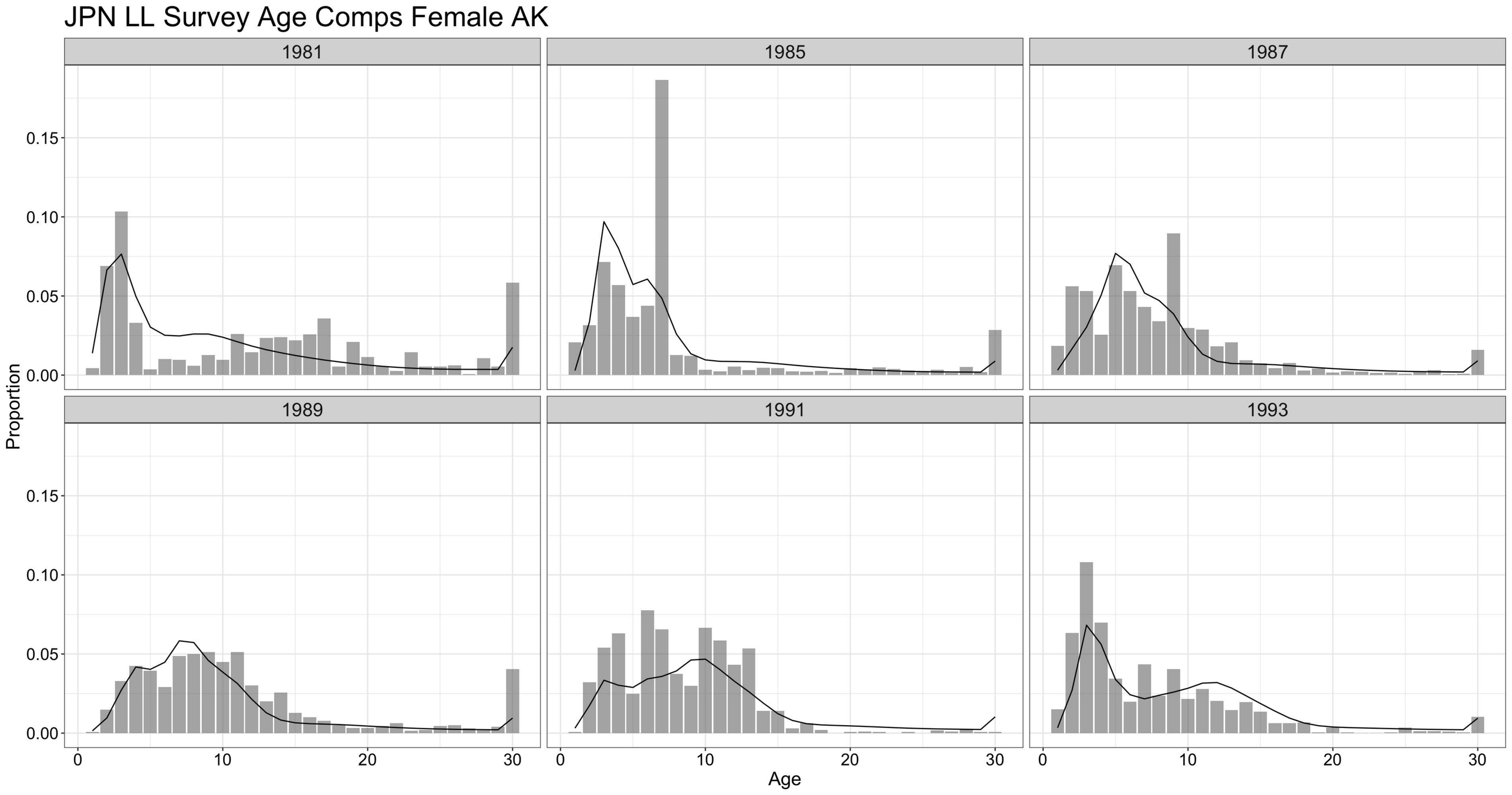

Model 26.1_1Reg
LLS
JPN LLS
RPNs (1000s)
Year
0
500
1000
1500
250
750
1980
1990
2000
2010
2020

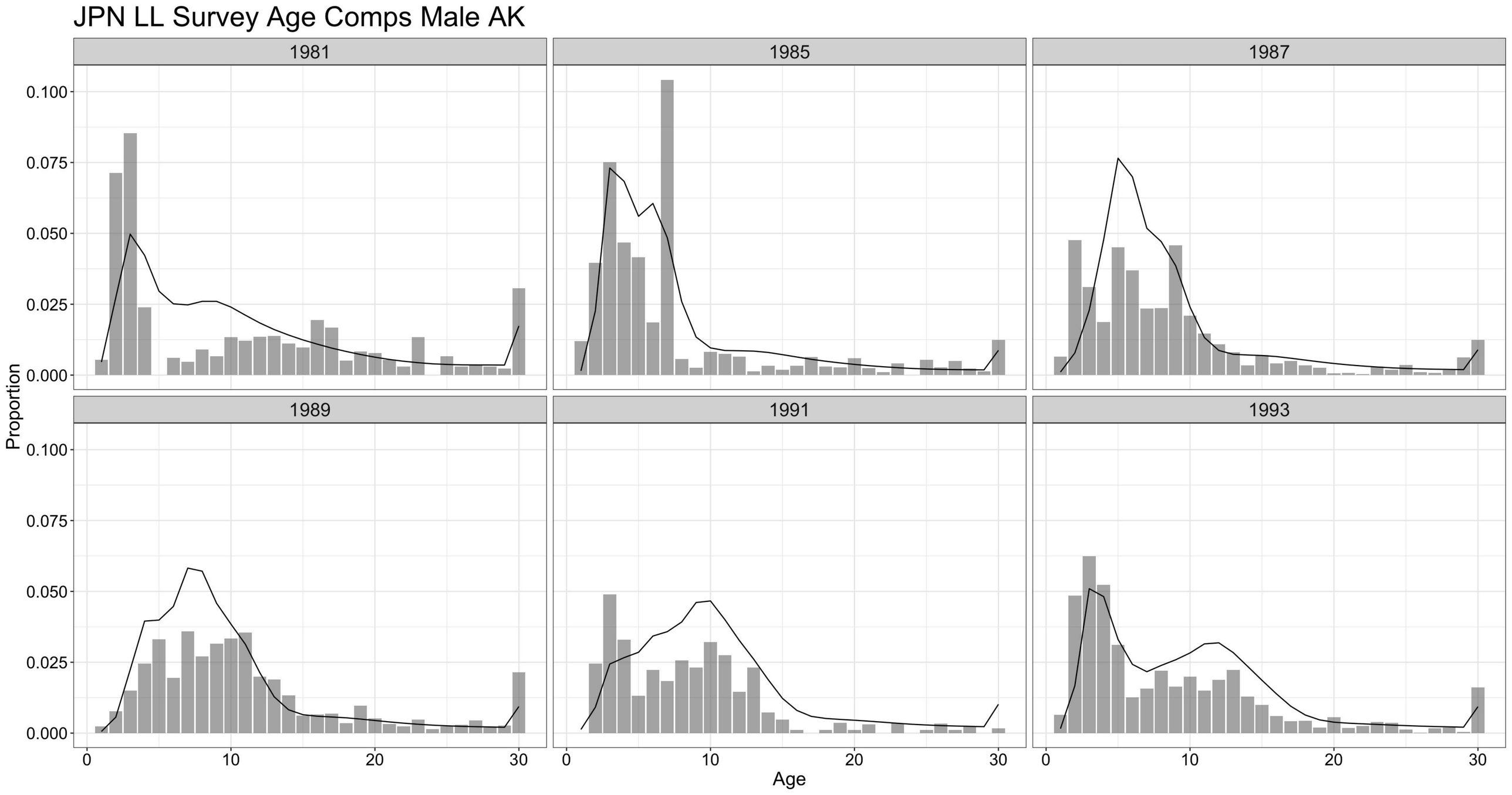
Jitter SSB and Recruitment



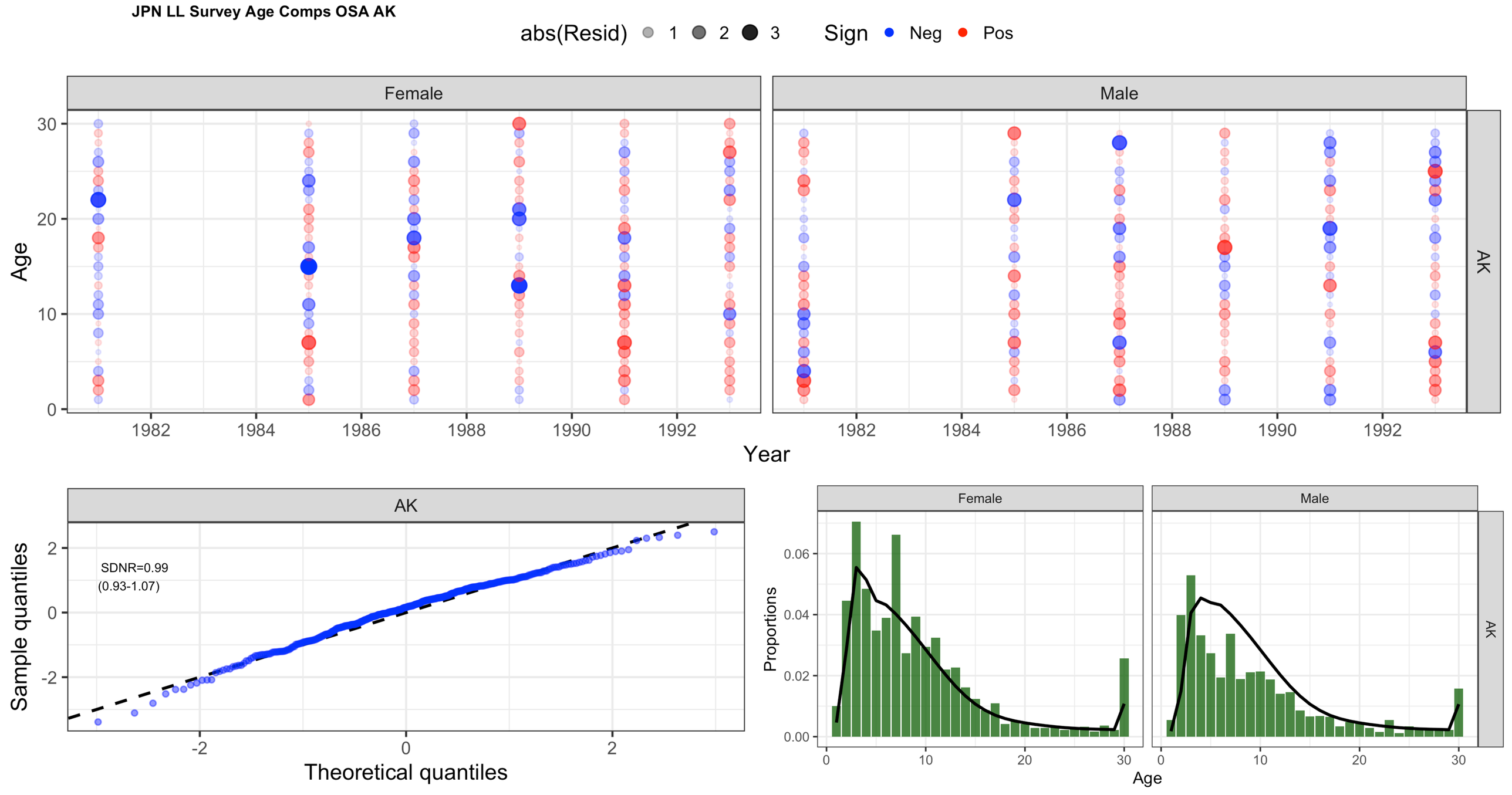
Jitter nLL and Convergence

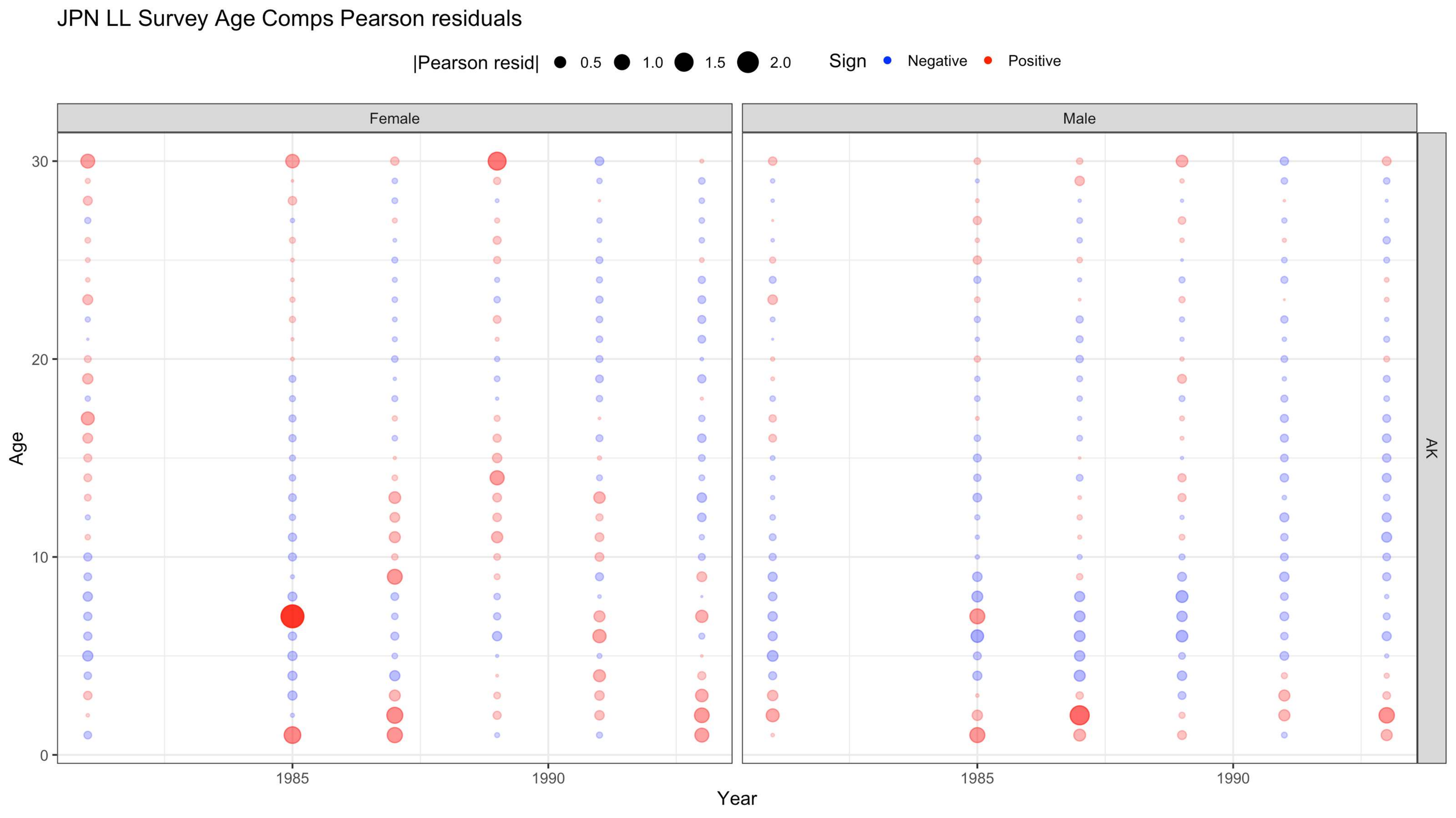


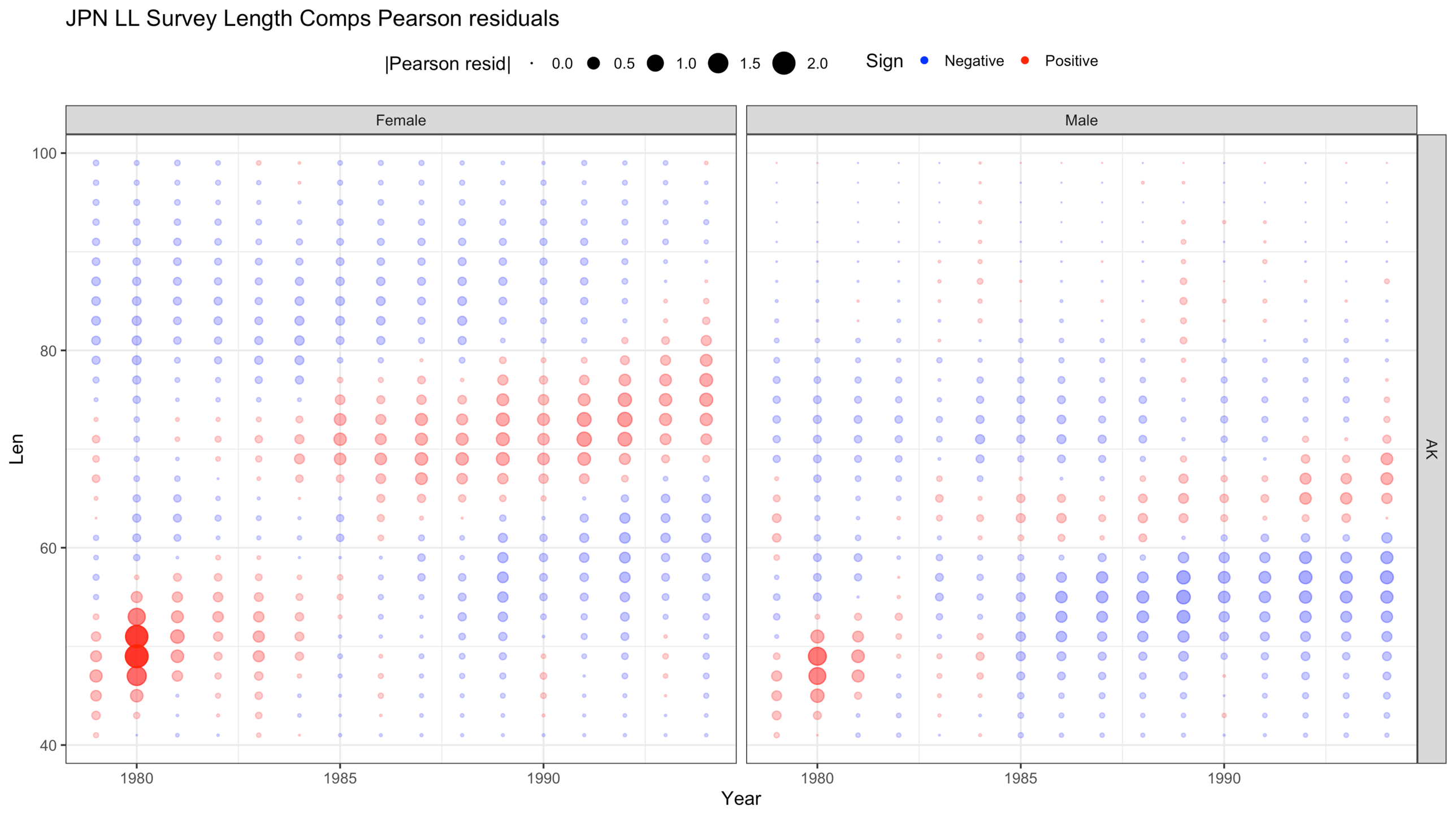

JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age

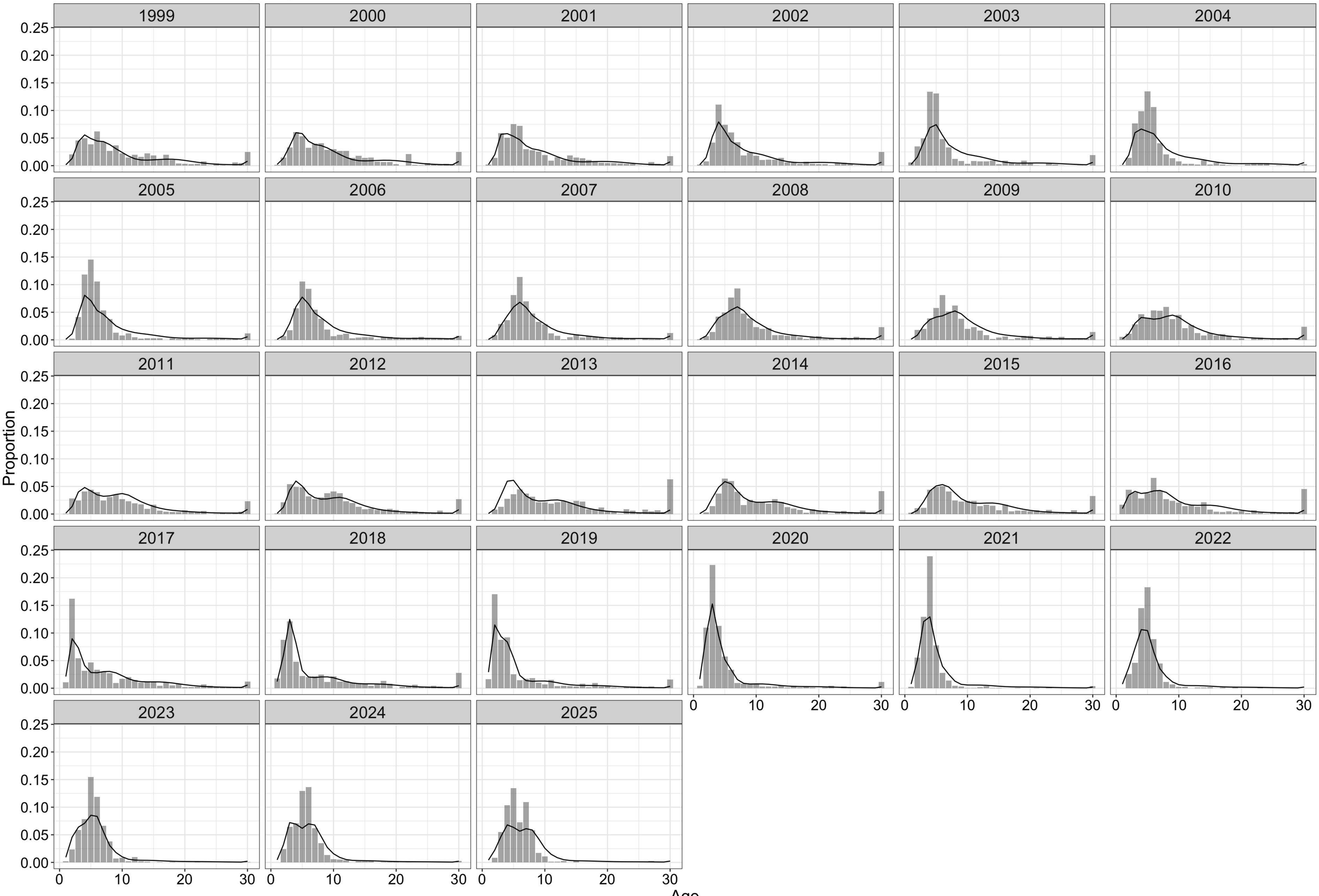
JPN LL Survey Age Comps OSA AK



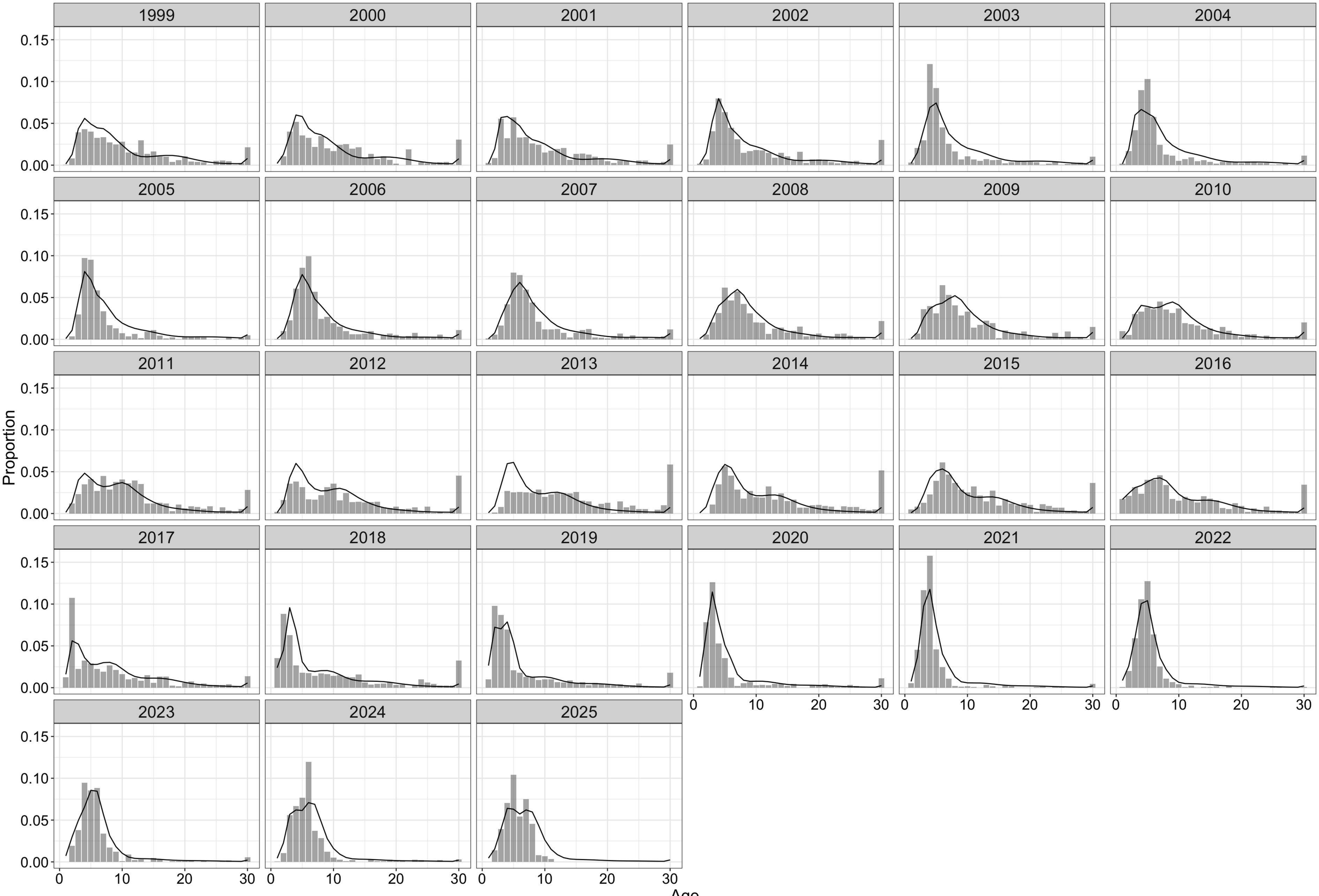

JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
Year


JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

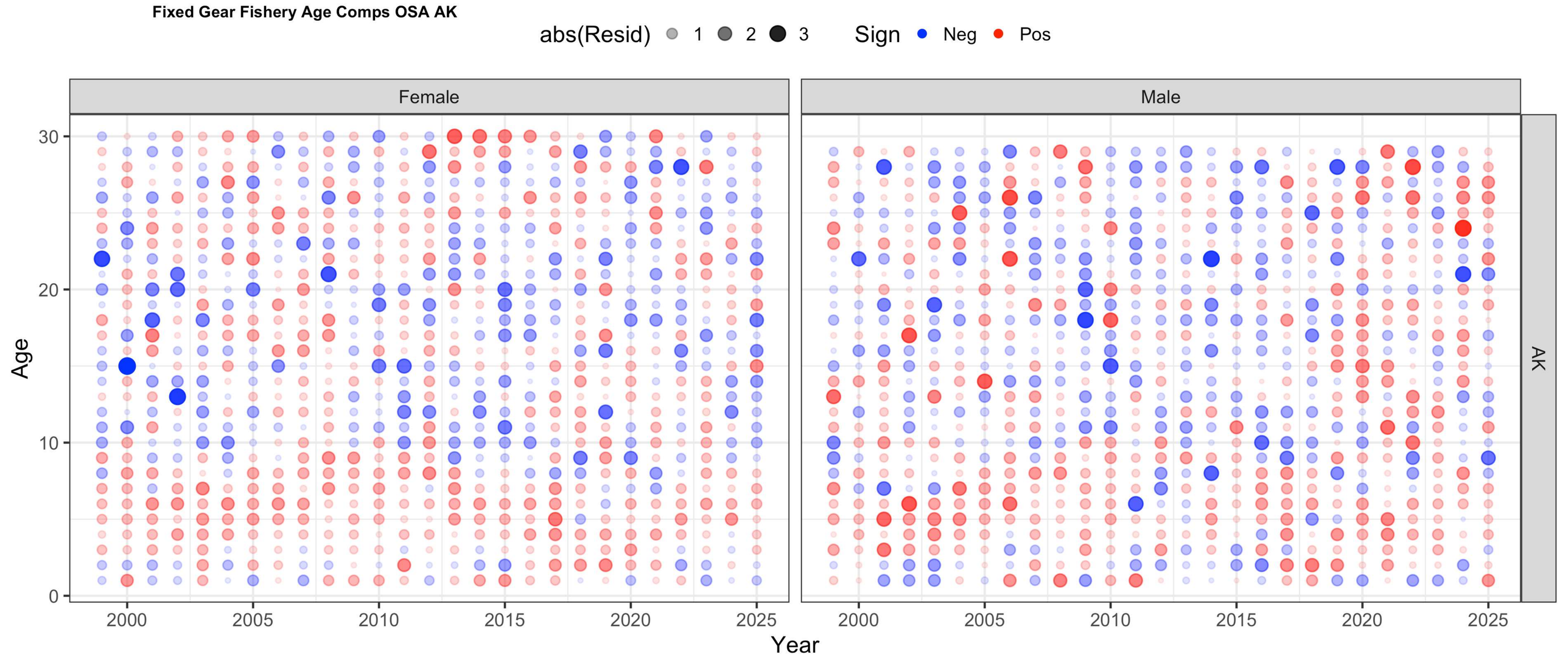
Fixed Gear Fishery Age Comps Female AK

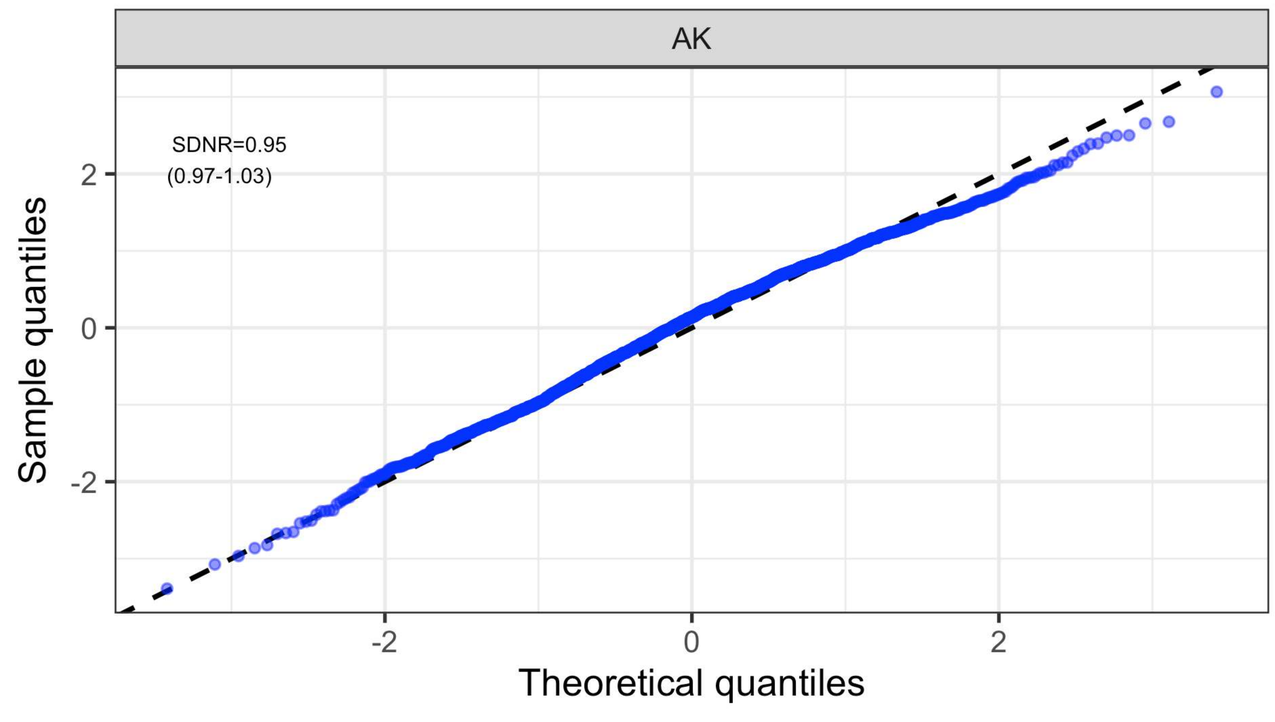


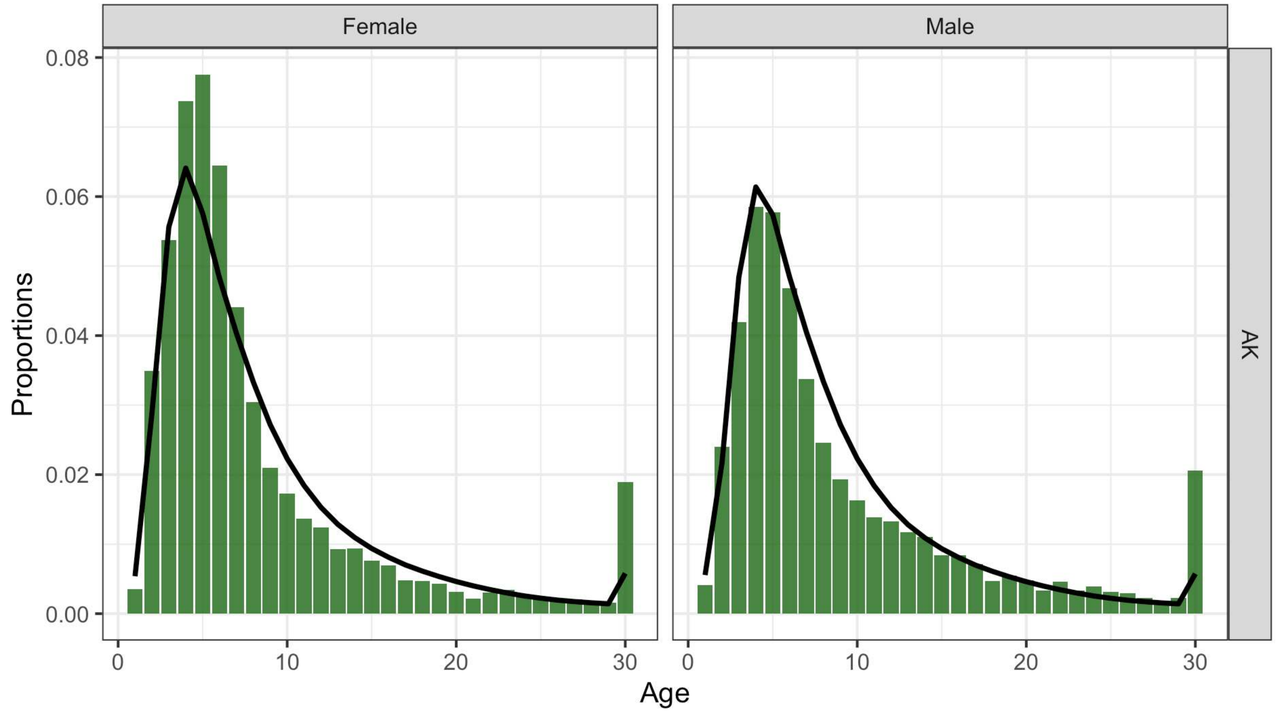
Fixed Gear Fishery Age Comps Male AK

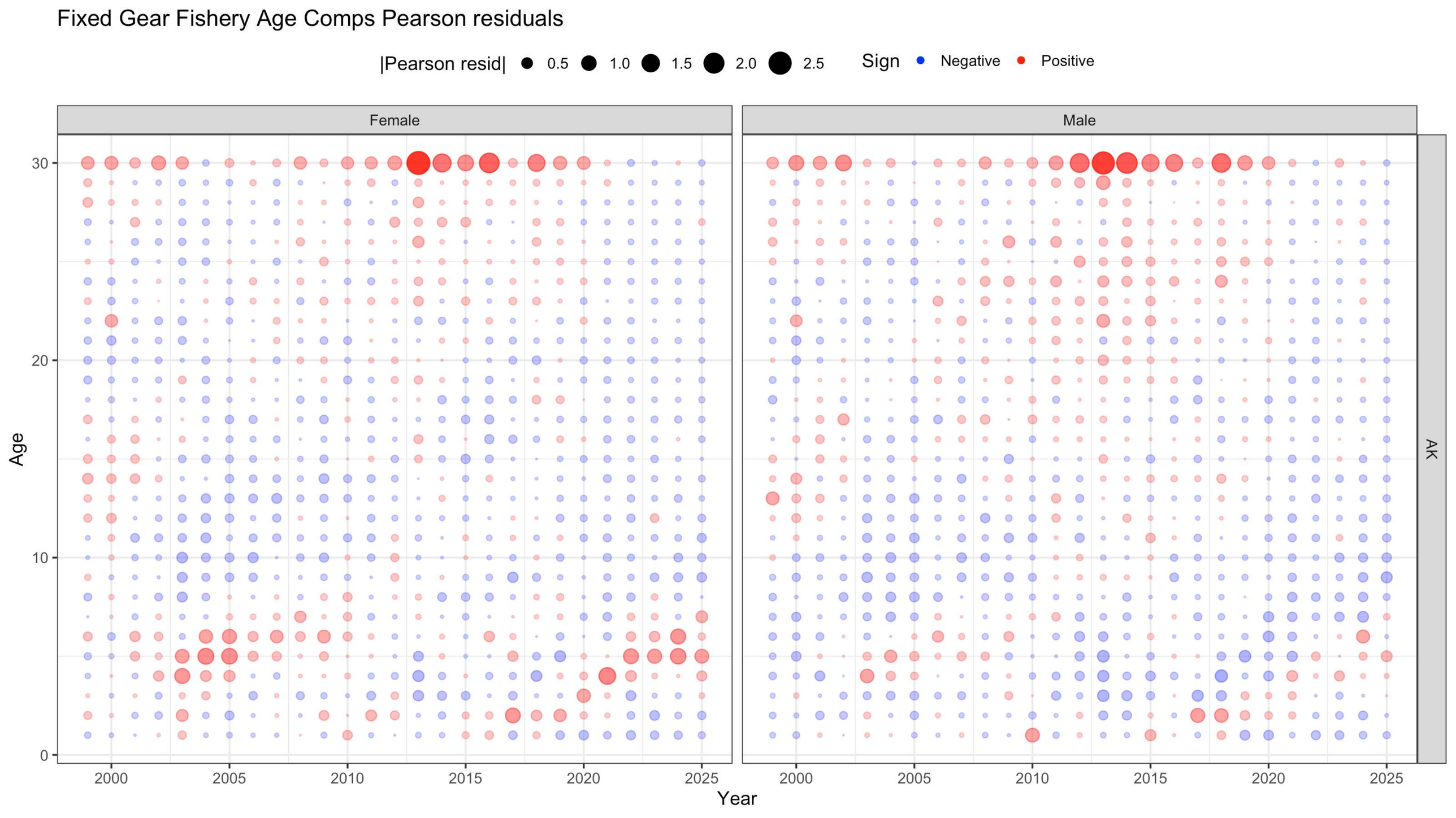


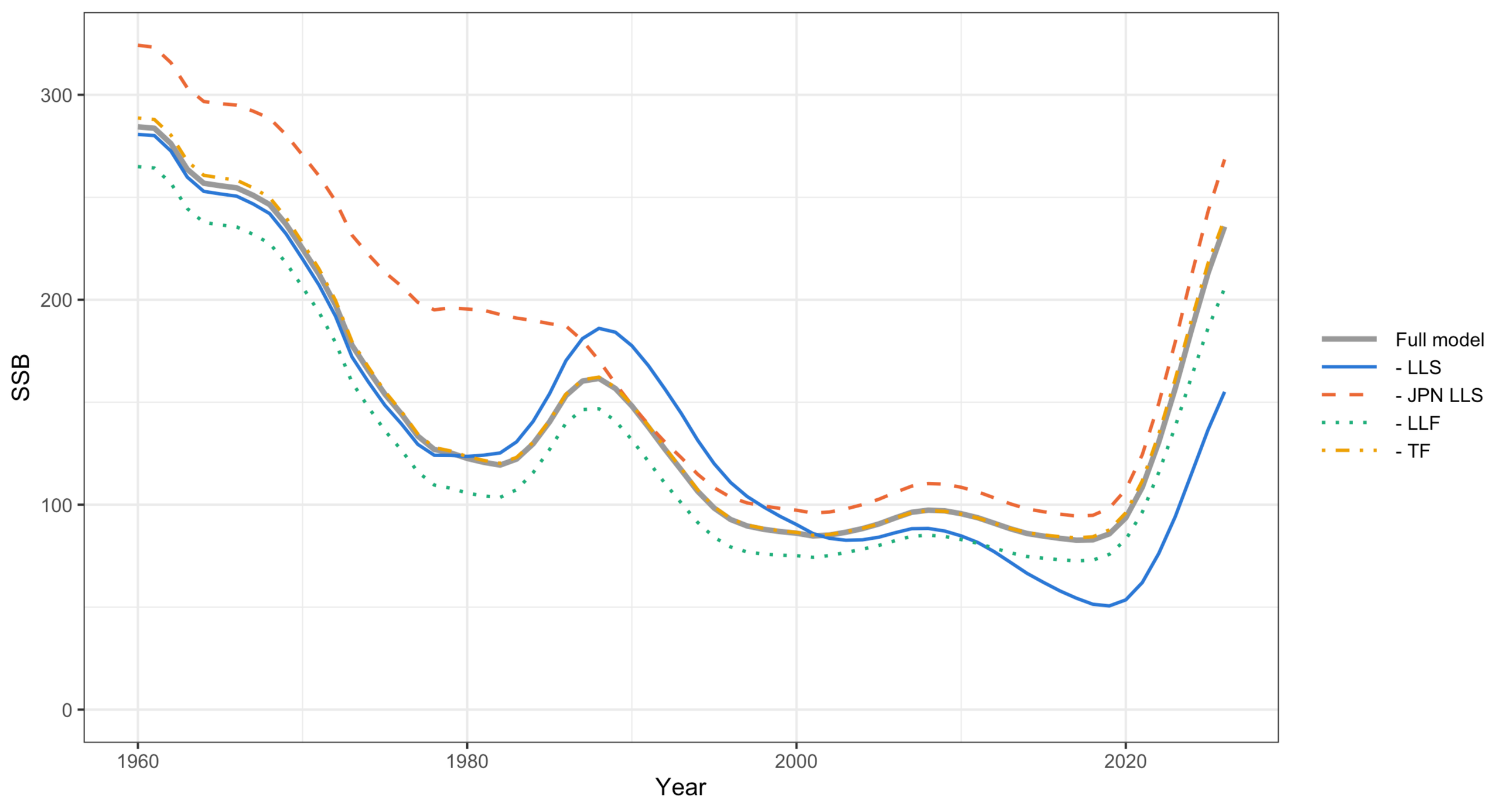
Fixed Gear Fishery Age Comps OSA AK

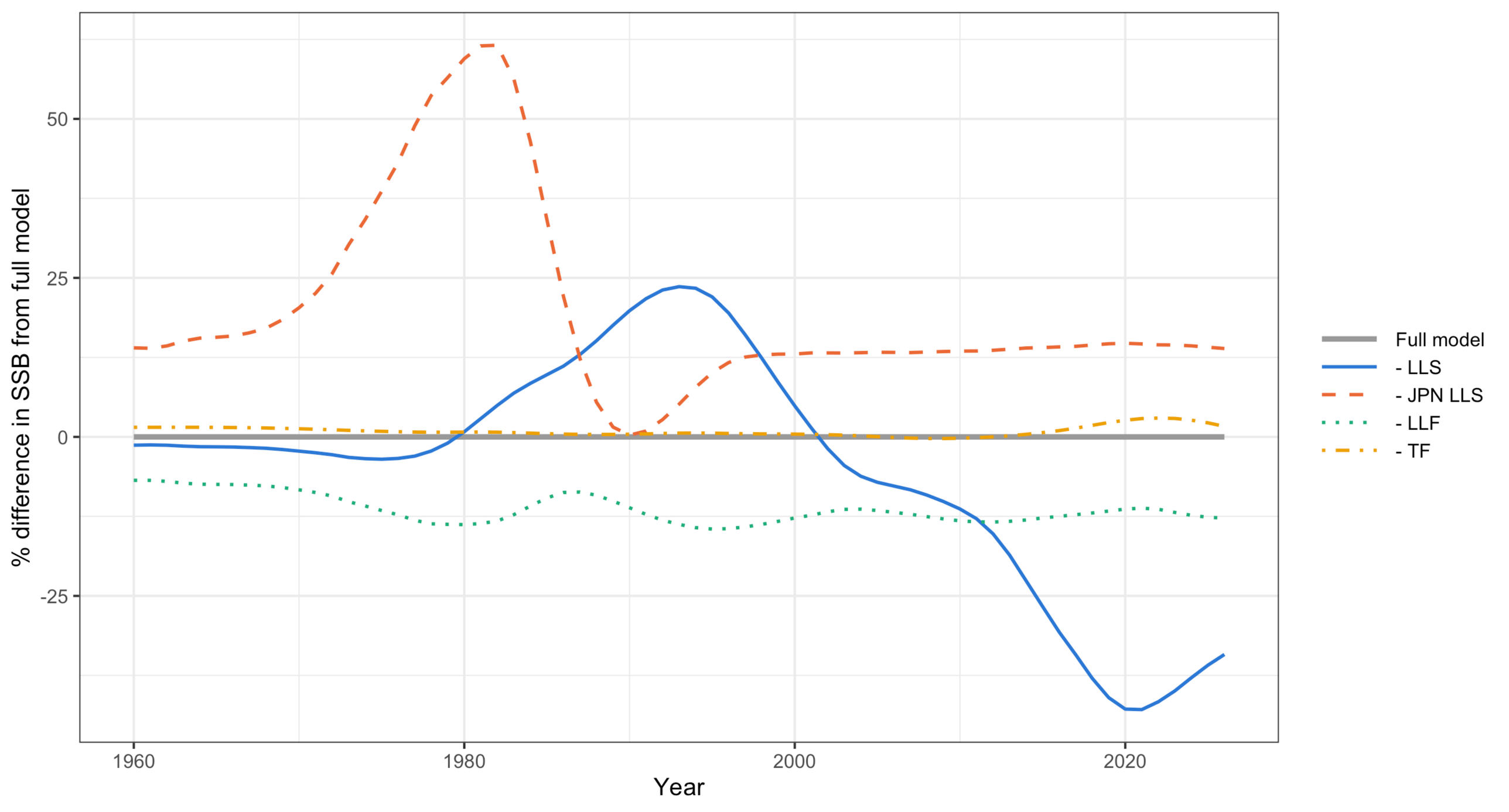


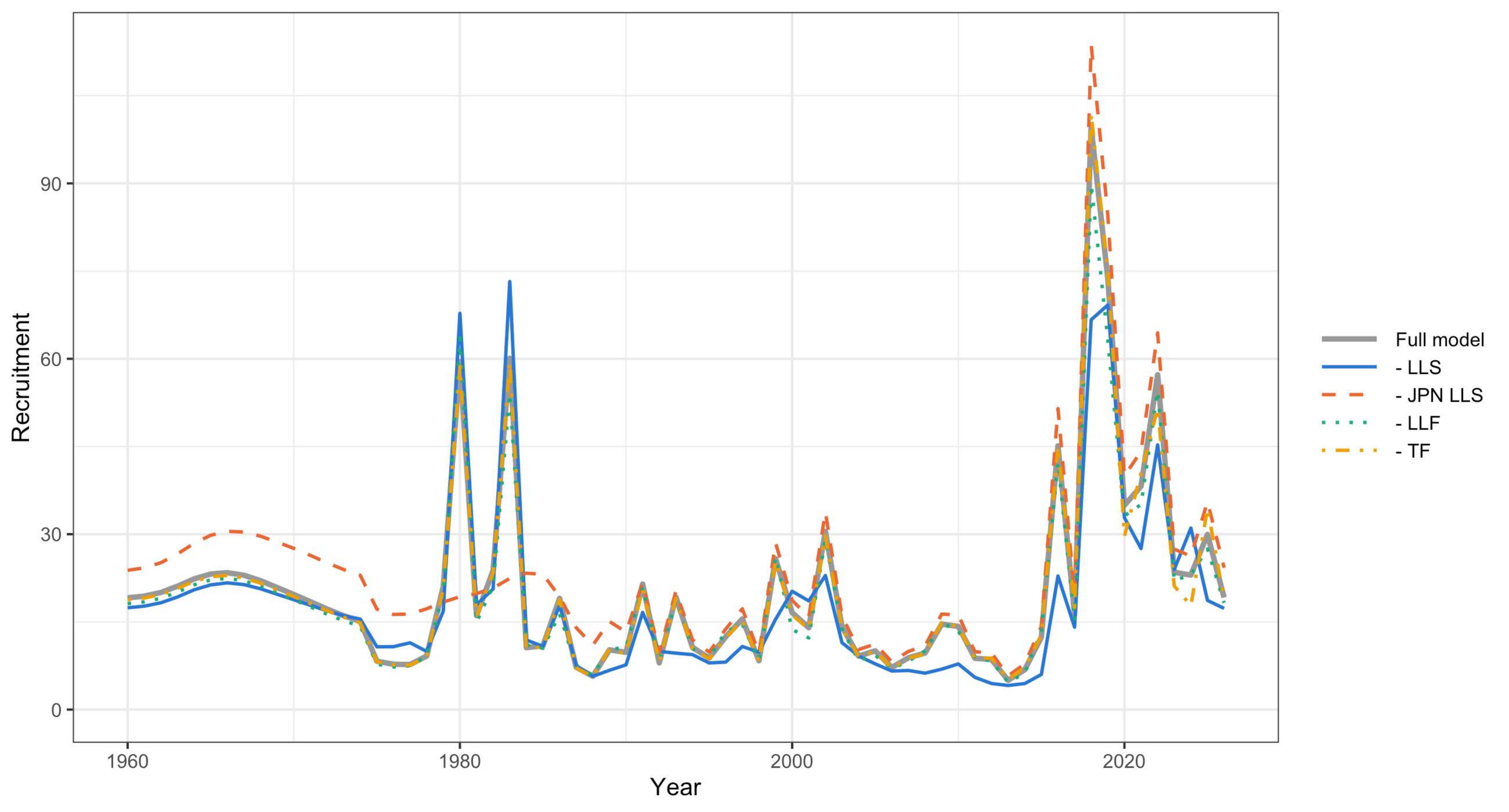


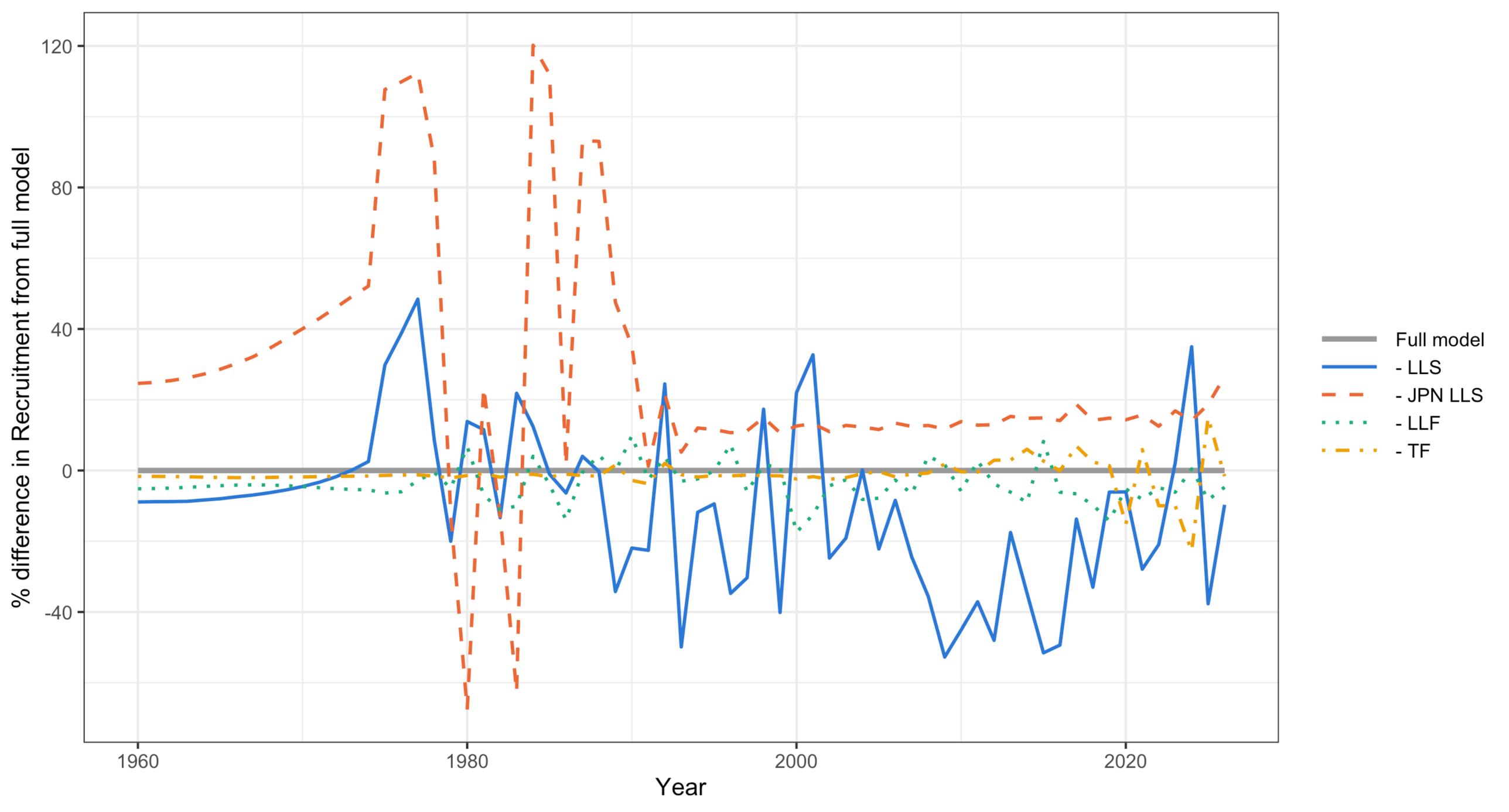



Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
2.5
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025


SSB
300
200
100
0
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF

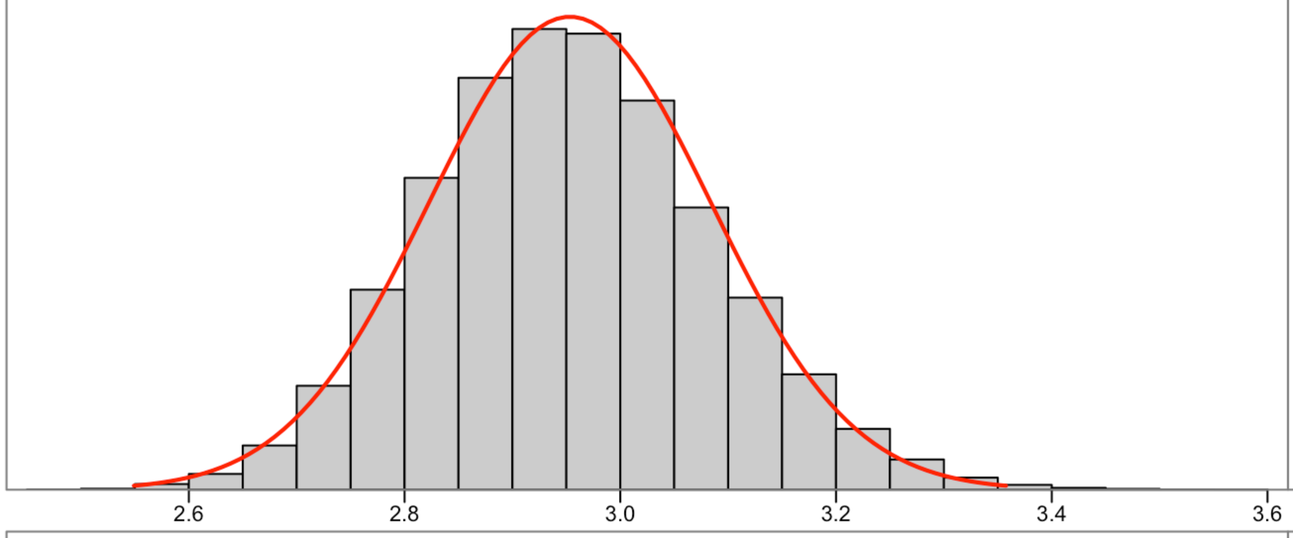

% difference in SSB from full model
50
25
0
-25
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF


Recruitment
90
60
30
0
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF


120
80
40
0
-40
% difference in Recruitment from full model
1960
1980
2000
2020
Year
Full model
- LLS
- JPN LLS
- LLF
- TF

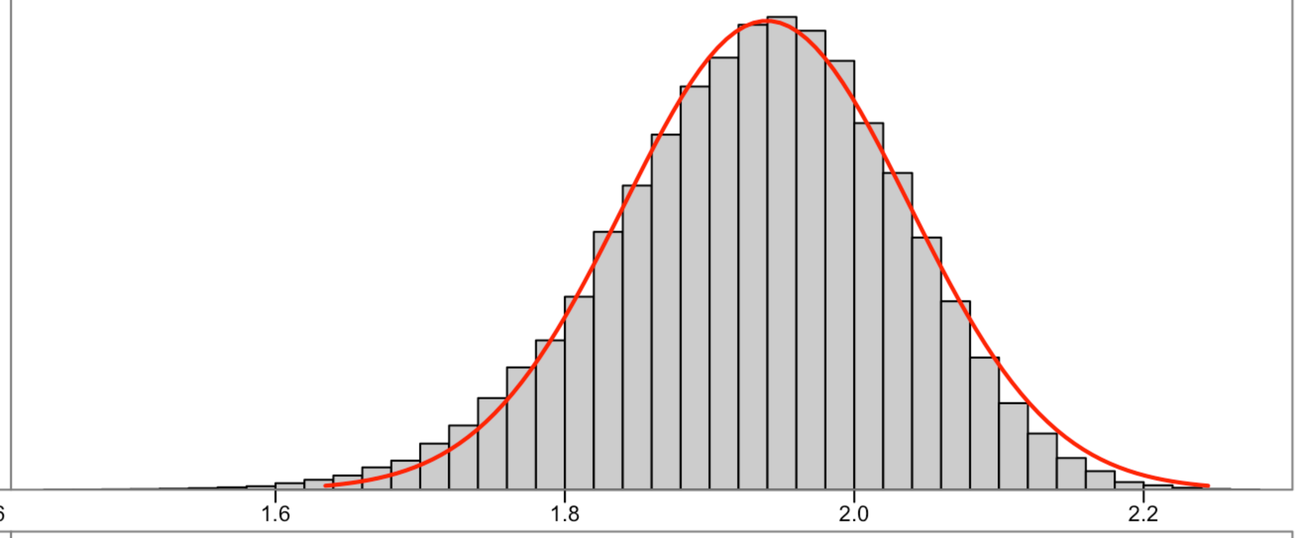
ln_global_R0

ESS=76123
Rhat=1



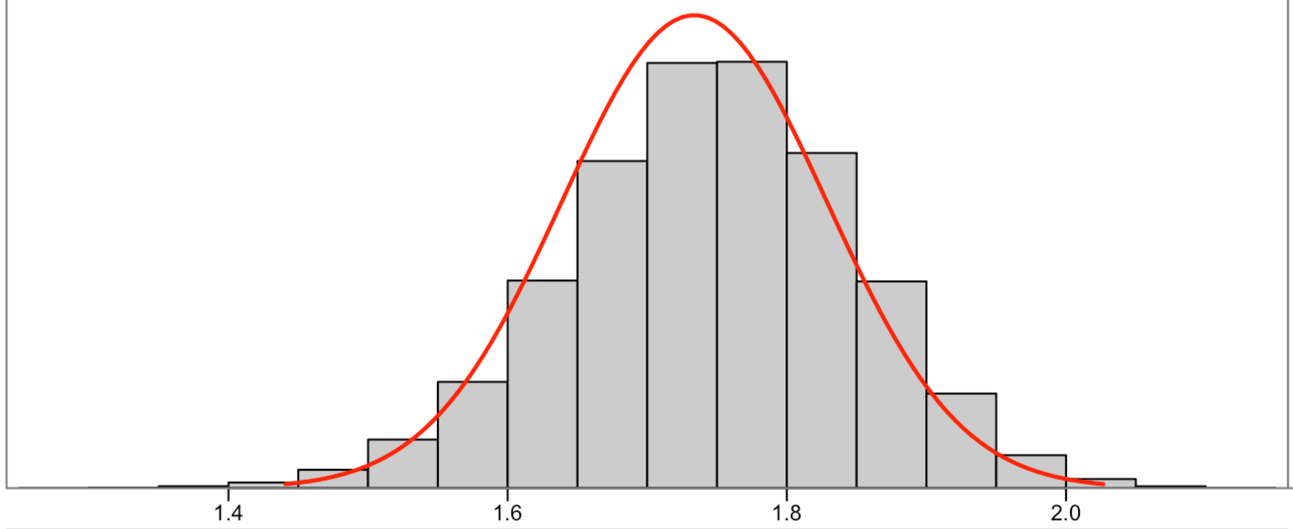
ln_srv_q[1]

ESS=72054
Rhat=1



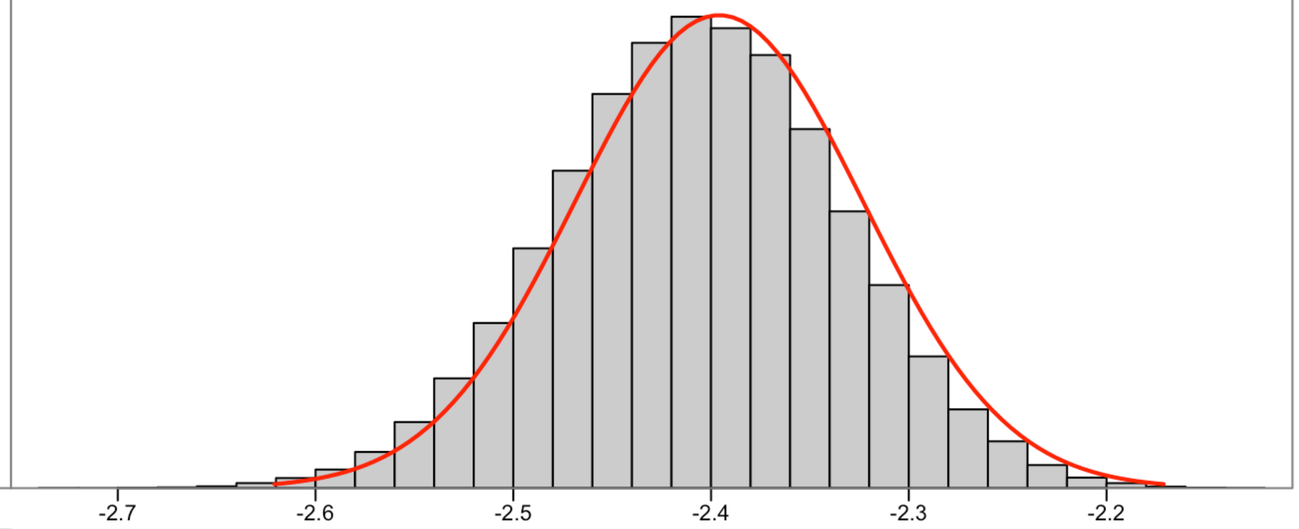
ln_srv_q[2]

ESS=40668
Rhat=1



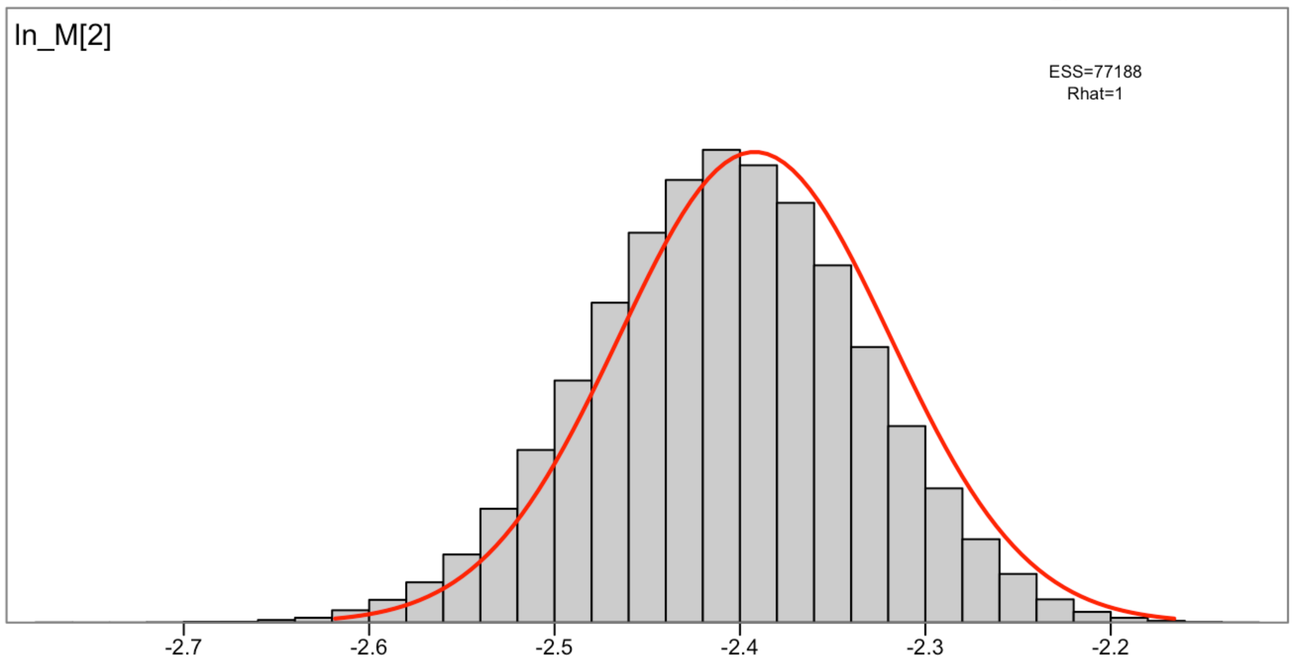
ln_M[1]

ESS=78452
Rhat=1



ln_M[2]

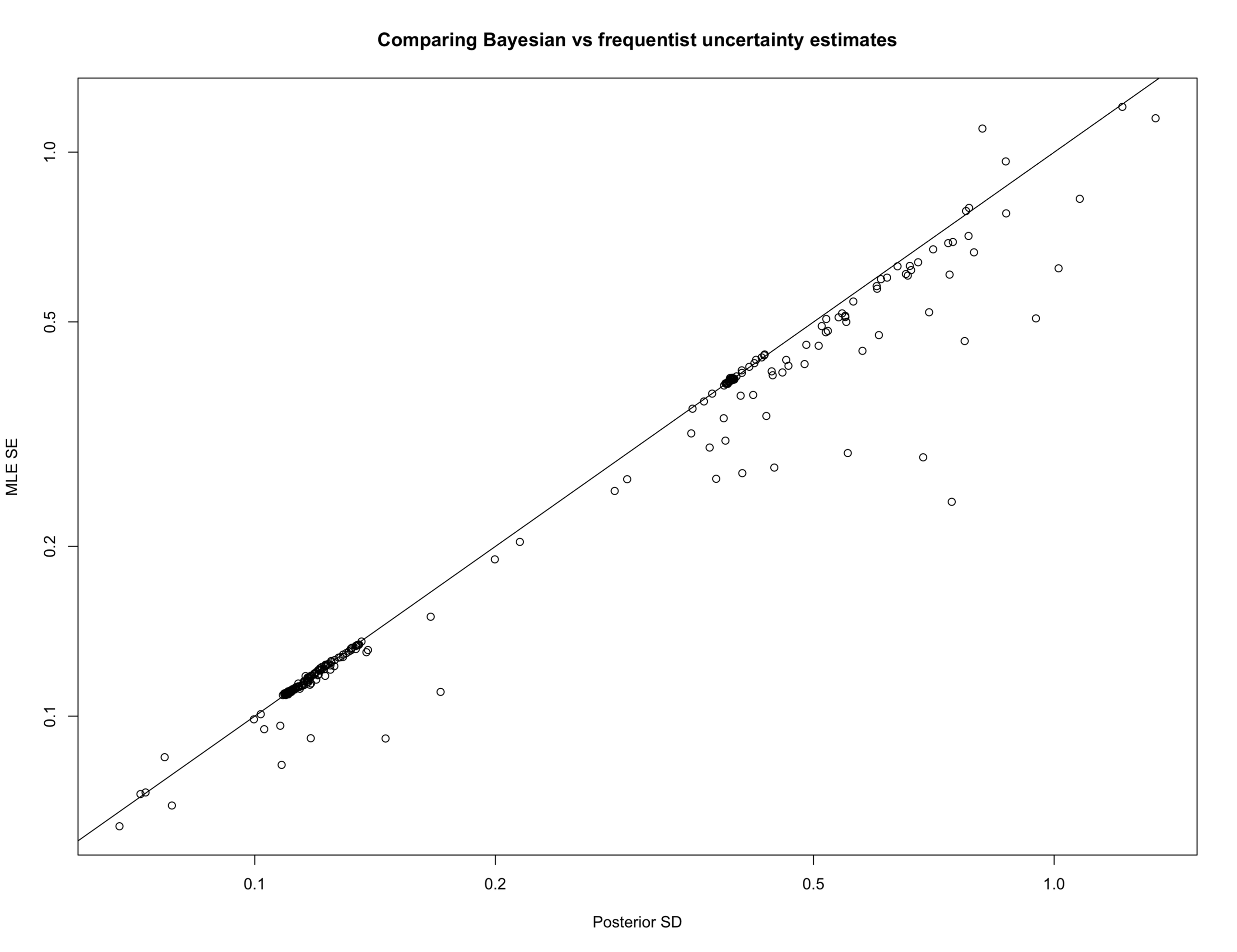
ESS=77188
Rhat=1

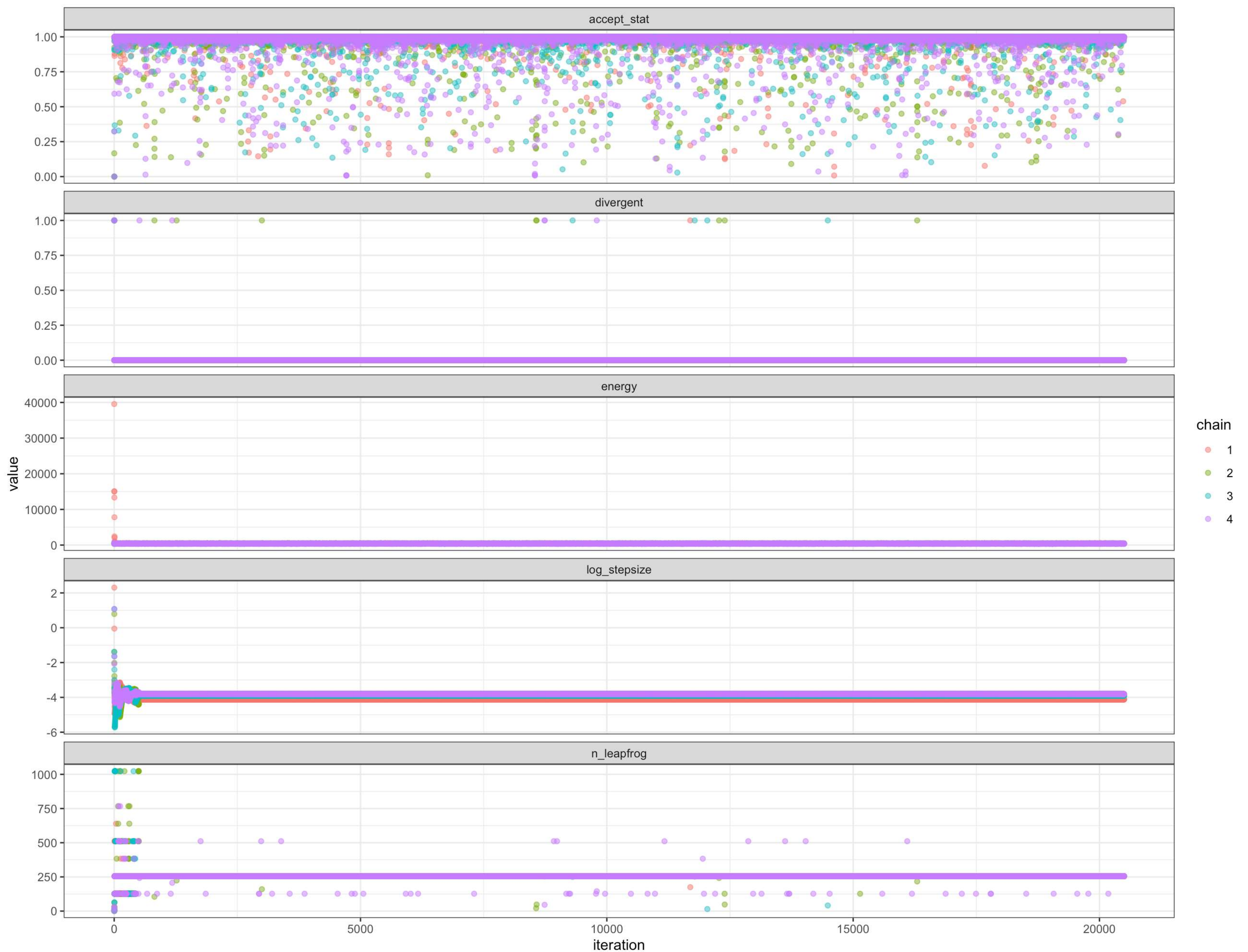


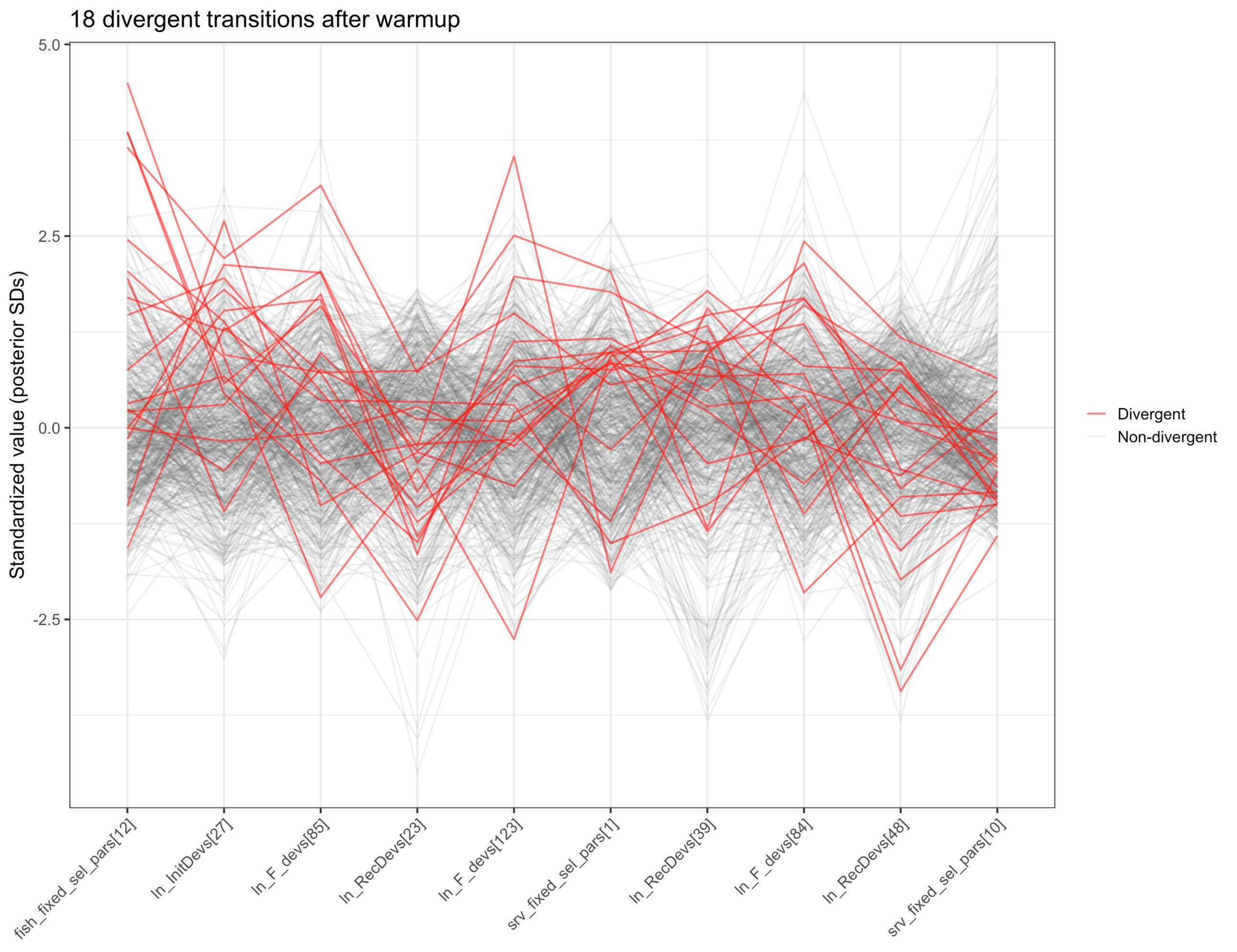
ln_global_R0	ln_srv_q[1]	ln_srv_q[2]	ln_M[1]	ln_M[2]
ESS=76123 Rhat=1.00	-0.74	-0.7	0.78	0.76
	ESS=72054 Rhat=1.00	0.86	-0.7	-0.69
		ESS=40668 Rhat=1.00	-0.64	-0.65
			ESS=78452 Rhat=1.00	0.75
				ESS=77188 Rhat=1.00

ln_global_R0 ln_srv_q[1] ln_srv_q[2] ln_M[1] ln_M[2]

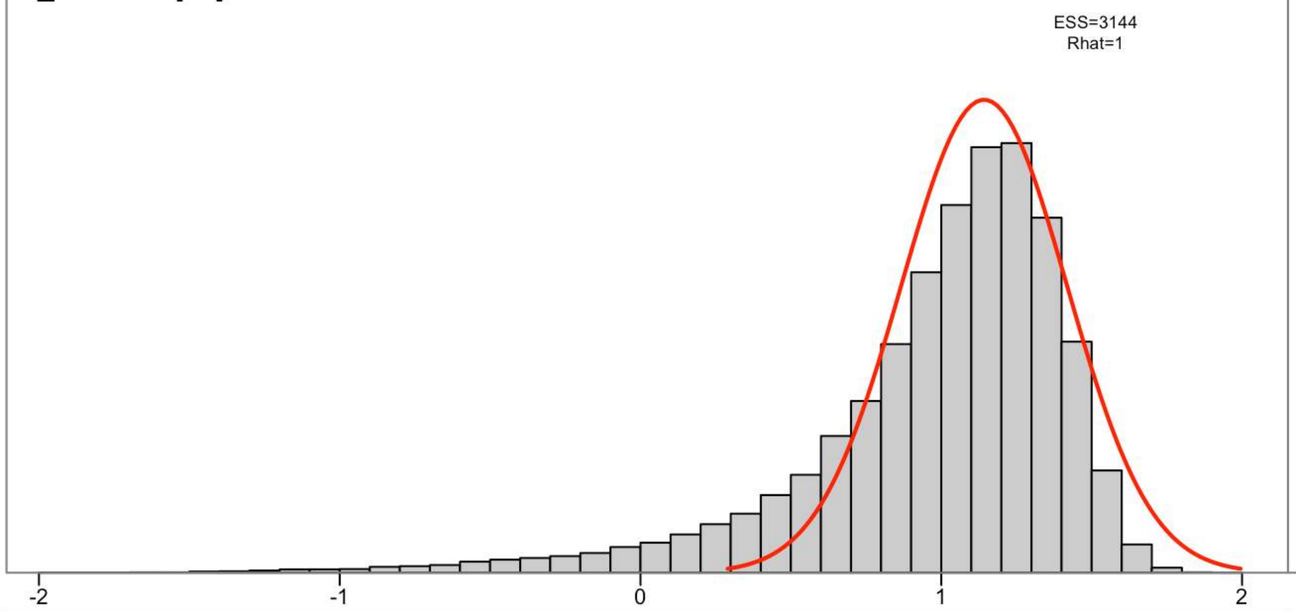
2.4 2.6 2.8 3.0 3.2 3.4 3.6 | 1.4 1.6 1.8 2.0 2.2 2.4 | 1.2 1.4 1.6 1.8 2.0 2.2 | -2.8 -2.6 -2.4 -2.2 | 5000 10000 15000 20000


Comparing Bayesian vs frequentist uncertainty estimates
MLE SE
Posterior SD
0.1
0.2
0.5
1.0

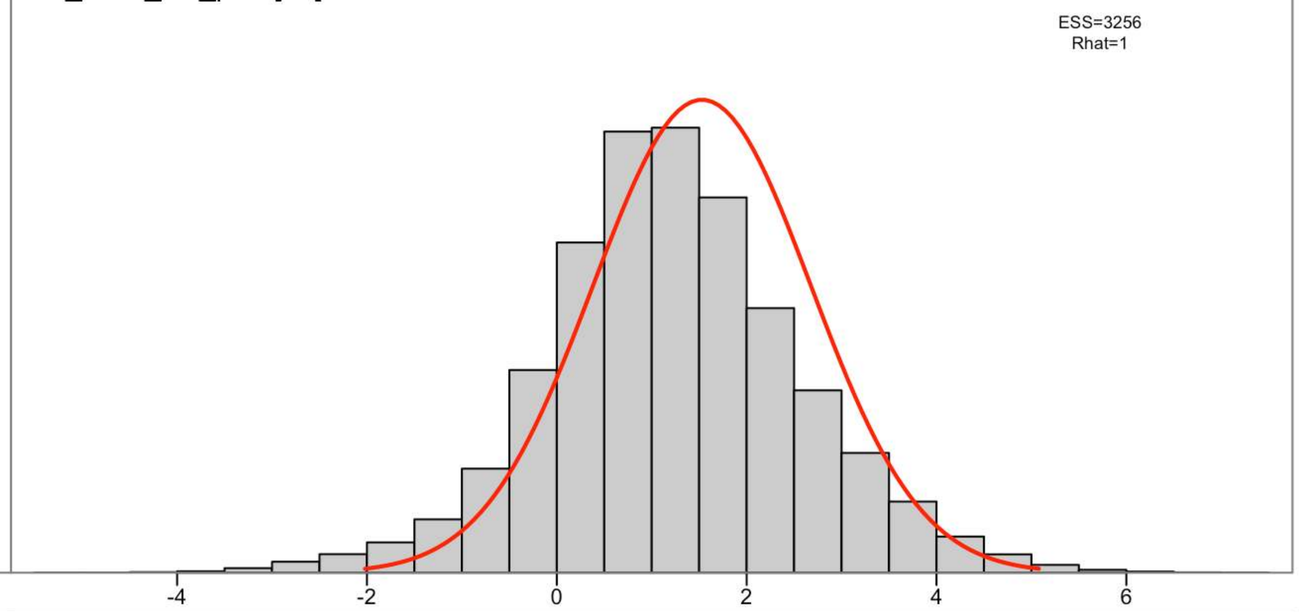

accept_stat
divergent
energy
log_stepsize
n_leapfrog
value
iteration
chain
1
2
3
4


18 divergent transitions after warmup
Standardized value (posterior SDs)
5.0
2.5
0.0
-2.5
fish_fixed_sel_pars[12]
ln_InitDevs[27]
ln_F_devs[85]
ln_RecDevs[23]
ln_F_devs[123]
srv_fixed_sel_pars[1]
ln_RecDevs[39]
ln_F_devs[84]
ln_RecDevs[48]
srv_fixed_sel_pars[10]
Divergent
Non-divergent

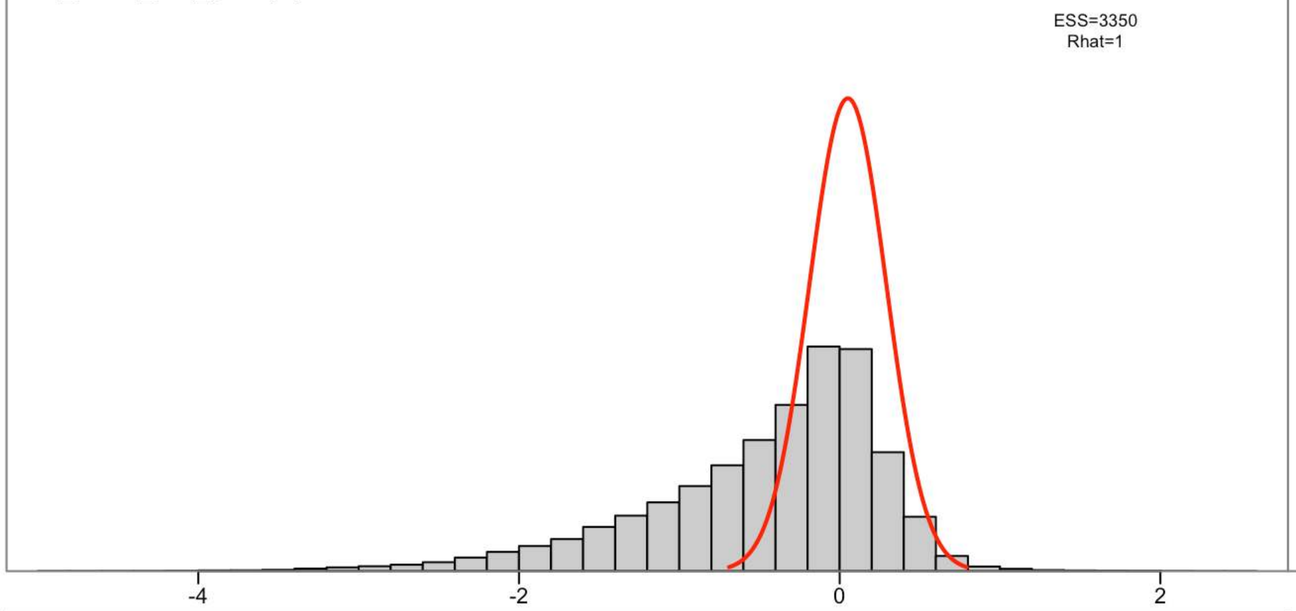
ln_RecDevs[24]



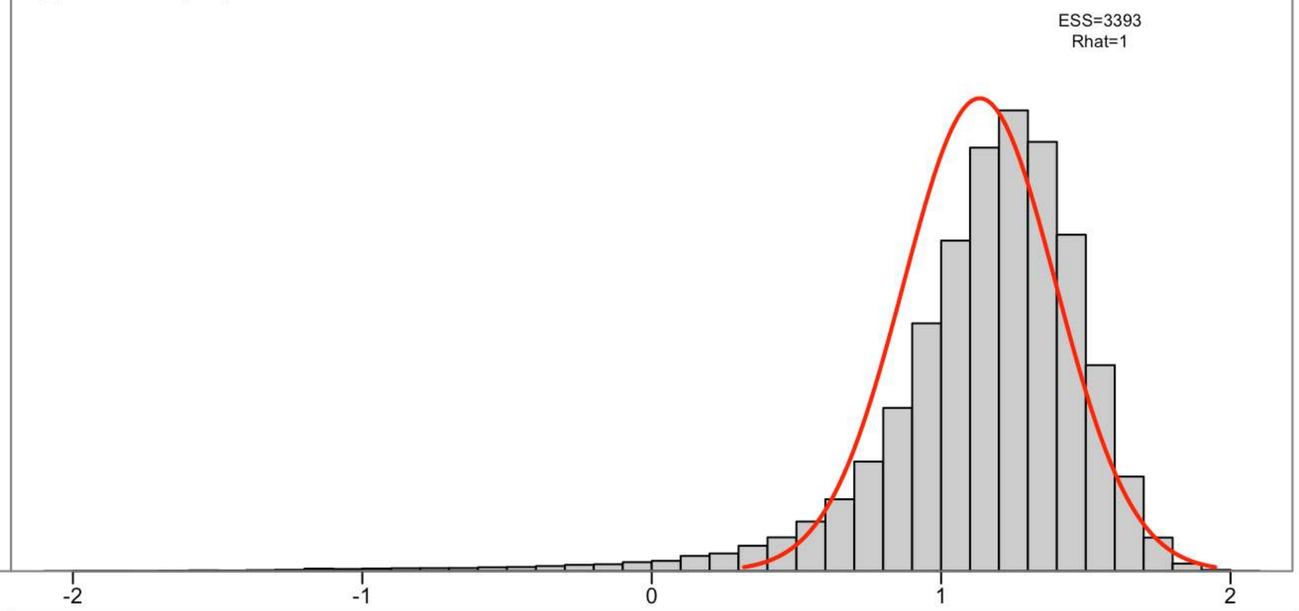
fish_fixed_sel_pars[10]



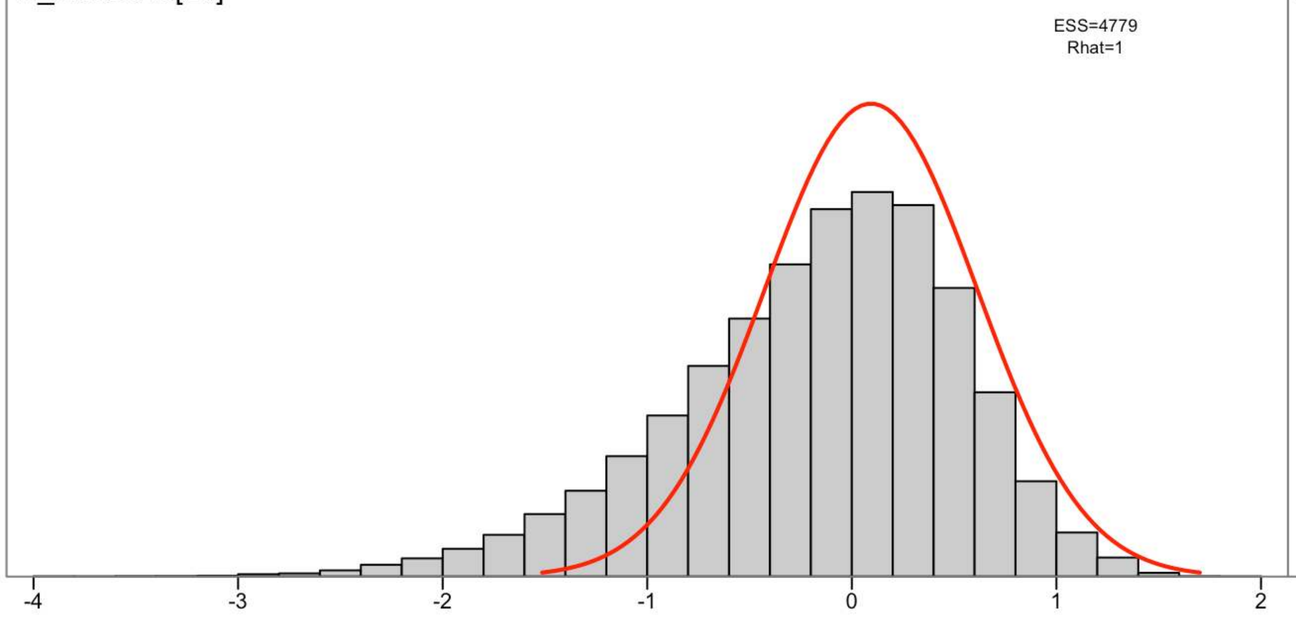
fish_fixed_sel_pars[9]



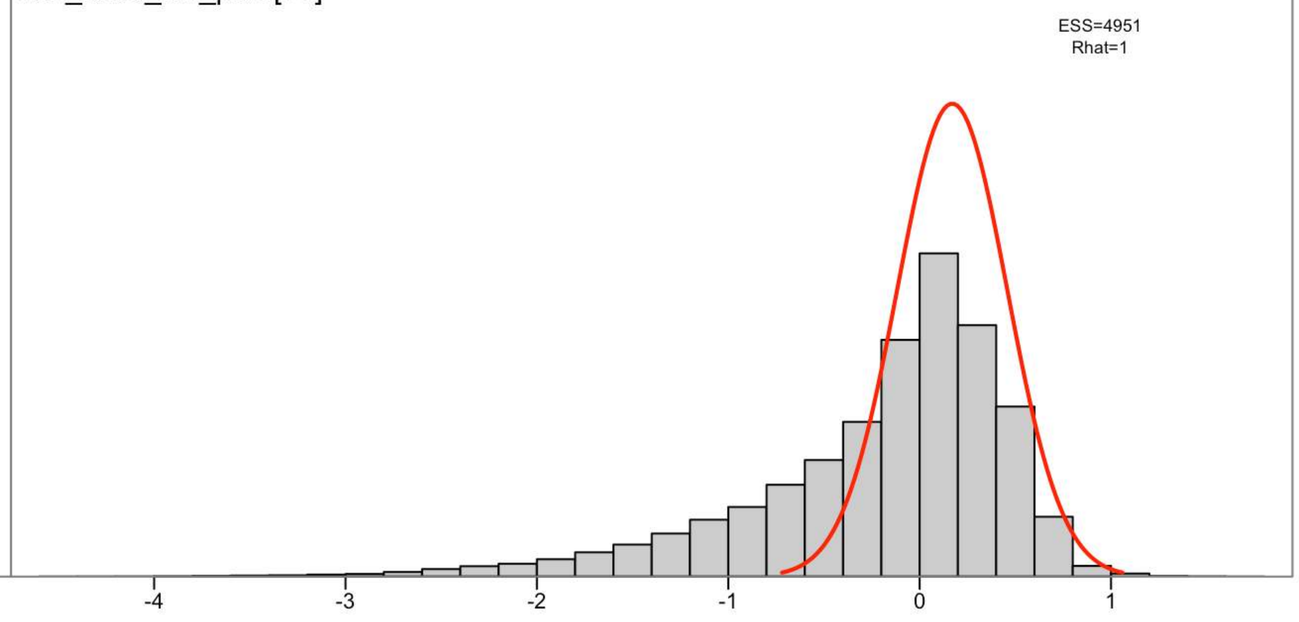
ln_RecDevs[21]



ln_RecDevs[20]



fish_fixed_sel_pars[11]



ln_RecDevs[24]
fish_fixed_sel_pars[10]
fish_fixed_sel_pars[9]
ln_RecDevs[21]
ln_RecDevs[20]
fish_fixed_sel_pars[11]

ESS=3144 Rhat=1.00

ESS=3256 Rhat=1.00

ESS=3350 Rhat=1.00

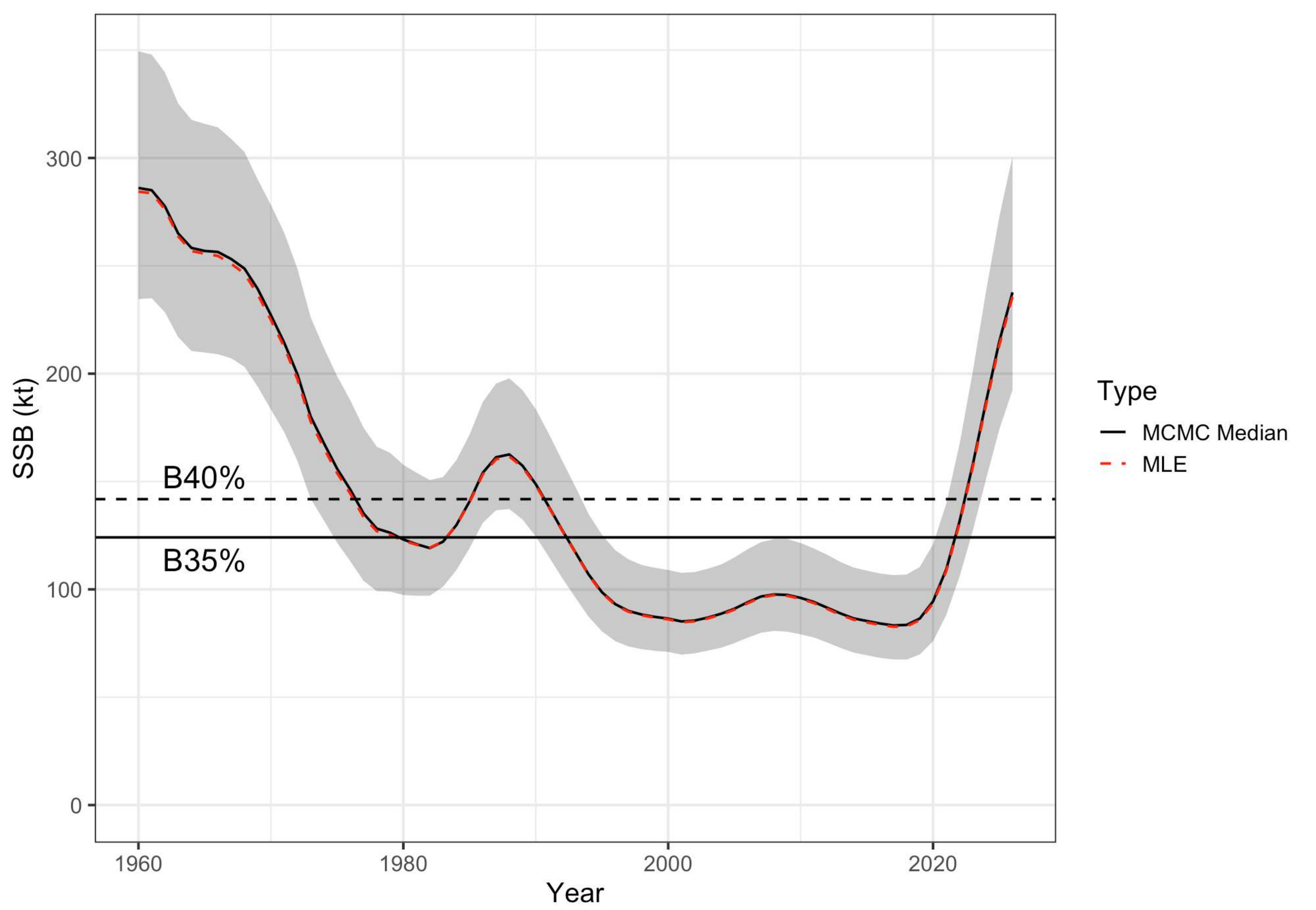
ESS=3393 Rhat=1.00

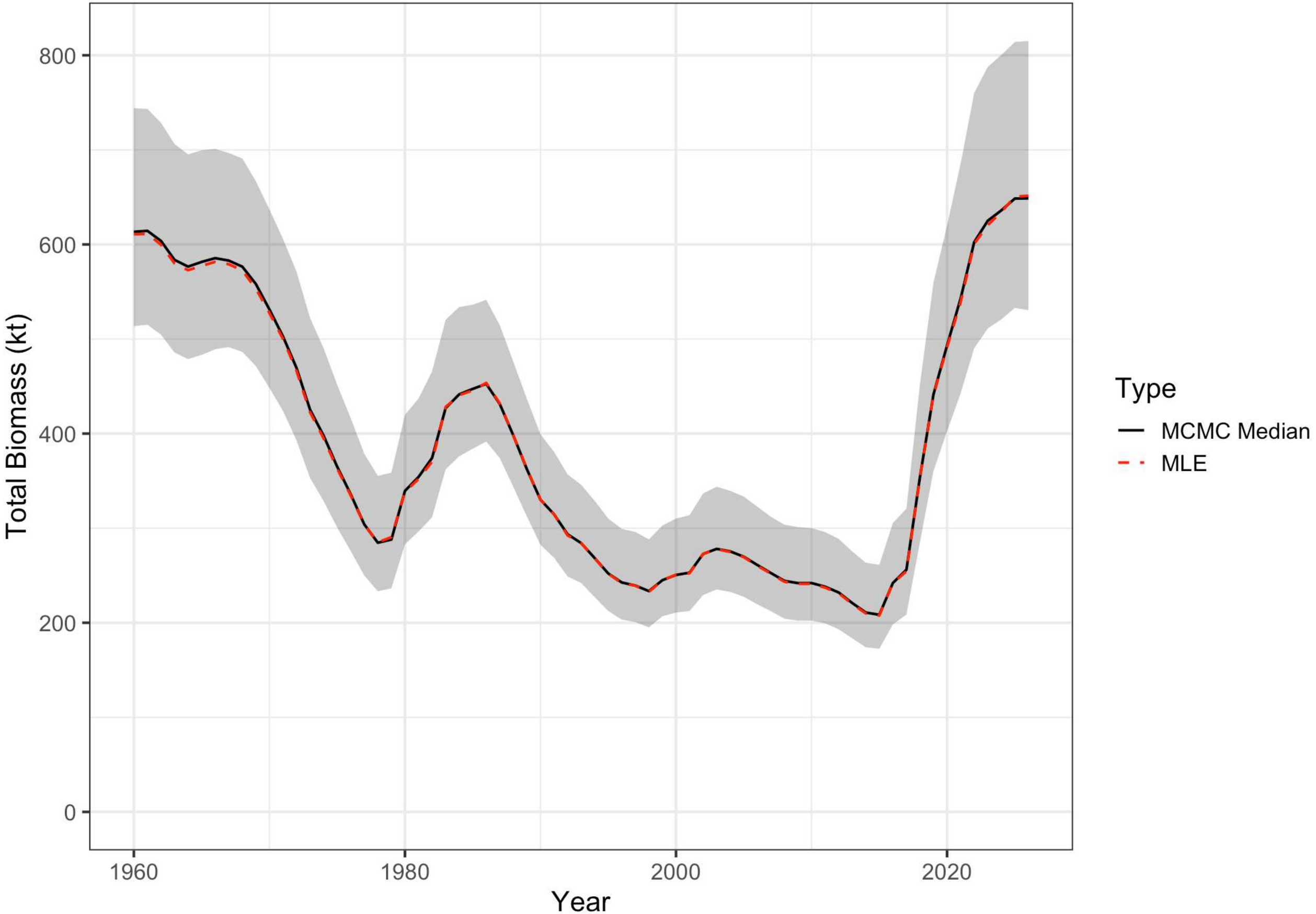
-0.54

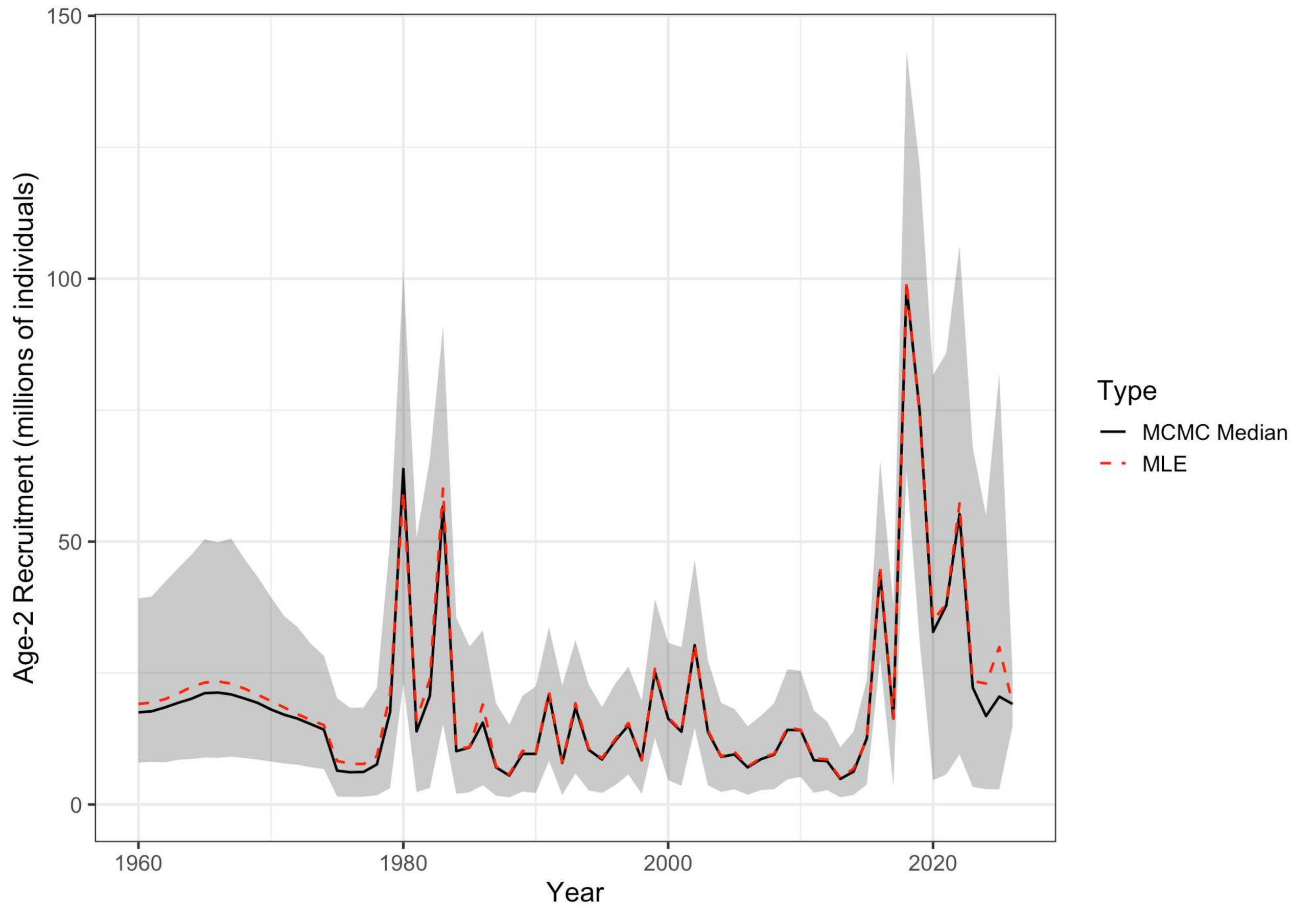
ESS=4779 Rhat=1.00

ESS=4951 Rhat=1.00

ln_RecDevs[24]
fish_fixed_sel_pars[10]
fish_fixed_sel_pars[9]
ln_RecDevs[21]
ln_RecDevs[20]
fish_fixed_sel_pars[11]


SSB (kt)
300
200
100
0
B40%
B35%
1960
1980
2000
2020
Year
Type
MCMC Median
MLE


800
600
400
200
0
Total Biomass (kt)
1960
1980
2000
2020
Year
Type
MCMC Median
MLE


150
100
50
0
Age-2 Recruitment (millions of individuals)
1960
1980
2000
2020
Year
Type
MCMC Median
MLE

Type: Age, Catch, Index, jnLL, Length, Penalty, Prior

26.1_1Reg

Likelihood

200
100
0
-100
-200

Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL

Likelihood Component

Numbers-at-Age

Sex — Female ···· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

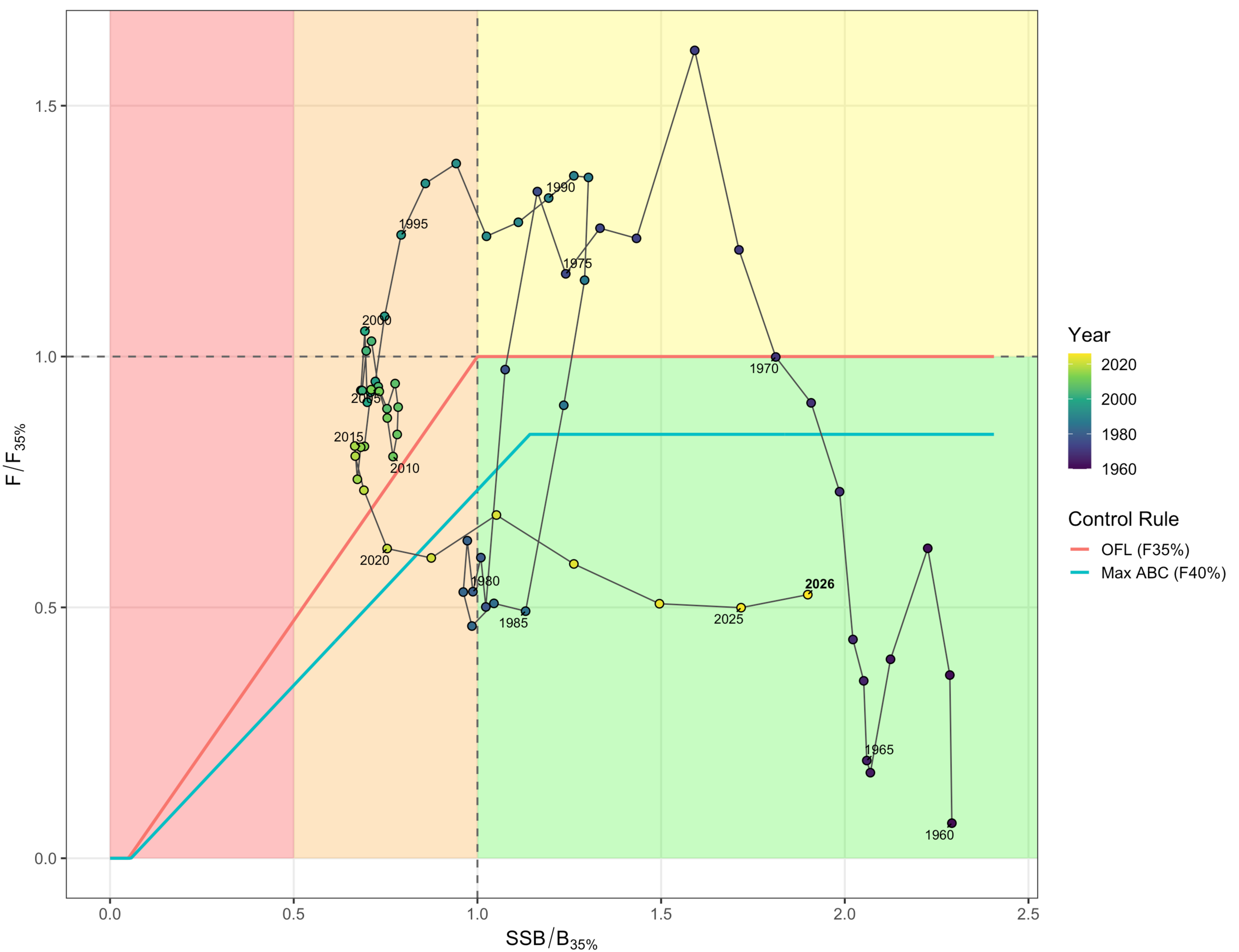
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

Year


1.5
1.0
0.5
0.0
F/F35%
0.0
0.5
1.0
1.5
2.0
2.5
SSB/B35%
1960
1970
1980
1990
1995
2000
2005
2010
2015
2020
2025
2026
1985
1975
1965
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Natural mortality profile

Scaled nLL

750

500

250

0

-5

-4

-3

-2

-1

ln(M)

Type

Age Comps

Catch

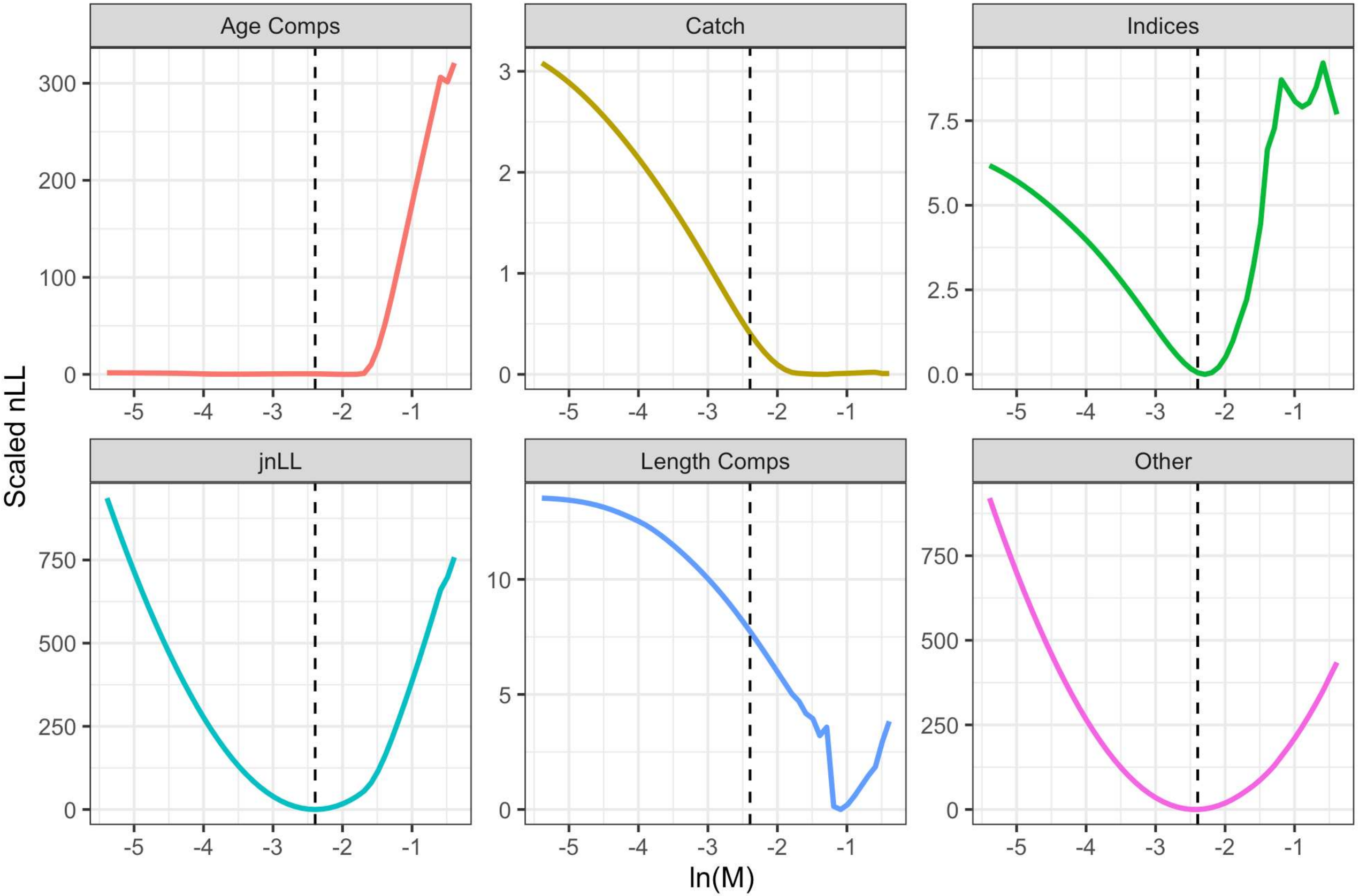
Indices

jnLL

Length Comps

Other

Natural mortality profile



Recruitment (R0) profile

Scaled nLL

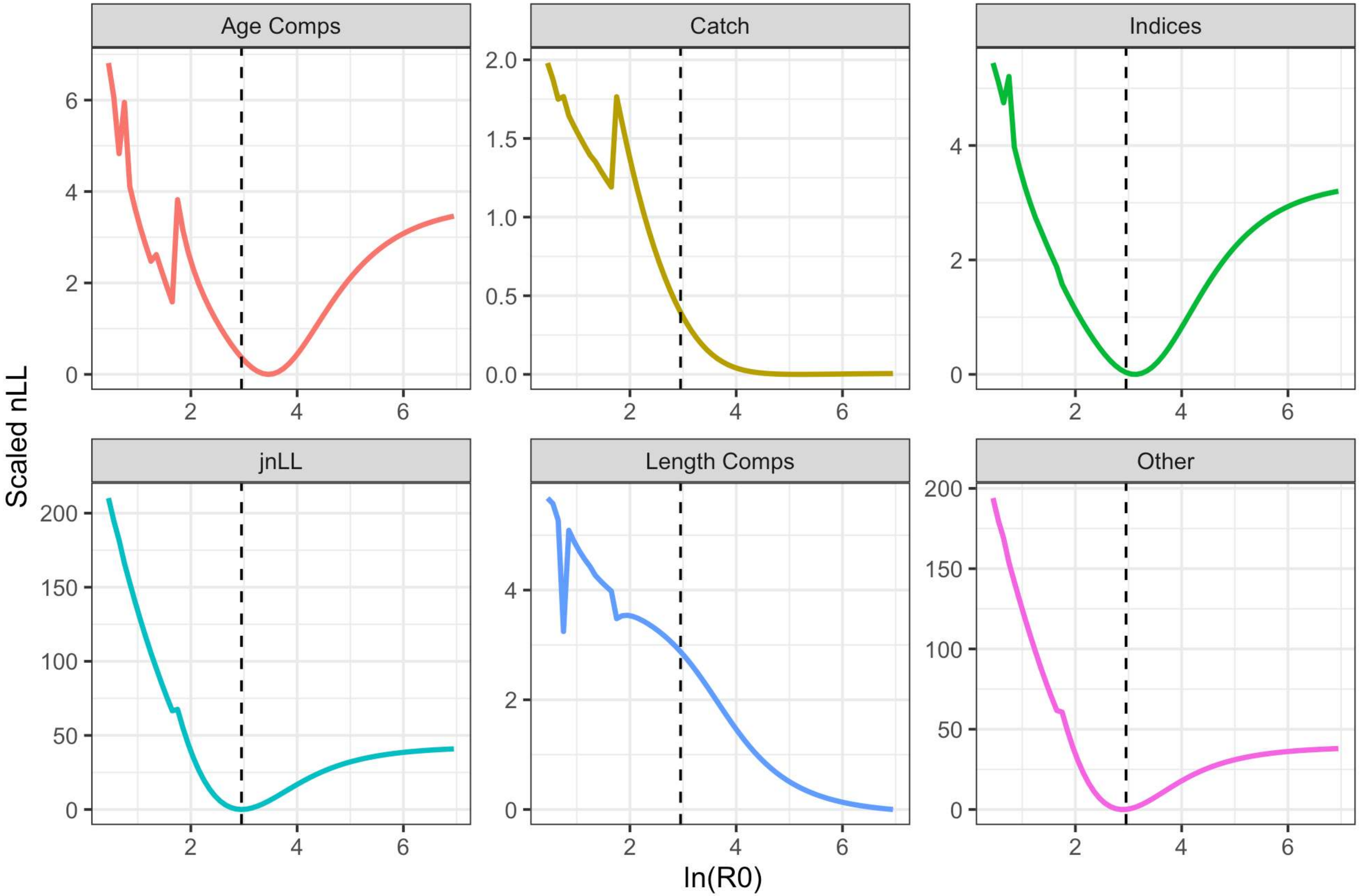
200
150
100
50
0

2
4
6

ln(R0)

Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

Recruitment (R0) profile



Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.453

SSB

Mohns Rho: 0.0195

AK

Relative Difference from Terminal Year

0.2
0.0
-0.2
-0.4
-0.6
-0.8

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

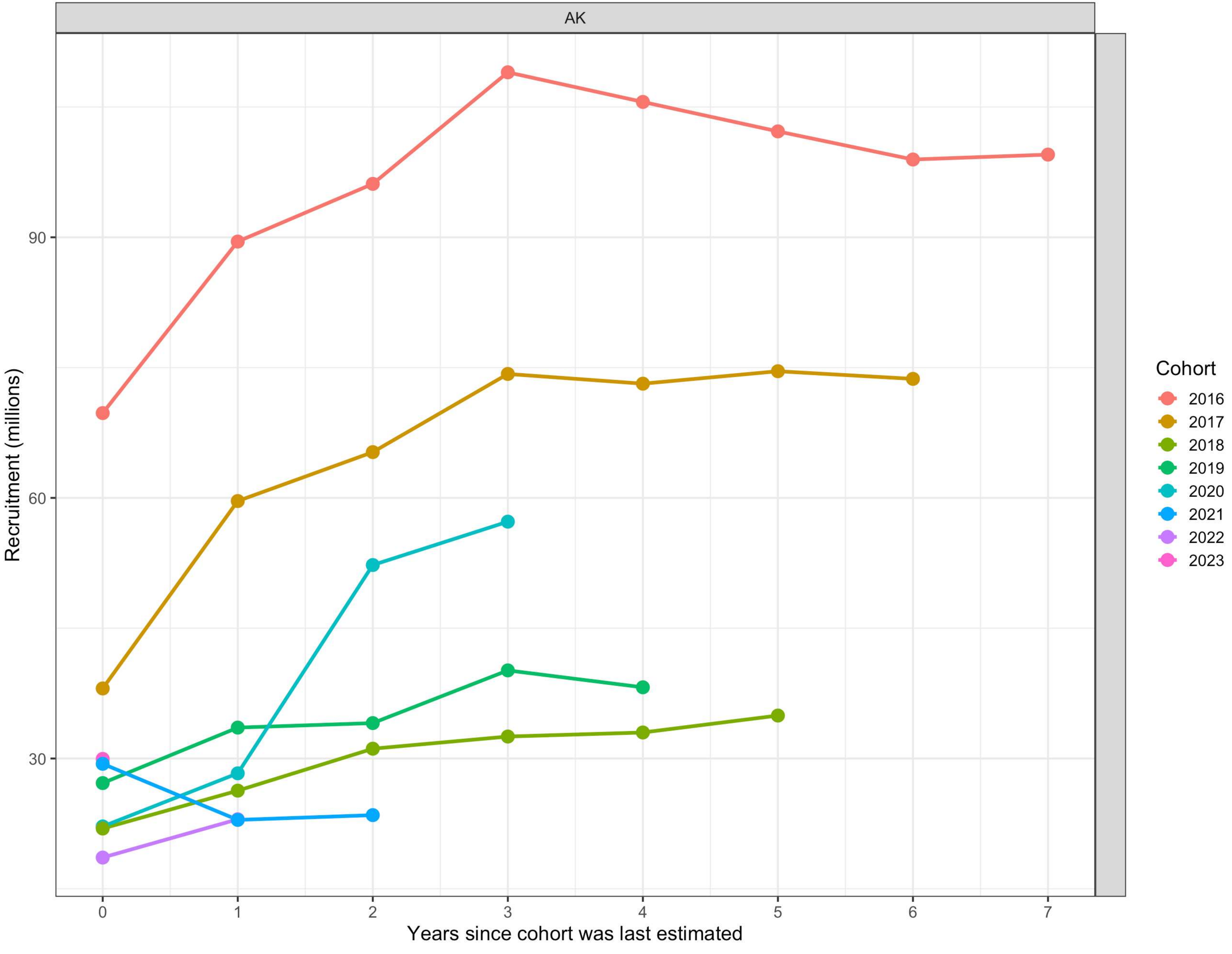
Value

90
60
30
0

200
100
0

1960
1980
2000
2020

Year


AK
90
60
30
Recruitment (millions)
0
1
2
3
4
5
6
7
Years since cohort was last estimated
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.4515

Mohns Rho: 0.0032

AK

Relative Difference from Terminal Year

2
1
0

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

100
75
50
25
0

200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

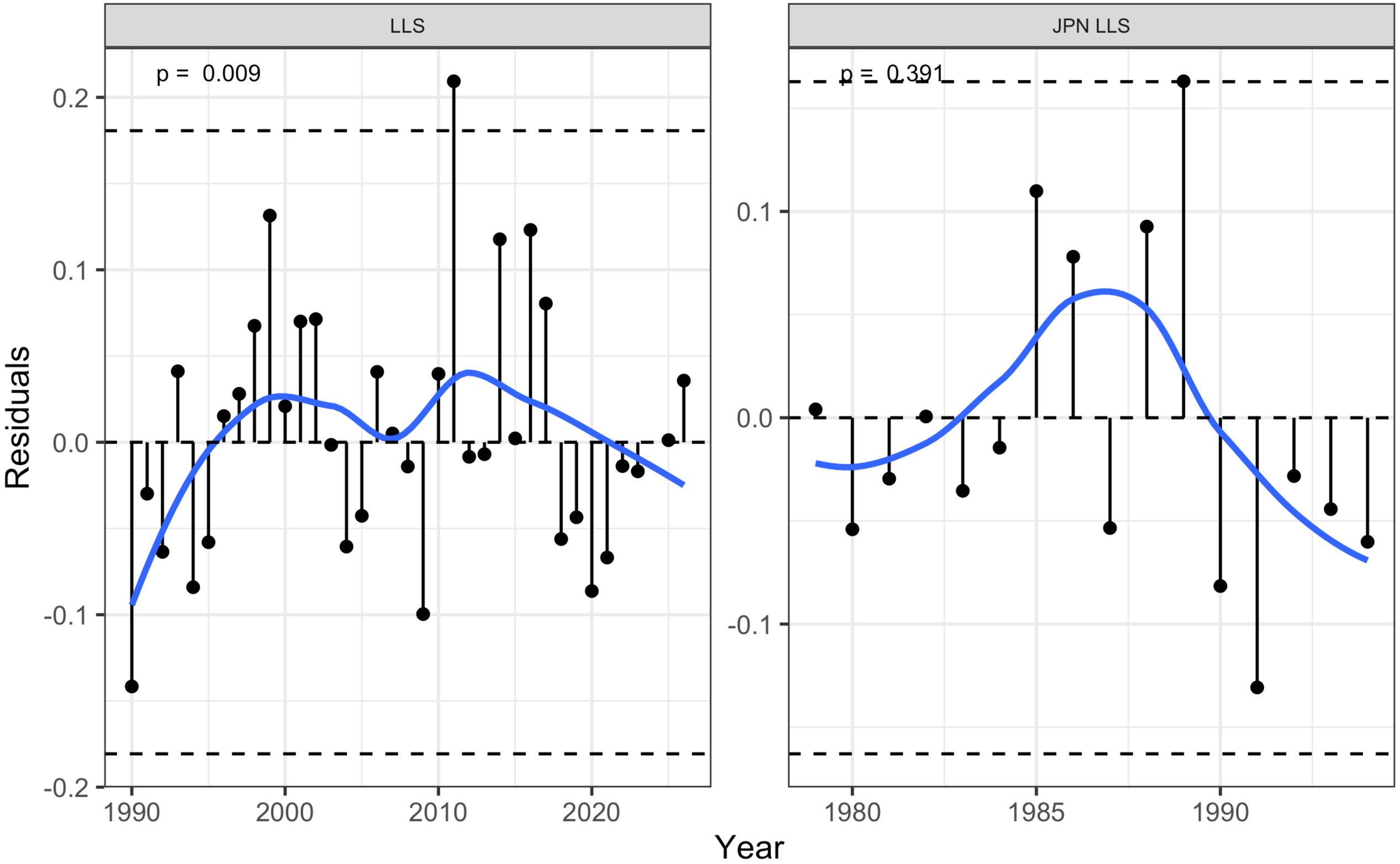
100
80
60
40
20

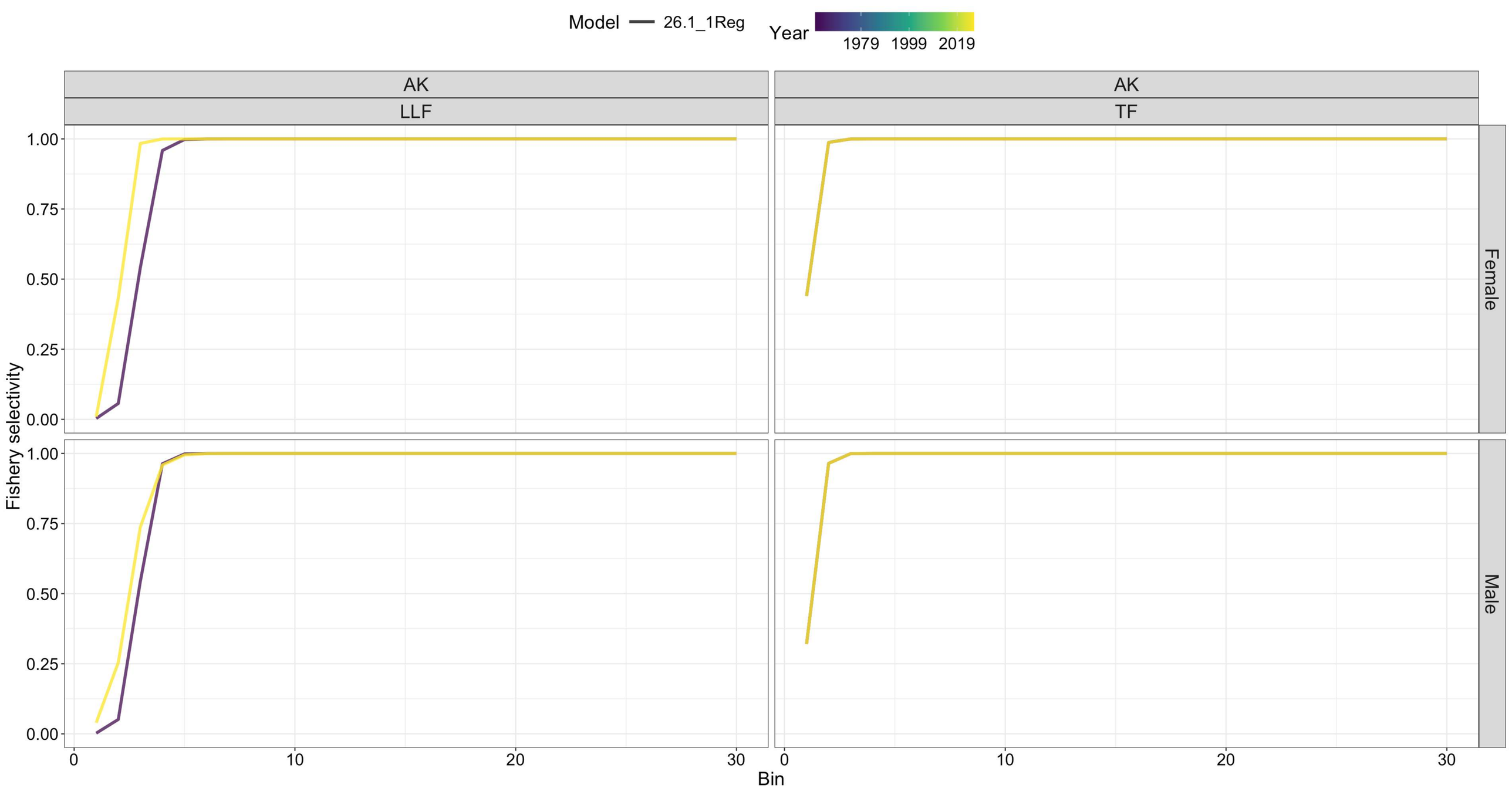
0
1
2
3
4
5
6
7

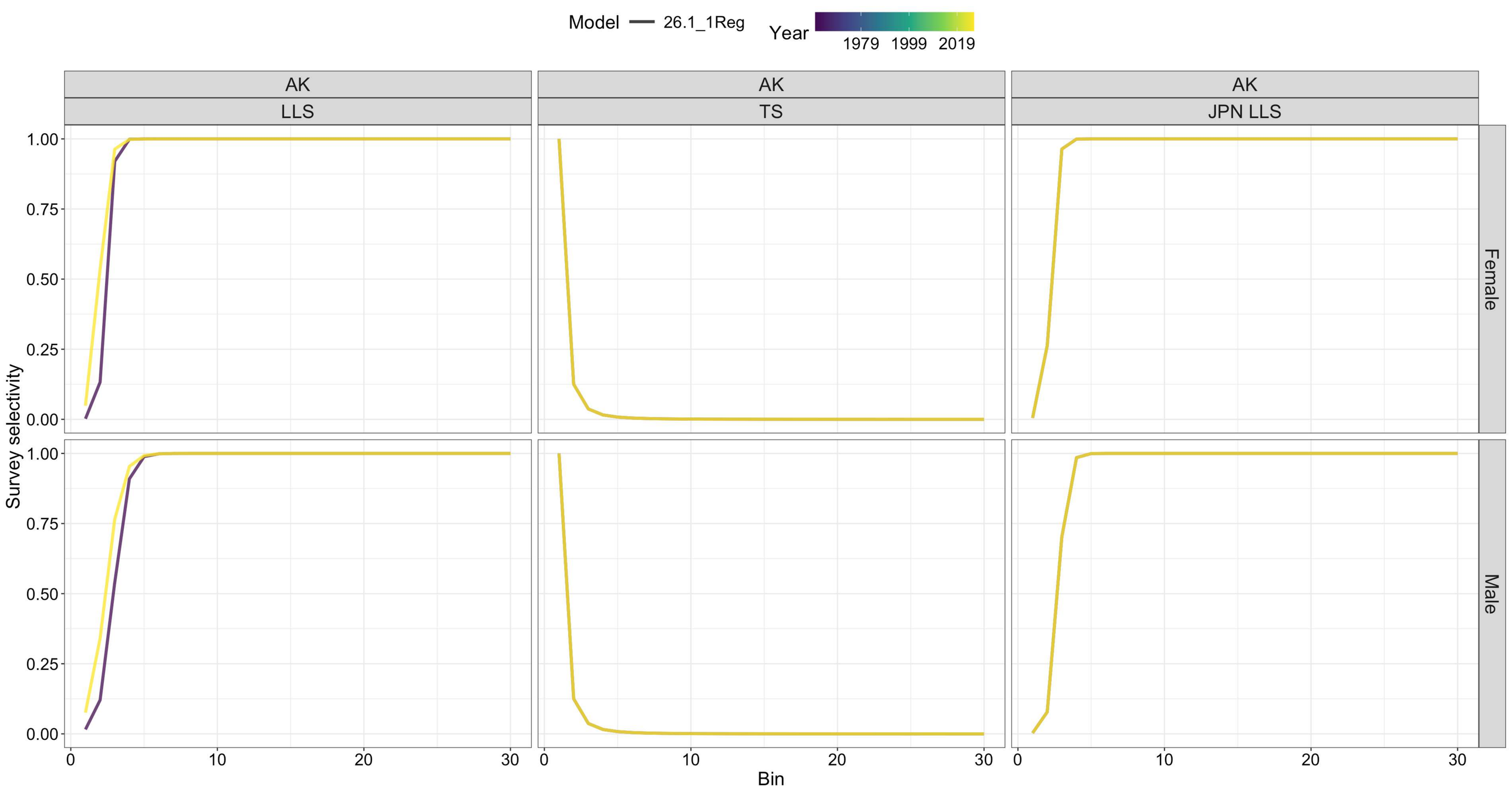
Years since cohort was last estimated

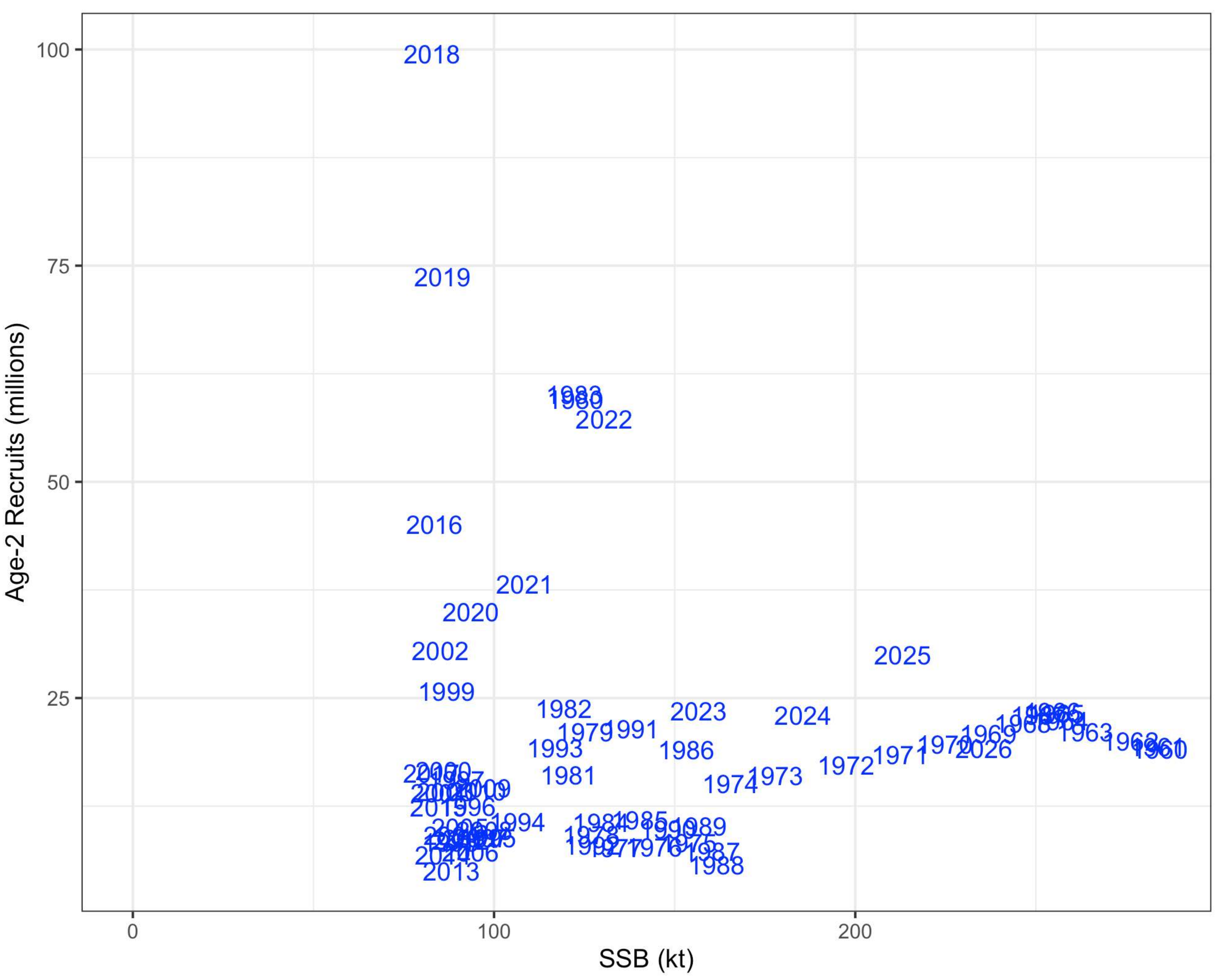
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Index Residuals Runs Test




Model
26.1_1Reg
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.1_1Reg
Year
1979
1999
2019
AK
LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Age-2 Recruits (millions)
100
75
50
25
0
100
200
SSB (kt)
2018
2019
2022
2016
2021
2020
2002
2025
1999
1982
2023
2024
1979
1991
1993
1986
1981
1974
1973
1972
1971
2026
1996
1994
1985
1989
1988
2013

Population and Catch Trends

Recruitment (millions of fish)

100
75
50
25
0

300
200
100
0

SSB or Catch (kt)

1960
1980
2000
2020

Year

Catch
Recruitment
SSB

Model 26.1_1Reg

AK

Dynamic SSB0

Recruitment

SSB

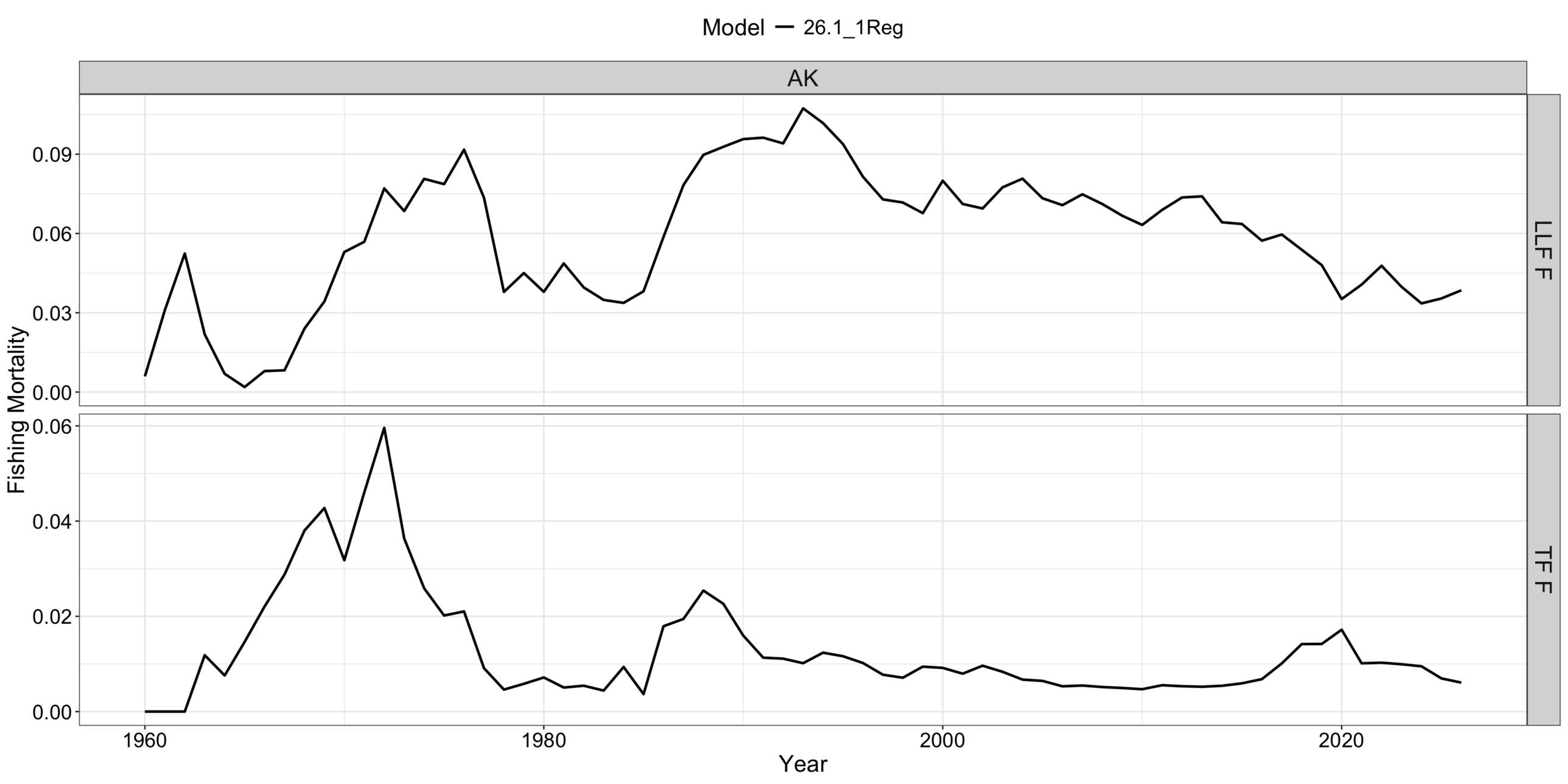
Total Biom

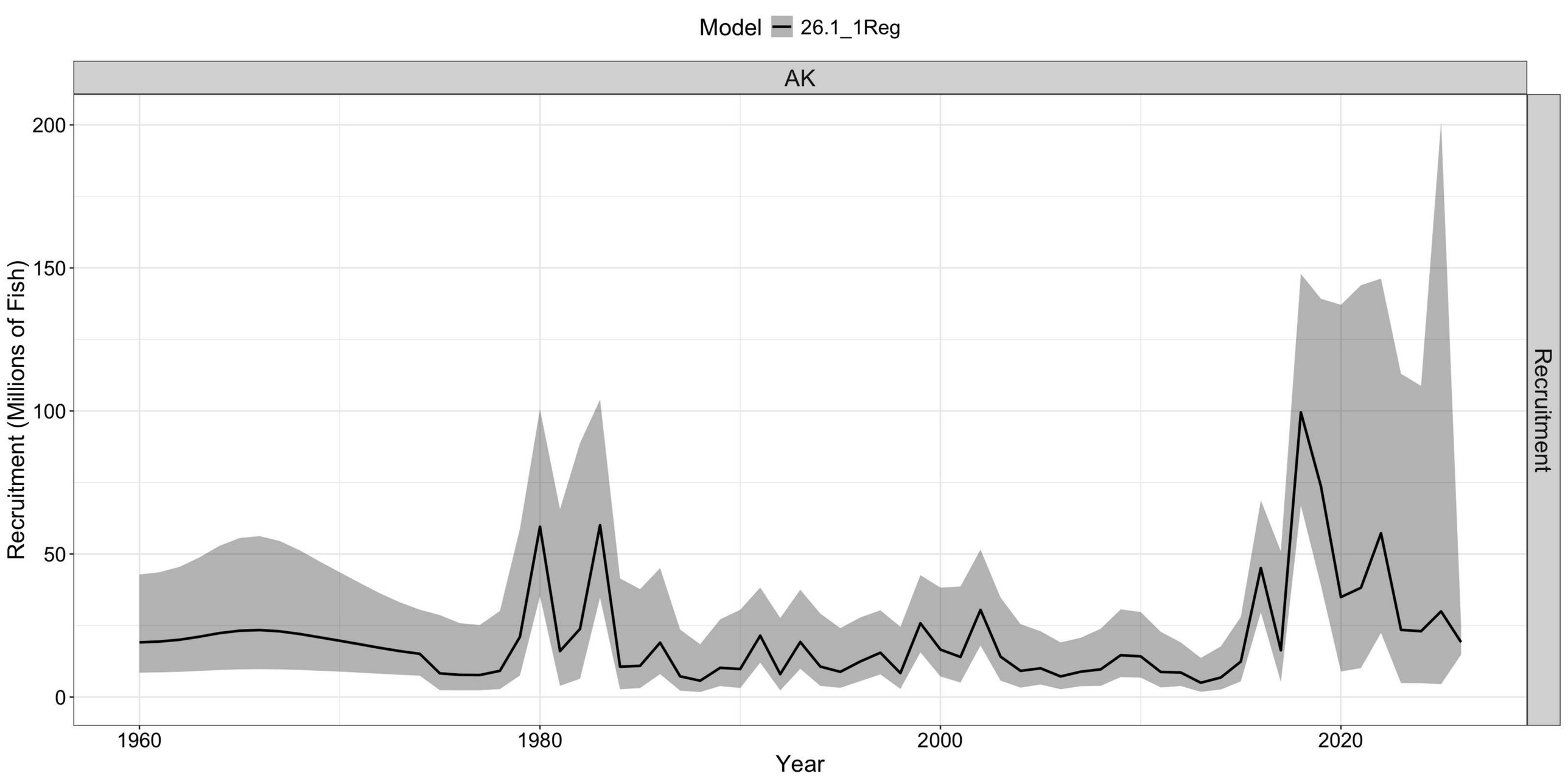
LLF F

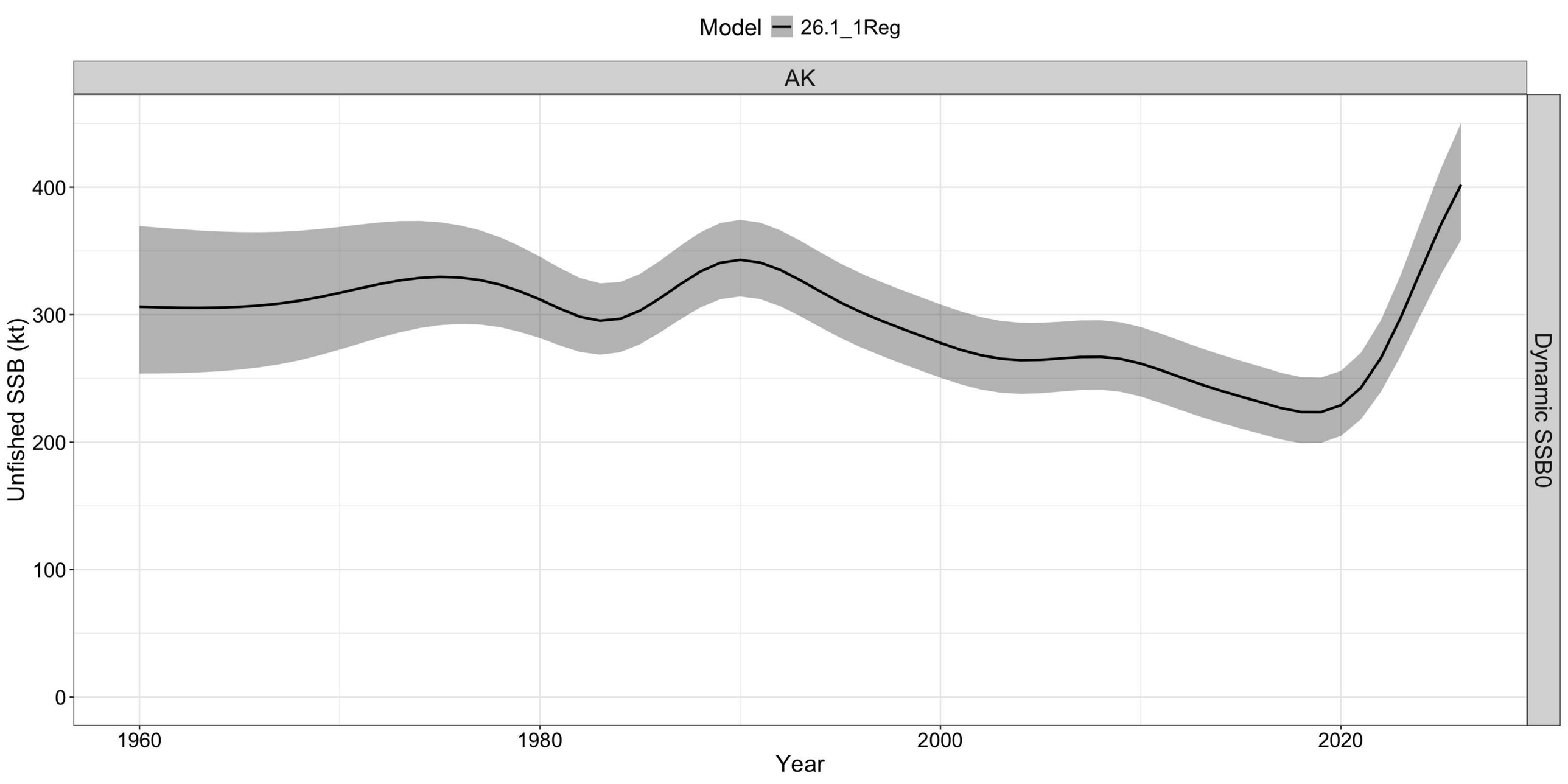
TF F

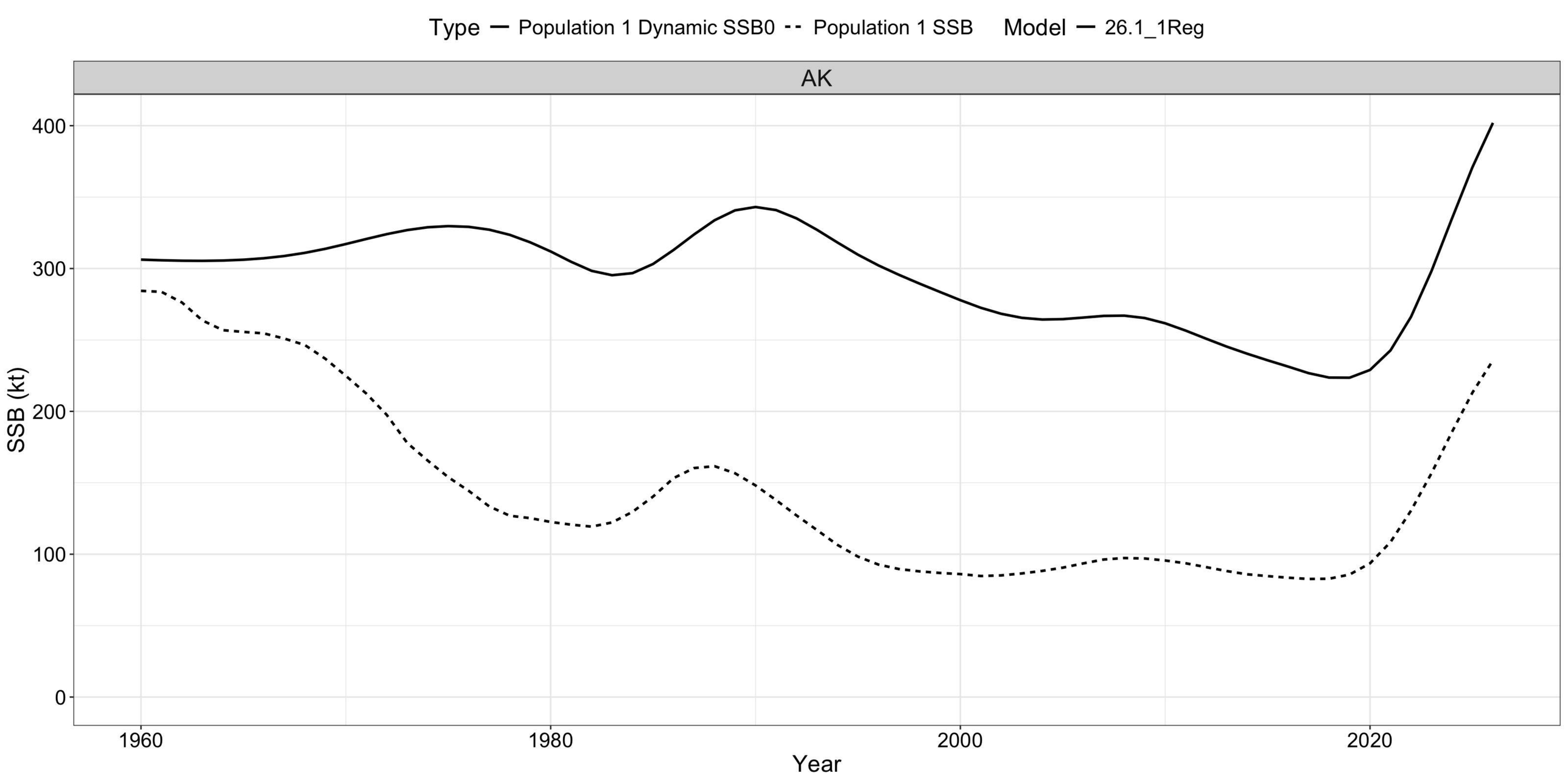
Value

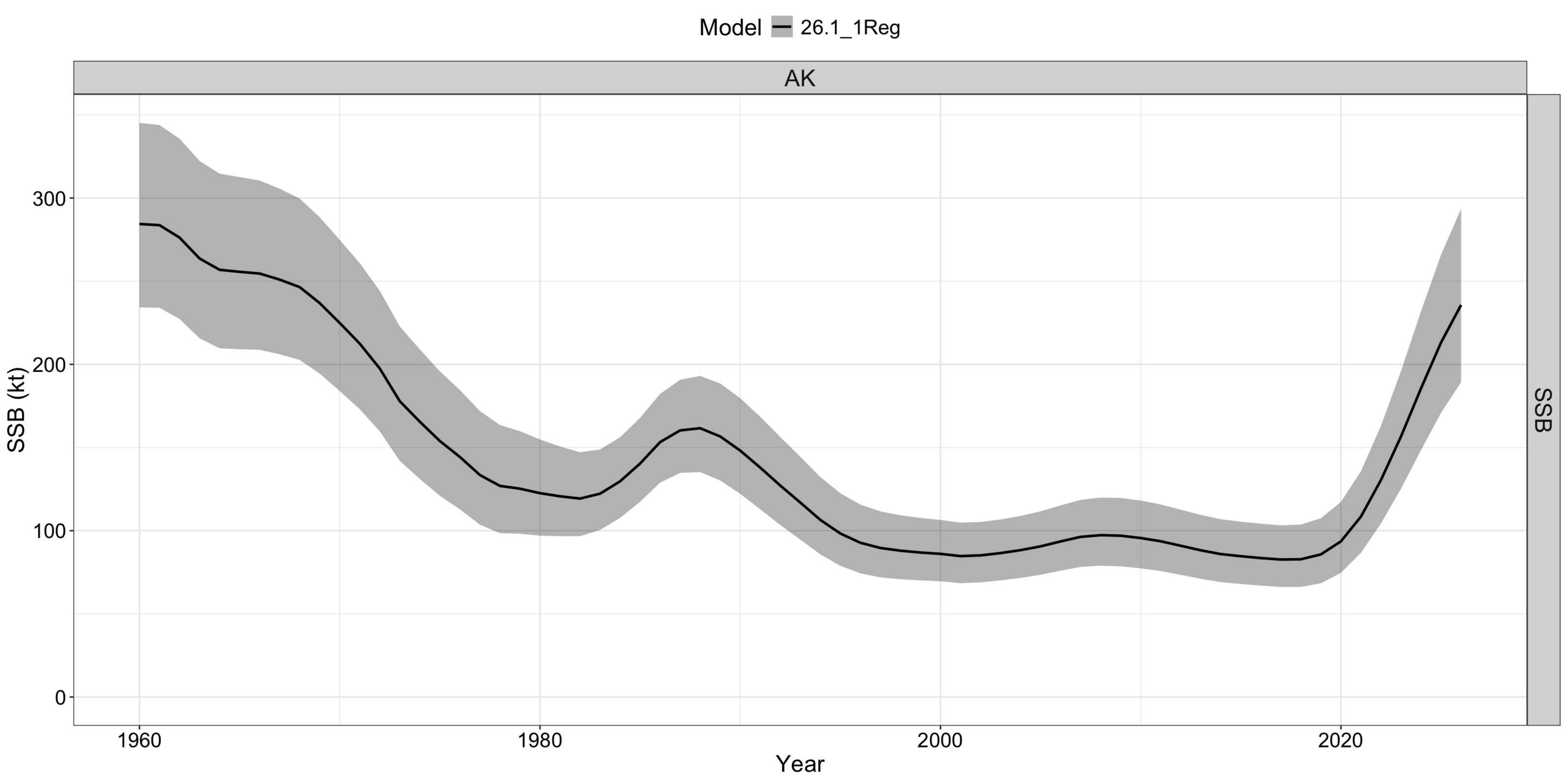
Year

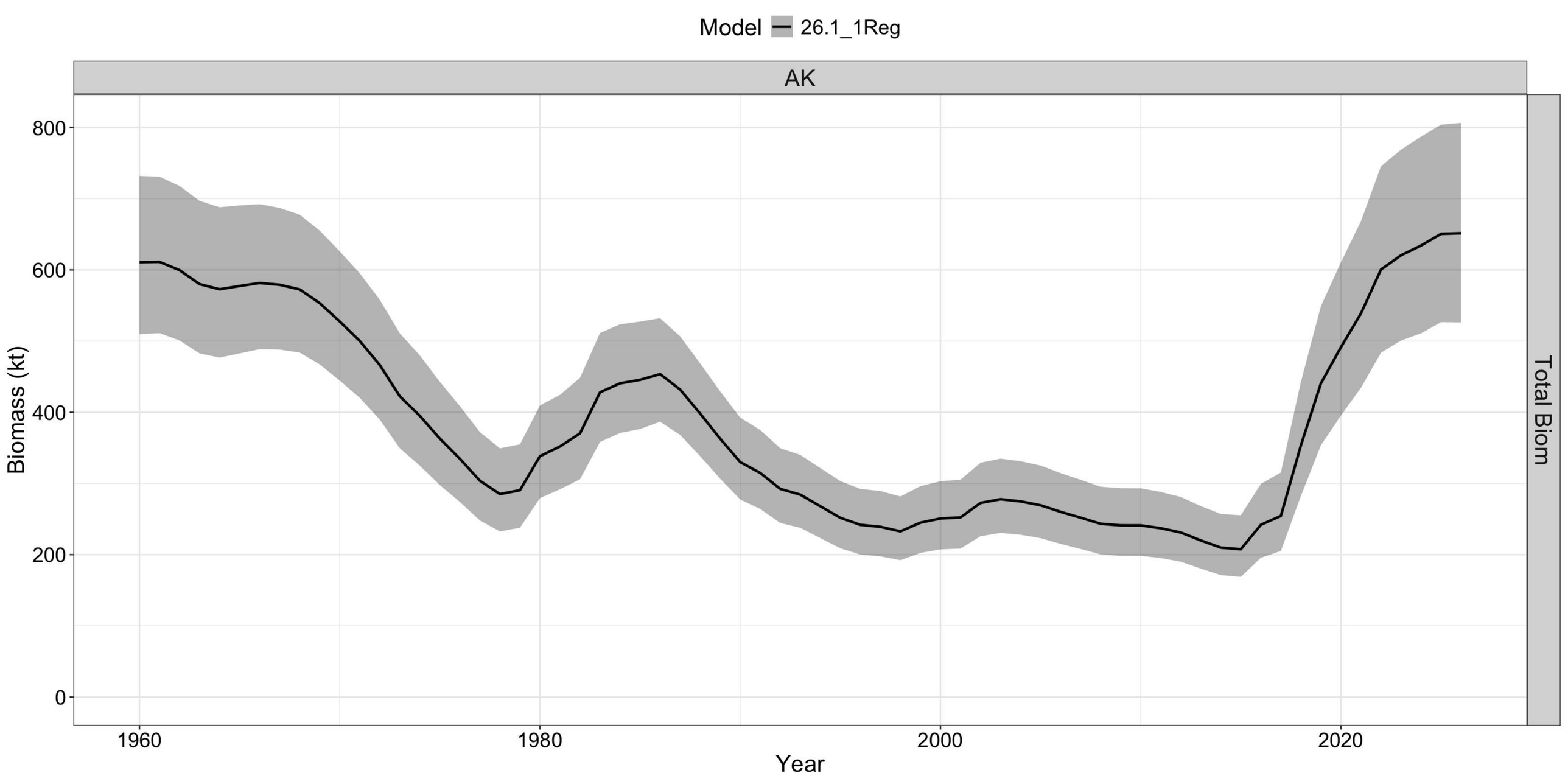

Model — 26.1_1Reg
AK
LLF F
TF F
Fishing Mortality
0.00
0.03
0.06
0.09
0.00
0.02
0.04
0.06
1960
1980
2000
2020
Year


Model
26.1_1Reg
AK
Recruitment (Millions of Fish)
200
150
100
50
0
1960
1980
2000
2020
Year
Recruitment

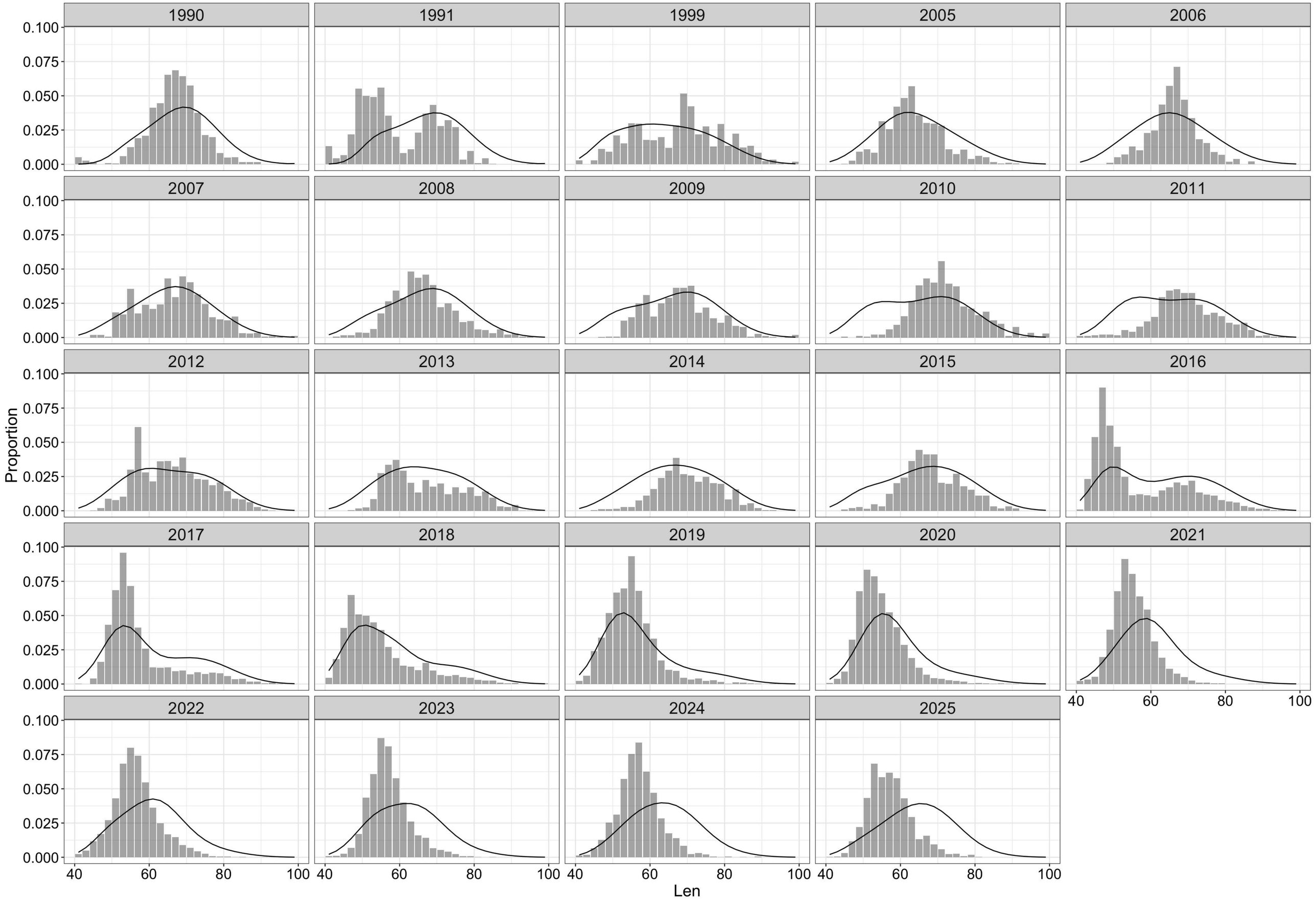

Model
26.1_1Reg
AK
Dynamic SSB0
Unfished SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year


Type — Population 1 Dynamic SSB0 -- Population 1 SSB
Model — 26.1_1Reg
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year

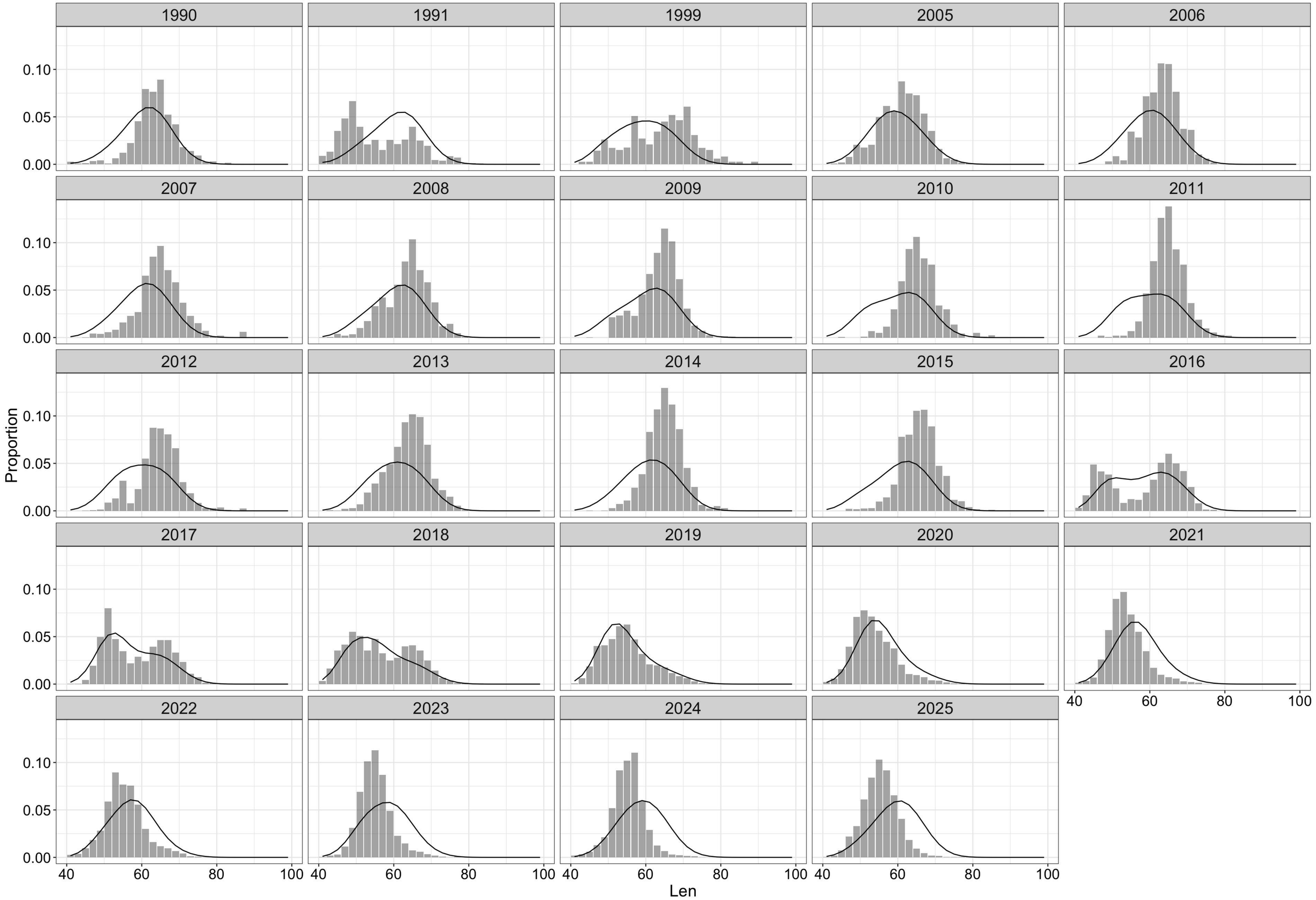

Model
26.1_1Reg
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model
26.1_1Reg
AK
Total Biom
Biomass (kt)
800
600
400
200
0
1960
1980
2000
2020
Year

Trawl Fishery Lengths Female AK



Trawl Fishery Lengths Male AK



Trawl Fishery Lengths OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Len

100
80
60
40

1990 2000 2010 2020

Year

AK

SDNR=0.98
(0.96-1.04)

Sample quantiles

2
0
-2

-2 0 2

Theoretical quantiles

Female

Male

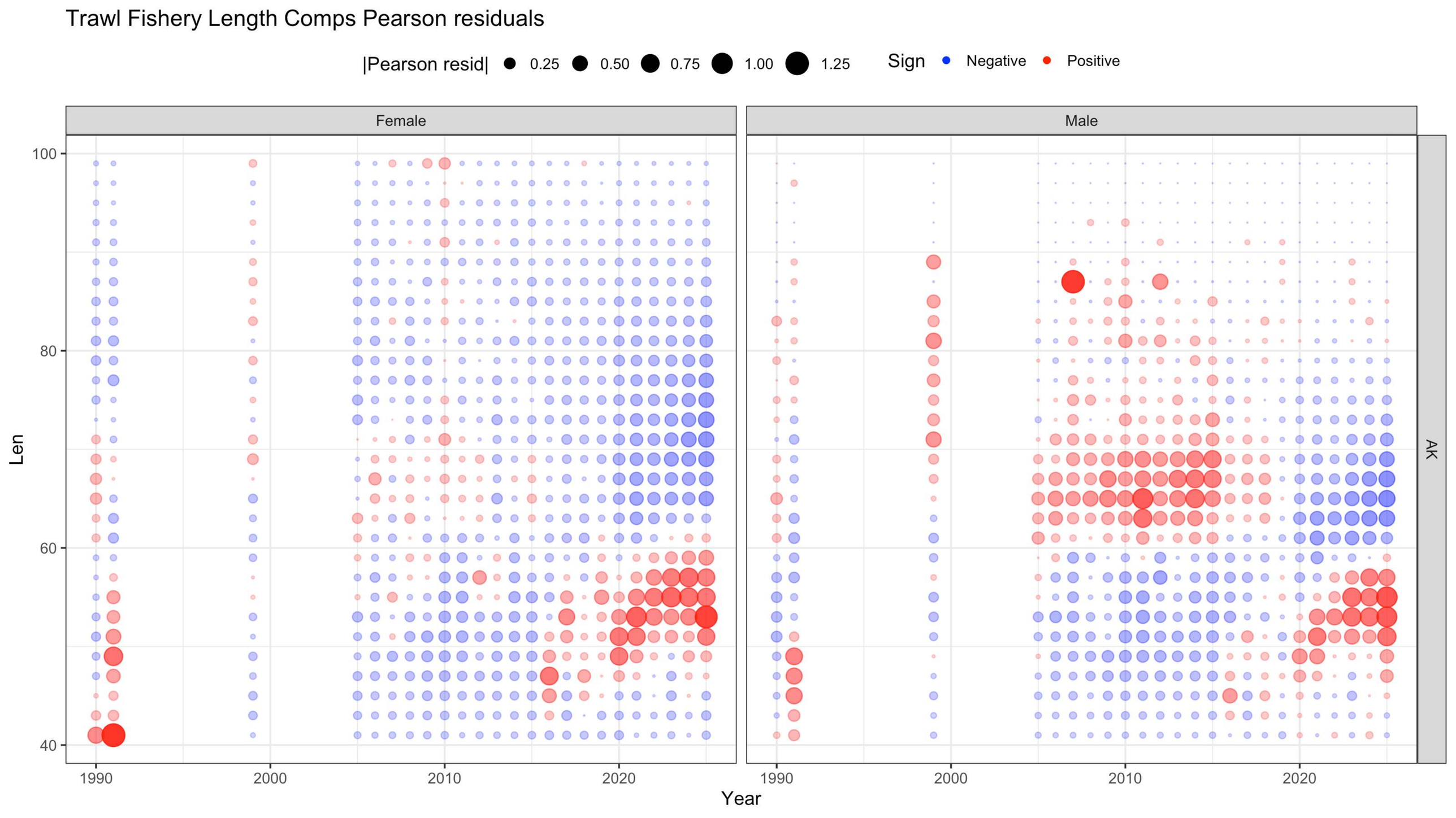
AK

Proportions

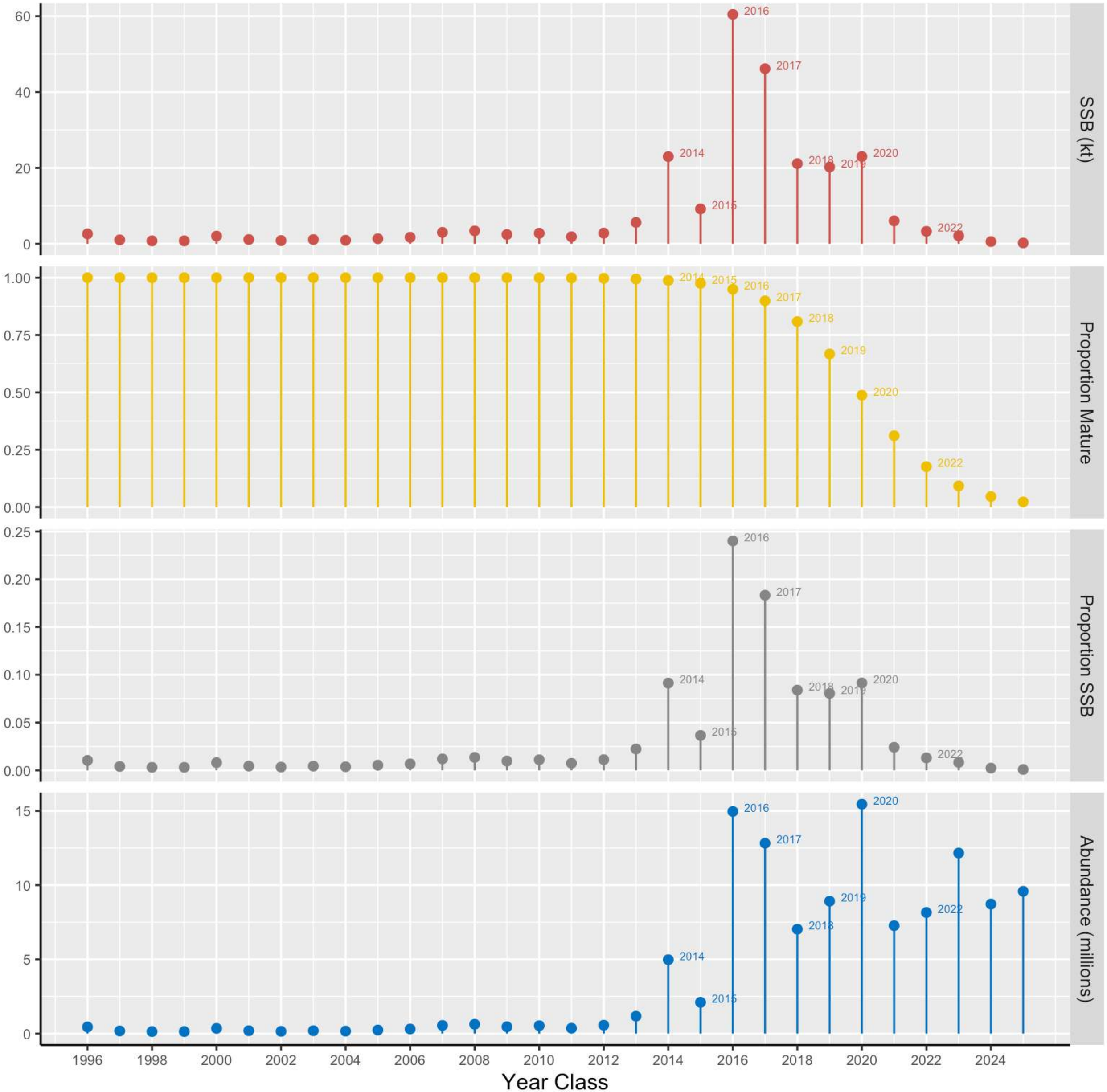
0.06
0.04
0.02
0.00

40 60 80 100

Len


Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.25
0.50
0.75
1.00
1.25
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

Contribution to 2027 SSB by Year Class



JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

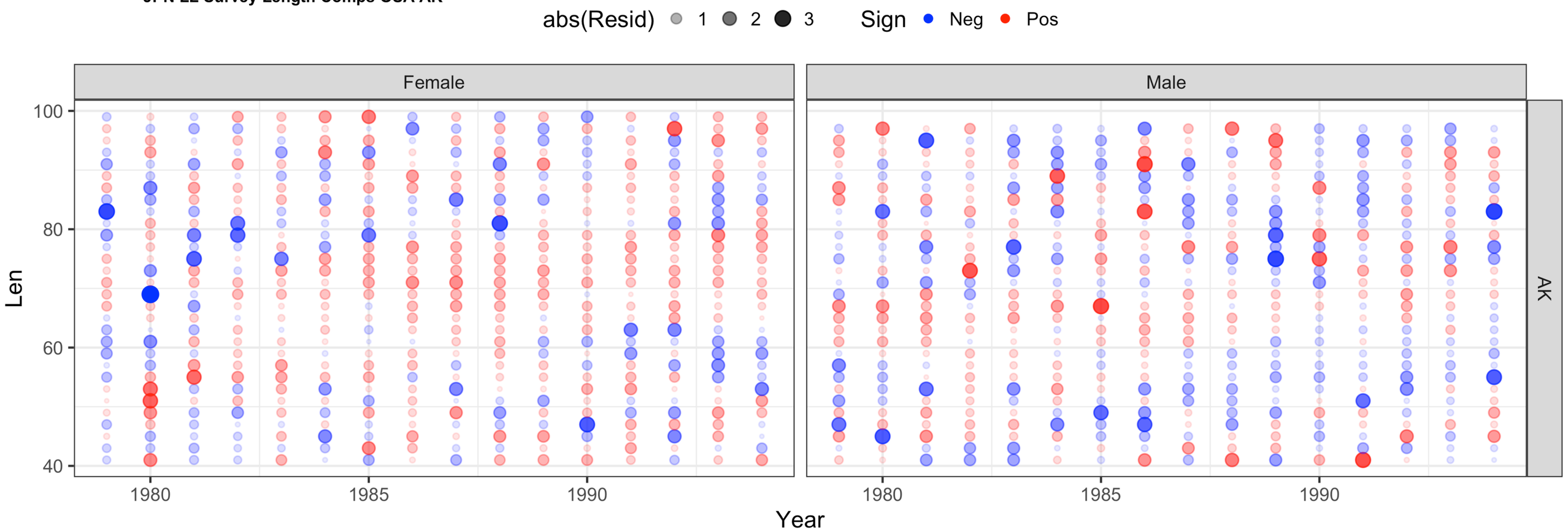
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

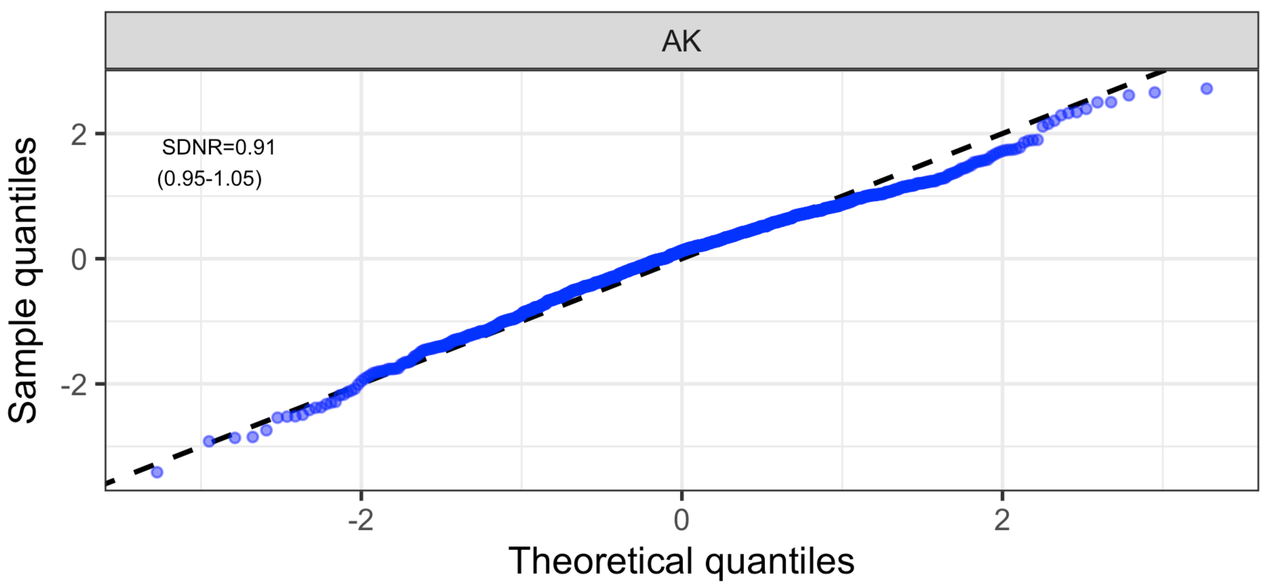
JPN LL Survey Length Comps Male AK

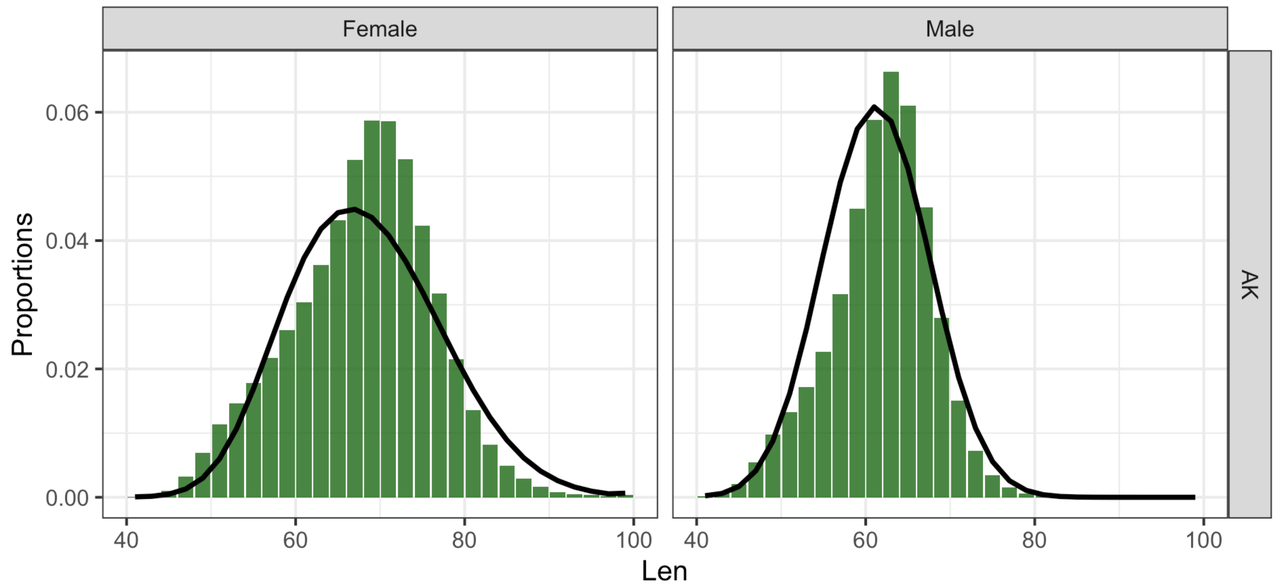
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

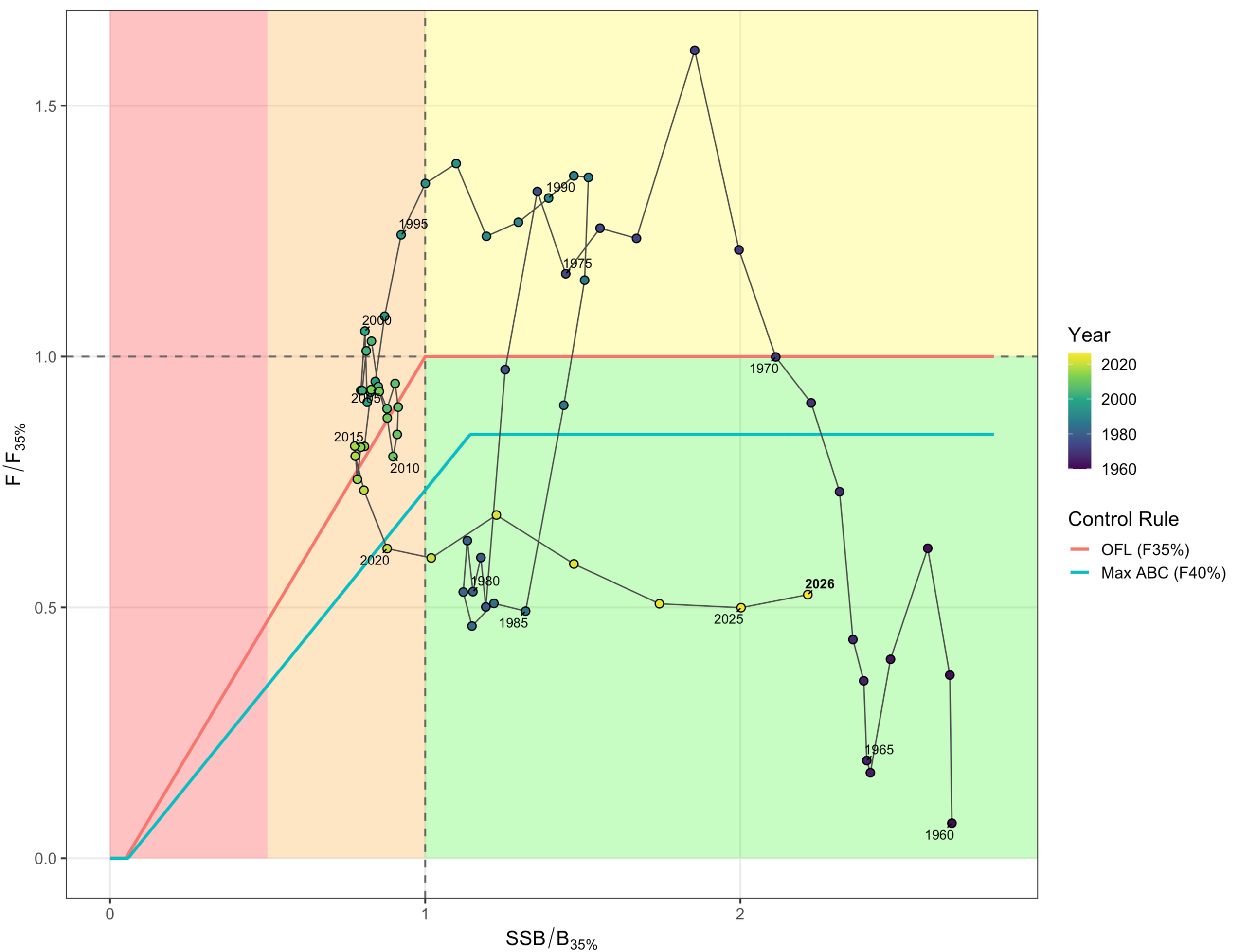






Source	SE	Data_Wt	ISS	ESS
LL Survey Index	0.07	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
LL Survey Ages	NA	0.29	164.21	47.22
JPN LL Survey Lengths	NA	0.34	54.62	18.64
JPN LL Survey Ages	NA	0.18	83.33	14.80
Fixed Gear Fishery Ages	NA	0.12	142.85	17.14
Trawl Fishery Lengths	NA	0.13	47.62	6.26

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.1_1Reg	1	235.7007	401.9635	0.04457618	40.17586	141.8276	0.07169	1.66188	0.58637	0.62182


1.5
1.0
0.5
0.0
F/F35%
0
1
2
SSB/B35%
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
2026
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.091	0.091
M Male (estimated)	0.114	0.114	0.091	0.091
Average Recruitment (millions)	25.9	25.9	21.4	21.4
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	647221	643401
Projected Female Spawning Biomass (t)	219,714	241,217	251836	263641
B100% (t)	302,672	302,672	354569	354569
B40% (t)	121,069	121,069	141828	141828
B35% (t)	105,935	105,935	124099	124099
FOFL	0.102	0.102	0.085	0.085
max FABC	0.087	0.087	0.072	0.072
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	47250	46922
OFLw (t)	58,532	57,797	47119	46791
max ABC (t)	50,283	49,651	40173	39895
ABC (t)	50,283	49,651	40173	39895
ABCw (t)	50,111	49,482	40042	39764
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.02

0.04

0.06

0.08

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
250
500
750
1000

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

1000

750

500

250

0

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
20
40

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0 10 20 30 40 50

2010 2020 2030 2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
25
50
75
100

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

25

50

75

100

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

2010

2020

2030

2040

Year

Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	12013	8007	20021	4405	8409	2403	4805	40042
2027	REMA_LLS_No_wAI	11212	9209	19621	4004	8409	2403	4805	40042
2027	Carryover	10811	10411	18820	4004	8008	2403	4405	40042
2027	REMA_LLS	10411	12012	17618	3604	7608	2002	4405	40042
2027	GOA_Fix	12013	9610	18419	3604	8008	2002	4805	40042
2028	Carryover_No_wAI	11929	7953	19882	4374	8350	2386	4772	39764
2028	REMA_LLS_No_wAI	11134	9146	19484	3976	8350	2386	4772	39764
2028	Carryover	10736	10339	18689	3976	7953	2386	4374	39764
2028	REMA_LLS	10339	11929	17496	3579	7555	1988	4374	39764
2028	GOA_Fix	11929	9543	18291	3579	7953	1988	4772	39764

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	-8	-30	-11	-6	-13	-9	-14	-15
2027	REMA_LLS_No_wAI	-14	-19	-13	-15	-13	-9	-14	-15
2027	Carryover	-17	-9	-17	-15	-17	-9	-21	-15
2027	REMA_LLS	-20	5	-22	-23	-21	-25	-21	-15
2027	GOA_Fix	-8	-16	-18	-23	-17	-25	-14	-15
2028	Carryover_No_wAI	-8	-30	-12	-7	-13	-10	-15	-15
2028	REMA_LLS_No_wAI	-15	-20	-14	-15	-13	-10	-15	-15
2028	Carryover	-18	-9	-17	-15	-17	-10	-22	-15
2028	REMA_LLS	-21	4	-22	-24	-21	-25	-22	-15
2028	GOA_Fix	-8	-16	-19	-24	-17	-25	-15	-15

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	12013	2028	BS	NA	11929
2026	AI	NA	11421	7440	632	2027	AI	NA	8007	2028	AI	NA	7953
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	20021	2028	GOA	NA	19882
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	4405	2028	WGOA	NA	4374
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	8409	2028	CGOA	NA	8350
2026	WY	NA	2652	2652	2271	2027	WY	NA	2403	2028	WY	NA	2386
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	4805	2028	SEO	NA	4772
2026	Total	57797	47008	38986	22161	2027	Total	47119	40042	2028	Total	46791	39764

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	279070	NA	20021	NA	NA
GOA	2028	275904	NA	19882	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	152813	NA	12013	NA	NA
BS	2028	151079	NA	11929	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	168234	NA	8007	NA	NA
AI	2028	166325	NA	7953	NA	NA